

COBORDISM MAPS AND ATIYAH–FLOER ISOMORPHISMS FOR FRAMED INSTANTON HOMOLOGY II

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ABSTRACT. We study the extension to integer coefficients of the cobordism comparison between framed instanton homology and the Lagrangian Floer homology of a Heegaard splitting. Orientations are transported from almost-complex orientations on the gauge side through moduli spaces of the mixed anti-self-duality and holomorphic curve equation. Assuming the unsigned analytic results of the companion paper, we obtain a projective natural isomorphism for connected cobordisms with a framed arc and a bundle trivialized at the ends. The symplectic maps are signed counts of the triangles, compression cups and caps used in the mod-two comparison. The principal additional arguments concern the constancy of sphere-gluing signs, reflection of coherent orientations, and the identification of the diagonal compression cup with continuation. We keep the grading signs in the gauge orientation convention explicit. The result allows one overall sign per cobordism; comparison with homology-oriented integral instanton maps and with relative-spin quilt orientations is left open.

Draft. The principal comparison assumes the unsigned analytic results of Paper I.

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1. INTRODUCTION

1.1. The integral comparison. Let M be a closed connected oriented three-manifold. The framed instanton construction replaces M by $M^\# = M \# T^3$ and uses an $\mathrm{SO}(3)$ bundle which is trivial on M and odd on a torus in the added summand. A Heegaard splitting determines two Lagrangians in the moduli space of flat connections on the stabilized surface. Daemi, Fukaya and Lipyanskiy construct an isomorphism between the resulting gauge and symplectic Floer homologies by counting solutions of a mixed equation [1]. The mixed equation couples the anti-self-duality equation on a four-dimensional region to a holomorphic curve equation on a surface; the two solutions have matching flat boundary values.

Paper I [19] concerns compatibility with four-dimensional cobordisms over $\mathbb{F}_2$. Its symplectic maps count holomorphic triangles and compression cups and caps. The two analytic comparisons there are a collapse of a compression body next to a holomorphic strip and an excision along a torus in a mixed domain. The purpose of this paper is to give the orientation arguments required for integer coefficients. We retain those geometric maps: the symplectic cobordism map is not defined by conjugating an instanton map through an isomorphism.

There are three distinct orientation questions. First, the quotient by the lifting class used in framed instanton homology must preserve orientations. Second, the collapse and torus-excision identifications must preserve the orientations of individual solutions. Third, the orientation system transported through a reflected mixed domain must be compared with the dual of the original system. The last comparison is needed to turn the tensor supplied by the compression cup into a map with the source complex used in Paper I. An unspecified sign on each generator would not suffice: the ambiguity in a cobordism map must be a single sign.

We use the almost-complex orientation construction of Kronheimer–Mrowka [8, Section 5.1]. It is invariant under bundle automorphisms. The almost-complex structure on a cobordism is fixed near the odd torus, but its restrictions at the two ends are allowed to vary. Conversion operators then identify the end orientation lines with fixed object lines. Their placement relative to the generator lines matters when a conversion is odd. We denote the resulting gauge functor by $I_J^\#$ to record this convention.

1.2. Statement. The category $\mathbf{B}$ is described in Section 2. An object is a closed connected oriented three-manifold with the framing data needed for its torus summand. A morphism is a connected oriented cobordism W equipped with a normally framed arc ν joining its ends and an $\mathrm{SO}(3)$ bundle E trivialized at the ends and along the arc. A presentation p consists of a Heegaard splitting, metric data, small primary holonomy perturbations, and regularizing data. The additional smallness and reflection conventions for integer coefficients are specified in Section 13.

Write $C_G(M, p)$ for the integral gauge complex with the almost-complex orientation convention, and $CF(M, p)$ for the symplectic complex with transported orientations. The graded homotopy category $\mathcal{K}(\mathbb{Z})$ and the notation $F \simeq \pm G$ are defined in Section 3. For an admissible decomposition d of a cobordism, $S_d(W)$ is the composite of signed holomorphic triangle maps, compression cups, compression caps, and continuation maps.

The geometric arguments use the unsigned compactness and gluing results of Paper I, together with two analytic inputs stated in Section 2.4 and proved in the appendices: the Fredholm and determinant-line foundations of Proposition 2.1, and relative preparation and shifting of compact families of mixed configurations (Proposition 2.2), the family version of the shifting lemmas of Daemi, Fukaya and Lipyanskiy on the domains used here. These

require more than the existence of the corresponding mod-two counts. The oriented torus-completion comparison is derived in Section 8 from these inputs and the unsigned analysis of Paper I.

The ordinary determinant-one self-gluing used in gauge excision is established in Proposition 2.3, for the regular cut data constructed in its proof. This is an application of ordinary instanton compactification and gluing, with the lifting groups and orientation convention retained explicitly.

Theorem A (Integral comparison). *Assume the unsigned analytic results of Paper I. With the orientation and admissibility conventions of this paper, the mixed maps*

$$N_p : C_G(M, p) \longrightarrow CF(M, p)$$

are isomorphisms of finite free integral complexes. For every morphism $(W, E, \nu) : M_0 \rightarrow M_1$, admissible endpoint presentations p_0, p_1 , and admissible decomposition d , one has

$$(1.1) \quad N_{p_1} I_J^\#(W, E, \nu) \simeq \pm S_d(W, E, \nu) N_{p_0} \quad \text{in } \mathcal{K}(\mathbb{Z}).$$

The sign is one sign for the entire map. Consequently the symplectic maps are independent of the decomposition up to sign and chain homotopy. On homology they define a projective functor and $H(N)$ is a projective natural isomorphism from $I_J^\#$ to that functor. Each symplectic homology map has the parity of the corresponding gauge map.

At objects with nonzero integral gauge homology the symplectic maps are the geometric maps of Paper I with the orientations constructed here. If the integral gauge homology vanishes, the complex is contractible; maps incident to it represent the zero morphism in $\mathcal{K}(\mathbb{Z})$, with the convention of Section 12.5. Reduction modulo two recovers the maps of Paper I, up to chain homotopy in the contractible case.

The qualifier “projective” has the usual literal meaning: maps are identified with their negatives. No system of signs making all composition identities exact is asserted. In particular, multiplying individual elementary maps by signs is permitted, whereas multiplying different matrix entries by unrelated signs is not.

1.3. The two orientation comparisons. The proof of Theorem A separates two further statements. They use the same assumptions as that theorem.

Theorem B (Orientations under collapse). *The bijection of regular index-zero solutions in the compression-body collapse of Paper I preserves the transported determinant-line orientations. This assertion uses the metric and the admissible linearizations specified in Section 5. It is compatible with ordinary gauge and strip gluing and with compact parameter families. The two-handle comparison therefore holds over $\mathbb{Z}$, up to the single sign arising from the auxiliary orientation data of the cobordism.*

The *sphere character* is the sign-valued character of $\pi_2(\mathbf{M}_h)$ recording the change of a coherent orientation when a sphere is attached to a strip configuration (Section 10). Its constancy does not compute its value, and the reflection statement does not assert that the character is trivial. Reflection interchanges the two compression bodies; $\bar{p}$ is the resulting presentation of $-M$. The admissible primary ball is a sufficiently small ball in the finite-dimensional perturbation space with the norm (10.2), chosen so that the perturbed compression-body Lagrangians remain smooth embeddings.

Theorem C (Reflection and the diagonal cup). *Fix a Heegaard splitting, metric data, and an almost-complex base structure at an object with nonzero integral gauge homology. In*

the admissible ball of primary perturbations, the sphere character of the transported strip orientation system is constant and agrees with that of the reflected presentation. There is consequently a reflection isomorphism

$$\Phi_p : CF(-M, \bar{p}) \longrightarrow CF(M, p)^\vee$$

determined by the configuration-level orientations up to one sign. With this pairing, the compression cup with no handle attached is chain homotopic, up to one sign, to the symplectic continuation map.

This distinction avoids imposing an unnecessary relative-spin orientation convention on the compression correspondence.

1.4. Strategy. The integral mixed map has a unit diagonal in the filtration used in Paper I. The diagonal configuration is a constant pair, and its orientation is the canonical orientation of an invertible operator. The resulting integral isomorphism is stronger than a mod-two isomorphism: an odd integer coefficient on the diagonal would not be enough.

For two-handles, determinant lines are transported through the collapse family. The relevant continuity is continuity over smooth configurations with all derivatives controlled on compact subsets and with the prescribed asymptotic behavior. Strong convergence of solutions alone does not identify all operator families used in an orientation argument. We state the needed realization and completion properties explicitly and prove the sign comparison from them.

For one-handles, a folded triangle gives a tensor in the product of the reflected source complex and the target complex. Excision identifies this tensor with a gauge handle state. This gives a cup formula before any reflection pairing is inserted. The remaining problem is to identify the pairing with the one for the actual symplectic continuation maps. We first prove that the sphere character is constant under small changes of primary data, then compare it across a fold. A rigidity argument for coherent systems identifies the diagonal cup with continuation. The cup formula now gives small-primary naturality, mixed duality, and the one-handle square. Reflection gives the three-handle square.

Finally, the algebraic descent uses chain homotopies throughout. Over $\mathbb{Z}$, equality on homology is generally insufficient to prove that two chain maps are homotopic. The proof therefore retains the homotopies supplied by the one-dimensional moduli spaces and uses the invertibility of the mixed maps only after the elementary squares have been established.

1.5. Scope of the draft. The almost-complex convention used for $I_J^\#$ is part of the statement. We do not identify it with Scaduto's homology-oriented integral cobordism maps [13]. An odd conversion can introduce a grading operator rather than an overall sign. Section 4 explains this point and the additional grading information needed for a comparison. We likewise do not identify N_p with the integral map of [1]; their reductions modulo two are compared in Paper I at common admissible data.

The orientations on holomorphic curves are transported from the gauge side. Comparison with the relative-spin orientations used in quilted Floer theory [16, 18] is a separate question. The underlying curves and correspondences are the same ones as in Paper I, but an equality of mod-two counts does not supply this integral orientation comparison. Disconnected objects, units, counits, closed pairings, and nontrivial bundles on the objects are outside the present statement. Bundles on cobordisms may be nontrivial relative to the specified trivialized ends.

2. GEOMETRIC DATA AND ANALYTIC FOUNDATIONS

2.1. Framed objects and cobordisms. Fix an oriented ball in M and form $M^\# = M \# T^3$ along that ball. Write $T^\circ = T^3 \setminus \text{int } B^3$. Choose a torus $T_0 \subset T^\circ$, a neighborhood $N(T_0)$, and an odd bundle $P_T \rightarrow T^3$ with $\langle w_2(P_T), [T_0] \rangle = 1$. The bundle $P_M \rightarrow M^\#$ is obtained by joining P_T to the trivial bundle on M . The lifting class

$$\alpha_M = \text{PD}[T_0] \pmod{2} \in H^1(M^\#; \mathbb{Z}/2)$$

acts on the determinant-one configuration space; the framed complex uses the quotient by $\phi_M = \langle \alpha_M \rangle$.

For (W, E, ν) as in the introduction, remove a tube around the normally framed arc and insert $T^\circ \times [0, 1]$. Denote the result by $(W^\#, E^\#)$. The lifting subgroup ϕ_W is the one induced by this torus tube, with the endpoint restrictions ϕ_{M_i} . At an intermediate disconnected cut we retain the full subgroup specified by the tube; it need not be the product of the groups on the components. This distinction is essential in the excision calculation in Section 8.

The category identifies isomorphic decorated cobordisms and uses gluing along the fixed endpoint data for composition. The connected source and target condition permits a handle decomposition with one-, two-, and three-handles, together with changes of presentation. As in Paper I, elementary words also include metric changes, Heegaard moves and frame changes. The presentation graph is connected and the gauge cobordism map is independent of such a word. These are the geometric inputs to the descent at the end of the paper.

2.2. Compression bodies and the symplectic complex. If $M = H_0 \cup_\Sigma H_1$ is a Heegaard splitting of genus g , adjoin $[0, 1] \times T^2$ to each handlebody. The resulting torus compression bodies C_i have boundary $\Sigma_h \sqcup T^2$, where $h = g + 1$. Let $\mathcal{M}_h$ be the moduli space of flat connections on the odd $\text{SO}(3)$ bundle over Σ_h , with the lift convention of [1, Section 2]. Its symplectic form is the usual Atiyah–Bott form. Restriction of flat connections on C_i gives a Lagrangian $L(C_i) \subset \mathcal{M}_h$.

Primary perturbations h_i are finite sums of averaged holonomy functions supported in C_i° , away from the fixed collars. Their critical equations are

$$(2.1) \quad *_3 F_B + \nabla_B h_i = 0.$$

We use a sign-sensitive trace convention for fixed-determinant $\text{U}(2)$ holonomy: on each solid torus choose a continuous square root of the determinant holonomy and take the trace of the resulting $\text{SU}(2)$ part. The choices are made once for the finite family. Equation (2.1) determines perturbed Lagrangians $P_i = L_{h_i}(C_i)$. At transverse data, their intersection points correspond to the gauge critical orbits.

For each intersection point x , let a_x be the corresponding gauge critical point and $L_G(a_x)$ its real determinant line. For a real line L , write $\mathfrak{o}(L)$ for the rank-one free abelian group generated by its two orientations with the relation $[-o] = -[o]$. The integral symplectic chain group is

$$(2.2) \quad CF(M, p) = \bigoplus_{x \in P_0 \cap P_1} \mathfrak{o}(L_G(a_x)).$$

The differential counts rigid strips with the transported orientation system of Section 6. The mod-two reduction of (2.2) is the chain group of Paper I.

2.3. Mixed configurations. A mixed domain consists of a gauge region and a symplectic surface meeting along a matching boundary. On the gauge side a connection satisfies a perturbed anti-self-duality equation. On the symplectic side a map to M_h , or to a product of such spaces, satisfies a perturbed Cauchy–Riemann equation. Along a matching boundary the surface restriction of the connection is flat and represents the value of the map. At ordinary ends we prescribe irreducible nondegenerate gauge limits or transverse Lagrangian intersection points. At mixed ends we retain the finite-dimensional stationary coordinates explicitly.

The mixed domains used below are the mixed vertex domain, mixed and folded triangles, their strip gluings, the closed-cut mixed domain, the finite-width torus domains, their completed limits, and the collapse families. We also use ordinary strips, continuation strips, and the disks obtained by doubling and capping a folded triangle. Statements about determinant lines apply on configuration spaces, not just on their solution subsets.

We use the smooth topology on these spaces: in fixed end trivializations all derivatives converge on compact sets, the asymptotic limits converge with their parameters, and the decaying parts converge in the chosen weighted norms at every derivative order. On a completed torus domain of Paper I, Definition 10.1(b), the torus tail $T \times [0, \infty)_r \times \mathbb{R}_\theta$ is one of these ends, with limit the odd flat connection α : in the fixed product trivialization of the tail, the difference of the connection from α is a decaying part, measured in H_δ^m with the weight $\tau_0(\theta)$ of Paper I, Definition 10.1. A continuous compact family means a family continuous in this topology. Sobolev completions are used locally to construct Fredholm operators and determinant lines. They do not replace the smooth topology when a holonomy family is differentiated repeatedly.

2.4. Analytic foundations and shifting. The unsigned moduli spaces, their energy-index estimates, and the low-dimensional compactifications are those of Paper I. The linear analysis used in addition is collected in the next proposition. It is proved in Appendix A from the local theory of [1, 2], the end models of Paper I, and standard results on pseudoholomorphic curves.

Proposition 2.1 (Analytic foundations). *For each domain listed above, and each compact family of smooth configurations and data used below, the following hold. The families may vary the almost-complex structures and perturbations smoothly, the metrics among those which near every matching seam are products up to a conformal factor depending on the base coordinates, the finite torus lengths and widths in compact ranges, and the collapse scale in compact intervals of positive scales; on the completed torus domains they are compact in the topology above.*

- (a) *Every smooth configuration with fixed asymptotics has local matching-boundary charts. The realized operators of Definition 5.1 and their formal adjoints on the dual weighted spaces satisfy local estimates and compact-remainder estimates uniform on the family. They are Fredholm, their kernels and cokernels consist of smooth elements, and they form continuous families. Changes of chart and the admissible changes of realization give the same determinant line. The fixed-positive-scale affine-center comparison, including nonsolutions and moving stationary limits, is Proposition 7.3.*
- (b) *At ordinary ends the weights lie in common spectral gaps; at mixed ends the stationary coordinates are separated from the decaying part. The scalar Neumann operator $d_A^* d_A$ is invertible on the weighted spaces at every configuration whose restriction to an odd surface or torus collar is irreducible, with bounds uniform on the family. Regular zero-dimensional solutions and the constant reference pairs with no secondary or*

inhomogeneous data have invertible realized operators, including the constant pairs on the finite, completed and joined torus domains and on the closed-cut piece; by openness the same holds for sufficiently small data. For the preglued families of Lemma 8.7 the estimates of (a) are uniform as the torus lengths tend to infinity, with the completed domains as limits. At positive collapse scale they are local on compact scale intervals bounded away from zero; no uniform bound for the full mixed inverse or a fixed chart radius as the collapse scale tends to zero is asserted, and Lemma 7.1 uses its stated logarithmic loss. The zero-scale limit domains are included.

- (c) *The weight comparison for physical solutions and homotopies is Proposition 5.3. The pure symplectic continuation, cap and strip-gluing moduli spaces have regular low-dimensional compactifications, with no additional bubbling boundary in the dimensions used here, and along the gluing paths from preglued configurations to regular solutions determinant transport preserves the canonical orientations. On a disc containing an interior node and separated from the matching and boundary collars, the real Cauchy–Riemann part of a mixed or pure operator glues at the node: the determinant lines form a continuous line bundle in the plumbing topology, the node contributes its complex fibre, canonically oriented and even, and node gluing is associative and commutes with ordinary strip gluing. In the collapse application the positive plumbing parameter is fixed first and the endpoint insertion lies in an external strip.*

The relevant asymptotic restriction is strict: the weights lie in the appropriate signed primal gaps; adjoints use the opposite signed dual weights. Every stationary and parameter column must belong to the weighted target. The affine comparison uses a common sufficiently small weight in its actual indicial gaps. It is not assigned the whole physical solution gap. Full-gap independence in part (c) concerns the physical solutions and counts. Uniformity is asserted for a fixed compact family, not over all Heegaard splittings or all primary perturbations.

The second input concerns relative preparation and shifting in families. Daemi, Fukaya and Lipynskiy state the one-input shift on their special quintuple and sketch its family version [1, Lemmas 4.62 and 4.64]; the domains, topology and families used here, and the several-input form, are not covered there. The following proposition is proved in Appendix B.

Proposition 2.2 (Relative preparation and shifting). *Let a compact family of smooth configurations be given on one of the domains below, with fixed asymptotic labels, where the data and geometry may vary as in Proposition 2.1.*

- (a) *On a one-input mixed domain (the vertex domain, a separate finite or completed torus domain, or the closed-cut piece with its ordinary input Z), relative preparation makes the original tails stationary beyond a common cutoff, preserving members already stationary there, and the subsequent shift deforms the family to gauge paths followed by a constant mixed pair C^c , where c is a representative of the critical point over the output label (Appendix B), keeping members which are already of this form in that subspace.*
- (b) *On the triangles and folded triangles at positive collapse scale, and on the finite-width joined torus domains, a deformation of the form described in Section 5 retains the ordinary cobordism and reaches the reference described there, taken for the torus domains at the closed cut of Paper I, Lemma 10.18; its relative form preserves that terminal subspace.*

These deformations are continuous in the topology of this section jointly in configurations and data, and coherent with ordinary end gluing. Limiting triangle orientations are defined by inverse collapse, not by shifting to a constant triangle.

The oriented comparison at torus completion is not an assumption. Section 8 proves it from Proposition 2.1, Proposition 2.2 and the unsigned analysis of Paper I, Section 10. There Proposition 2.2 is used on the separate finite and completed torus domains and on the closed-cut piece, directly and through the localized shift on the joined domains, in the topology of Section 2 in which the torus tail is an end with limit α . The determinant lines of the torus domains receive orientation maps which are natural under the lifting groups, compatible with gauge and strip gluing and with the conversion line at the closed cut, and equal to the product of the canonical orientations at split constant pairs. The gluing paths and the one-dimensional comparison families then give the integral chain homotopy of Theorem 8.17, with every term an actual signed count.

Proposition 2.3 (Ordinary determinant-one self-gluing). *Consider the nonsingular gauge cobordisms and designated closed cuts in the torus-excision construction of Section 8. Fix the ordinary Floer data on the original ends, including their trajectory perturbations. The end critical sets are finite, irreducible and nondegenerate, the low-dimensional trajectories are regular, and the restricted lifting groups act freely on these sets. There are compatible data on the new torus ends and invariant secondary perturbations, supported in fixed compact end collars, for which the cut moduli spaces and the required parameter families are regular. For these choices the following hold.*

- (a) *For fixed regular cut data and all sufficiently large neck lengths, the index-zero solutions are exactly the gluings of matched determinant-one cut solutions on the prescribed bundle. The gluing fiber is the torsor of Lemma 8.2, and one neck threshold works for the entire count.*
- (b) *Determinant gluing uses the graded symmetry of Section 3. Its signs are determined by the ordered degrees and the fixed orientation choices. In the nonseparating torus case the determinant-one count is twice the graded partial trace, before the larger lifting-group quotient is taken.*
- (c) *Relative regular families give the integral composition and change-of-data homotopies. In a family the data parameter is shared by the pieces and occurs once in the linearization.*

The proof is in Section 8.3. It applies the ordinary instanton compactification of [8, Section 3.9] and its signed self-gluing formula [8, Proposition 5.5]. Regular cut data are essential: nondegeneracy of the end critical points alone does not regularize a cut solution. Part (c) is a statement about universal moduli spaces, rather than a pointwise gluing bijection in every fiber of a universally regular family. Neither Proposition 2.1 nor Proposition 2.2 is used in this proposition.

3. COMPLEXES, GRADED SIGNS AND DETERMINANT LINES

3.1. The homotopy category. A complex is a finite free $\mathbb{Z}/2$ -graded abelian group $C = C^0 \oplus C^1$ with an odd differential d_C . For a homogeneous map $F : C \rightarrow D$ of degree f , set

$$(3.1) \quad \delta F = d_D F - (-1)^f F d_C.$$

A closed map satisfies $\delta F = 0$. Thus an odd closed map anticommutes with the differentials. We allow sums of the two degrees when discussing homotopies of the composites below.

The category $\mathcal{K}(\mathbb{Z})$ has these complexes as objects and closed maps modulo boundaries as morphisms. In particular

$$F \simeq G \iff F - G = \delta K.$$

For homogeneous F, G of degree f , the homotopy K can be chosen of degree $f + 1$. The notation $F \simeq \pm G$ means that one of $F - G, F + G$ is a boundary, with a single choice of sign.

If A and B are closed and B has degree b , then

$$(3.2) \quad (\delta K)A = \delta(KA), \quad B\delta K = (-1)^b \delta(BK).$$

These identities justify composition of every homotopy used below. They also show that projectivizing by the relation $[F] = [-F]$ is compatible with composition.

Lemma 3.1. *An acyclic finite free $\mathbb{Z}/2$ -graded complex over $\mathbb{Z}$ is contractible. An even quasi-isomorphism between finite free complexes is a chain homotopy equivalence.*

Proof. Let $B^i = \text{im}(d : C^{i-1} \rightarrow C^i)$. Each B^i is free as a subgroup of a finitely generated free group. If the complex is acyclic, the exact sequence $0 \rightarrow B^i \rightarrow C^i \xrightarrow{d} B^{i+1} \rightarrow 0$ splits. Relative to these splittings the differential identifies the complementary summand of C^i with B^{i+1} . Its inverse on this summand, and zero on the complement, defines h with $dh + hd = 1$. Apply this to the mapping cone of a quasi-isomorphism to obtain the second assertion. $\square$

The lemma does not say that a map inducing zero on homology is null-homotopic. Nor does it identify arbitrary free complexes with their homology, which may contain torsion. We use the lemma only for acyclic complexes and actual quasi-isomorphisms.

3.2. The tensor convention. Generator lines precede the determinant lines of trajectories and cobordisms. With this ordering the tensor differential is

$$(3.3) \quad d(x \otimes y) = (-1)^{|y|} dx \otimes y + x \otimes dy.$$

For homogeneous maps F, G we use

$$(3.4) \quad (F \otimes_R G)(x \otimes y) = (-1)^{|F||y|} Fx \otimes Gy.$$

Closed maps tensor to a closed map of degree $|F| + |G|$ with this rule. The usual left tensor differential is related to (3.3) by the even chain isomorphism

$$T(x \otimes y) = (-1)^{|x||y|} x \otimes y.$$

Thus a formula imported in a left tensor convention is conjugated by T at each tensor end before it is used here.

Lemma 3.2. *For a disconnected gauge end $Y_1 \sqcup Y_2$, with the generator line ordered as $L(a_1) \otimes L(a_2)$, gluing index-one trajectories on the two components gives (3.3).*

Proof. A trajectory on Y_1 has an odd determinant line. To place it next to $L(a_1)$, move it past $L(a_2)$; this gives $(-1)^{|a_2|}$. A trajectory on Y_2 is already adjacent to its generator line. Summing the two contributions gives the formula. $\square$

Put $\varepsilon_C x = (-1)^{|x|} x$. If a homogeneous map F of degree f commutes strictly with the differentials, the map $F^r = F \varepsilon_C^f$ is closed. For strictly commuting maps F, G of degrees f, g , respectively,

$$(3.5) \quad (FG)^r = (-1)^{fg} F^r G^r.$$

The discrepancy is one sign per composite. By contrast the factor ε_C is not generally an overall sign.

For example, on $C^0 = C^1 = \mathbb{Z}$ with zero differential, the map interchanging the two generators and its product with ε_C are not equal up to sign. The difference persists on homology. This elementary example explains why an integral comparison must specify the convention for odd cobordism maps.

3.3. Duals, evaluation and tensors of maps. The graded dual has

$$d^\vee f = -(-1)^{|f|} f \circ d, \quad F^\vee(f) = (-1)^{|F||f|} f \circ F.$$

The canonical even double-dual map is

$$\text{ev}_C : C \longrightarrow C^{\vee\vee}, \quad \text{ev}_C(x)(f) = (-1)^{|x||f|} f(x).$$

The sign is necessary: the unsigned double-dual identification does not commute with the differential in this convention. There is an unsigned chain isomorphism

$$(3.6) \quad \beta : C^\vee \otimes_R D \longrightarrow \text{Hom}(C, D), \quad \beta(f \otimes y)(x) = f(x)y.$$

Indeed, substituting the dual differential in (3.3) gives exactly (3.1) after evaluation on a homogeneous x . For a tensor in $C \otimes_R D$ we write

$$B_C = \beta \circ (\text{ev}_C \otimes_R 1) : C \otimes_R D \longrightarrow \text{Hom}(C^\vee, D).$$

If e_i is a homogeneous basis of C , the tensor corresponding to a homogeneous map $F : C \rightarrow D$ is

$$\text{ten}(F) = \sum_i e_i^\vee \otimes F(e_i).$$

This expression is independent of the chosen basis. It is a cycle if and only if F is closed. For $F = 1$, it is the coevaluation tensor in the present convention.

3.4. Determinant lines. For a real Fredholm operator $D : X \rightarrow Y$ set

$$\det D = \Lambda^{\max} \ker D \otimes (\Lambda^{\max} \text{coker } D)^*, \quad |\det D| = \text{ind } D \pmod{2}.$$

We use the graded determinant convention of [20]. Stabilization by a finite-dimensional map $V \rightarrow Y$ with $D \oplus V$ surjective gives the local trivializations. Enlarging the stabilization gives compatible transition maps. This construction applies to continuous families of Fredholm operators without requiring constant kernel dimension.

For direct sums the determinant lines are ordered in the order of the summands. Interchanging two summands gives $(-1)^{\text{ind } D_1 \text{ind } D_2}$. The dual of a tensor reverses the order of its factors and uses nested evaluation. An invertible operator has the canonical determinant orientation of $\mathbb{R}$; a continuous path of invertible operators preserves this orientation.

Lemma 3.3 (Ordinary gluing). *Suppose two Fredholm problems have matching ordinary ends with invertible asymptotic operator. Gluing with a sufficiently long neck gives a determinant-line isomorphism*

$$\det D_1 \otimes \det D_2 \longrightarrow \det(D_1 \# D_2)$$

well defined up to a positive scalar. It is continuous in compact parameter families, natural under bundle automorphisms, and associative for disjoint necks. On invertible operators it is the canonical identification.

Proof. Choose finite-dimensional stabilizations supported away from the ends so that the two stabilized operators are surjective. Patch their right inverses using cutoffs. The error tends to zero with the neck length, so a Neumann series corrects the patched operator to a right inverse. Projection of the patched kernels to the kernel of the glued stabilized operator gives

an isomorphism for long necks. Its top exterior power and the stabilization isomorphisms define the asserted map.

Enlarging stabilizations commutes with this construction. Changing cutoffs, projections or neck length gives a path of kernel isomorphisms, so the sign does not change. For two disjoint necks use stabilizations supported away from both; the two successive patching maps are homotopic to the direct three-piece patching map. This proves associativity, with only the graded signs caused by an actual reordering. When both operators are invertible, no stabilization is needed and the map is $\mathbb{R} \otimes \mathbb{R} \rightarrow \mathbb{R}$. The gauge version is the determinant-line construction accompanying ordinary Floer gluing [4, Proposition 5.11]; the strip version uses the same stabilization argument for real Cauchy–Riemann operators [5]. $\square$

Gluing a constant cylinder is the positive identity: its operator is invertible, and the patched kernel map converges to the identity as the neck grows. Weighted gluing is treated by conjugating with the exponential weight. The resulting asymptotic operator must be invertible, and matched ends must use compatible weights. We will never use ordinary gluing at a mixed end without separately keeping its stationary coordinates.

4. THE INTEGRAL GAUGE CONVENTION

4.1. Almost-complex orientations with free ends. Choose an S^1 -invariant almost-complex germ J_T near $S^1 \times N(T_0)$, with T_0 an almost-complex torus in its specified orientation. On an object, such a structure is equivalently specified by a nonvanishing vector field on $M^\#$, normalized near T_0 . Choose a base J_M^0 for each object. On a cobordism require J_W to agree with the fixed germ near $N(T_0) \times [0, 1]$ and to be translation invariant on its end collars. Its end restrictions J'_0, J'_1 are not prescribed in advance.

Lemma 4.1. *Every decorated cobordism admits an almost-complex structure of this kind.*

Proof. The fiber of the bundle of compatible oriented almost-complex structures is $\mathrm{SO}(4)/\mathrm{U}(2) \cong S^2$. Let $A = N(T_0) \times [0, 1]$. The primary relative obstruction lies in $H^3(W^\#, A; \mathbb{Z})$ and maps to $W_3(W^\#) = 0$. Here an oriented four-manifold admits a spin^c structure; equivalently its second Stiefel–Whitney class has an integral lift. The remaining relative class is in the image of the connecting map from $H^2(A; \mathbb{Z})$. The restriction $H^2(W^\#; \mathbb{Z}) \rightarrow H^2(A; \mathbb{Z})$ is surjective: the torus summand supplies a class pairing once with T_0 , and this class extends by the Mayer–Vietoris sequence across the three-sphere where the tube is attached. Thus the connecting map vanishes.

The secondary obstruction lies in $H^4(W^\#, A; \mathbb{Z})$. Since $A \simeq T^2$, its third cohomology vanishes; since $W^\#$ is connected with nonempty boundary, $H^4(W^\#; \mathbb{Z}) = 0$. The relative group therefore vanishes. Homotopy extension makes the structure equal to the prescribed germ, and a collar reparametrization makes it translation invariant near the ends. $\square$

Prescribing both end structures would change this obstruction problem. We use Lemma 4.1 only with free end restrictions. Conversion maps, rather than an extension assertion with fixed ends, implement the fixed object convention.

The almost-complex deformation of the anti-self-duality operator has a complex-linear endpoint. The complex orientation of its kernel and cokernel orients its determinant line. This construction is invariant under the full group of bundle automorphisms, which act complex linearly at that endpoint [8, Section 5.1]. Ordinary determinant gluing preserves these orientations. In particular the action of ϕ_M on the orientation lines has no additional sign twist.

4.2. The comparison character. Let v, v' be the normalized vector fields corresponding to J, J' on $S^1 \times Y$, where $Y = M^\#$. Their difference class $\delta(v, v') \in H^2(Y; \mathbb{Z})$ satisfies

$$2\delta(v, v') = c_1(\xi_{v'}) - c_1(\xi_v).$$

The difference class is defined by the obstruction to homotoping the vector fields on the two-skeleton, rather than by choosing an arbitrary half of the displayed Chern-class difference. This is relevant when $H^2(Y; \mathbb{Z})$ has two-torsion.

Proposition 4.2. *If $\alpha \in H^1(Y; \mathbb{Z}/2)$ is the reduction of an integral class, the change of almost-complex orientation around a loop with lifting class α is*

$$(4.1) \quad \xi_{J, J'}(\alpha) = (-1)^{\langle \delta(v, v'), \text{PD } \alpha \rangle}.$$

In particular it is 1 on ϕ_M when J and J' agree near T_0 .

Proof. Represent the loop by a path of connections whose endpoints are identified by a bundle automorphism of class α . Closing up gives a bundle P_α on $S^1 \times Y$ with

$$w_2(P_\alpha) = \text{pr}_Y^* w_2(P) + [dt] \smile \alpha.$$

The determinant-line comparison for this closed operator is divided by the comparison for the constant loop. The P -independent orientation factor therefore cancels.

For completeness, the bundle-dependent comparison can be computed on a reducible model $\mathbb{R} \oplus L$. The anti-self-dual operator on the L -summand has both the complex structure induced by J and that of L . On the summand where these structures are opposite, conjugation changes the determinant orientation by $(-1)^{\text{ind}_\mathbb{C}}$. The index formula gives, for two almost-complex structures, the ratio

$$(-1)^{\langle c_1(L) \smile \delta(v, v'), [S^1 \times Y] \rangle}.$$

This is the relative form of the usual almost-complex orientation calculation [3, Section 3(c)]. Changing the instanton number does not change the ratio: instanton addition preserves both complex orientations.

Because α has an integral lift, so does the displayed Stiefel–Whitney class. The reducible model therefore computes the desired ratio. Only the term $[dt] \smile \alpha$ survives after division by the constant-loop comparison; integration over the circle gives (4.1). The closing-up identification is made in pair form $\det D \otimes (\det D_{\text{const}})^*$, so it introduces no choice of orientation at the closed end. It is the real excision identification, independent of J and J' . Finally α_M is dual to T_0 , and the difference class restricts to zero there when the structures have the same germ. $\square$

Only integral lifting classes are used in this proposition. The lifting classes in the framed objects have that property. No assertion for arbitrary mod-two classes is needed.

4.3. Conversion maps and their ordering. For a critical point a , let Q_a interpolate the complex operators for J and J' . Choose a weight for which this operator is Fredholm and choose an orientation q_a of its determinant line. Transport q_a along paths of connections. Proposition 4.2 says that this transport is consistent on the framed quotient. It is unique up to one sign after a choice at one reference connection.

We place q_a after the generator line and obtain

$$(4.2) \quad \Theta_{J \rightarrow J'}^r : C_G(Y, J) \longrightarrow C_G(Y, J'), \quad x \longmapsto \gamma(x \otimes q_a).$$

Its parity $t = \text{ind } Q_a \pmod{2}$ is independent of a . Moving the odd trajectory line past q_a shows that

$$d\Theta^r = (-1)^t \Theta^r d.$$

Thus it is a closed isomorphism of degree t . If Θ^1 denotes the convention with q_a placed before the generator, then

$$(4.3) \quad \Theta^r = \Theta^1 \varepsilon^t.$$

Composition of conversion maps agrees with the direct conversion up to one sign. The sign is constant because the orientation comparison is defined over the connected space of connections, not chosen separately at its critical points.

Definition 4.3. For (W, E, ν) choose an admissible J_W with end restrictions J'_0, J'_1 . Define

$$(4.4) \quad I_J^\#(W) = \Theta_{J'_1 \rightarrow J_{M_1}^0}^r I(W^\#, E^\#, J_W)^{\phi_W} \Theta_{J_{M_0}^0 \rightarrow J'_0}^r.$$

At the induced end structures the middle map uses the almost-complex orientations and the stated ordering of its determinant line. Tensor ends use (3.4).

Proposition 4.4. *The homotopy class of (4.4) is independent of J_W up to one sign. These classes define a projective gauge functor, and their reduction modulo two is the framed gauge functor of Paper I.*

Proof. Compare two choices of J_W after inserting their endpoint conversions. They orient the same real determinant line over the space of connections with fixed endpoint limits. Their ratio is continuous and takes values in $\{1, -1\}$, so it is constant on that space. Gluing a path at either endpoint shows that the ratio does not change when the limiting critical point is changed. The character calculation makes this assertion invariant under changing the lift of an end critical point. Thus the ratio is one sign for all entries of the map.

For composition choose an admissible almost-complex structure on the composite and make it cylindrical at the intermediate cut. The restrictions then have identical structures there, and the ordinary gauge composition theorem applies [8, Proposition 5.4]. Insert the conversions to the chosen object bases. The conversions at the common end cancel up to sign, and (3.5) accounts for their ordering. Independence of J_W gives the result for the original choices. The product cobordism gives the identity class. All orientation signs disappear modulo two. $\square$

4.4. Reflection and the relation to other conventions. Choose the base on the oppositely oriented object in pairs:

$$J_{-M}^0 = \Psi_* \rho_* J_M^0.$$

Here ρ reverses the cylindrical direction and Ψ is the fixed identification of the reflected torus-summed presentation with $(-M)^\#$. Take $T_0 = \{\theta_0\} \times T^2$ and the product germ there; require Ψ to preserve the odd torus, its germ, and its lifting class. Gauge reflection then gives an even isomorphism

$$(4.5) \quad \Phi_{G,M} : C_G(-M) \longrightarrow C_G(M)^\vee, \quad \Phi_{G,-M} = \Phi_{G,M}^\vee \text{ ev}.$$

The transpose formula for a reversed cobordism follows by bending its cylindrical ends and applying the same ordered determinant gluing. All constant end lines are even; any remaining change is one orientation sign of the cobordism. This also gives the transpose formula for gauge continuation.

If the conversion lines are instead placed on the left, put $I_1(W) = \Theta_1^1 I(W) \Theta_0^1$. In the induced end gradings the unconverted gauge map $I(W)$ is even. If the two conversions have parities t_0, t_1 , then

$$(4.6) \quad I_J^\#(W) = (-1)^{t_0 t_1} I_1(W) \varepsilon^{t_0 + t_1}.$$

This is an identity of maps in the stated conventions, rather than an assertion that the two maps differ only by a sign. The degree-dependent factor matters on an object whose homology has both parities.

A $\mathbb{Z}/4$ grading compatible with all the orientation data would provide a further regrading $\eta(x) = (-1)^{g(x)(g(x)-1)/2}x$. The elementary identity

$$\eta(g+d)\eta(g)^{-1} = (-1)^{dg+d(d-1)/2}$$

explains how an odd grading operator can then be converted into a conjugation and one sign per arrow. Identifying such a lift with Scaduto's data is not part of this draft. We therefore keep $I_J^\#$ in the theorem and make no integral identification on the basis of the mod-two equality alone.

5. DETERMINANT LINES ON MIXED CONFIGURATION SPACES

5.1. Admissible operator realizations. Let c be a smooth configuration on a mixed domain, with its specified limits. Denote the mixed equation by $\mathcal{F}(c) = 0$. A chart Φ centered at c incorporates the exact matching condition and the normal seam condition. Its differential identifies a fixed linear boundary problem with the tangent space to the configuration space.

Definition 5.1. An admissible realization of the linearized problem at c is an operator of the form

$$(5.1) \quad L_c = (\mathcal{S} d\mathcal{F}_c, \lambda^2 d_{A'}^{*,g'} \pi_A) d\Phi_0.$$

It acts on the prescribed weighted spaces with stationary mixed-end coordinates included. Here π_A denotes the connection component of a physical tangent vector. At mixed ends we use spatially Coulomb stationary tangent representatives and matching lifts whose remaining divergence decays, so that the scalar row takes values in the weighted target. Both rows are transported by the chart differential. A change of stationary lifts likewise transports their finite scalar columns as well as the equation columns. The map $\mathcal{S}$ is a continuously chosen invertible target identification, equal to the identity outside a compact set; A' is a smooth slice center differing from the connection in c only on a compact core and sufficiently little in C^0 ; $g' = f^2 g$ with $f = 1$ off the collapse region and f depending only on base coordinates near a seam; and $\lambda > 0$ equals one outside a compact set. The chart has the same end and seam conventions as the original mixed problem. Changes of all these choices are taken inside their fixed components. In a comparison of weights, admissibility also requires that the full stationary and parameter columns belong continuously to every intervening weighted target. Membership of the weights in a spectral gap alone does not impose this condition on a nonsolution with slower asymptotic decay.

The restriction on the target identification prevents an arbitrary reversal of orientation from being included as a change of realization. The reference identification is the one induced by the geometric charts. Its admissible component is fixed before transporting orientations.

For the collapse contraction we also use the extended realization of Proposition 7.3. Its center and geometric target identification can have stationary limits different from those of the configuration and the identity, respectively. Its source representatives are Coulomb for the varying center, both rows are transported, and differentiation of the moving equation target contributes the residual term specified there. This is a separate fixed-positive-scale family, not a compactly supported change under Definition 5.1.

Lemma 5.2 (Realization comparison). *Under the Fredholm and continuity statements of Proposition 2.1(a), admissible realizations at a fixed configuration have canonically identified orientation double covers. The identification is continuous in compact families and commutes with ordinary gluing. At a regular zero of index zero it preserves the canonical determinant orientation.*

Proof. Changes of charts act on the entire operator in (5.1). Together with target identifications, they are bounded invertible changes of source and target and induce the standard naturality maps on determinant lines. An admissible weight interval gives a continuous Fredholm path after conjugation to fixed spaces, and hence an identification of orientation covers. At a solution, improved asymptotic decay identifies the kernels, cokernels and common stabilized determinant lines throughout the permitted gap. We do not assert this improved decay for an arbitrary nonsolution. For a change of slice center at a solution, split the tangent space into infinitesimal gauge directions and a fixed complement. Gauge invariance gives $d\mathcal{F}_c dA_c = 0$. The gauge-fixed operator is then block triangular. Its gauge block is invertible by the Poincaré estimate and the smallness of $A' - A_c$; its other block is the intrinsic linearized equation on the quotient. Thus invertibility at a regular zero of index zero is unchanged.

Conformal changes in the specified unperturbed region do not change the anti-self-duality equation in four dimensions. They do not change the surface Cauchy–Riemann equation or the normal seam subspace. The same block argument applies to the gauge row. Positive row multiplication does not alter an orientation. These paths give the comparison. A homotopy between two such paths gives an orientation cover over a square, so its endpoint identification is unchanged. The gluing statement follows by choosing the changes away from the ordinary gluing collars and then using continuity. At a regular zero of index zero every operator in these paths is invertible, so the canonical orientation is preserved. $\square$

The lemma uses the Fredholm families of Proposition 2.1(a), including those on the completed domains. In particular, moving a Coulomb center at a configuration which is not a solution is not justified by the block-triangular argument above. We will use a two-parameter determinant-line comparison when such a configuration occurs in the collapse argument. Likewise, a reference path in a small weighted space need not itself lie in every larger weighted space. For weight comparisons of its endpoint orientations, truncate its decaying coordinates on a common far end while retaining the exact stationary family. The truncation homotopy is taken at the initial small weight. The truncated path, together with the endpoint truncation segments at solutions or stationary references, admits the larger weights. Determinant transport around this homotopy gives the comparison; it does not improve the decay of the original nonsolution path.

5.2. Comparison of physical end weights. Reverse incoming ends to outward coordinates. Write an autonomous end operator as $\partial_s - \mathcal{A}_e$. Its stationary kernel is the primary tangent space on a mixed end; at an ordinary end it is zero. Choose $\gamma_e, \gamma_e^{\text{opp}} > 0$ so that the negative spectrum is at most $-\gamma_e$ and the positive spectrum is at least γ_e^{opp} . They need not be equal. The permitted positive primal weights satisfy $r_e < \gamma_e$; the physical adjoint is paired at weight $-r_e$, not r_e . Mixed-end stationary coordinates remain a separate finite summand.

Proposition 5.3. *For the autonomous ends of the mixed domains and ordinary strips used here, physical solutions, their allowed gauge arrows, and their determinant lines are independent of the weights within these open gaps. At ordinary ends zero weight is also permitted. The identifications are transitive, respect the lifting actions and ordered gluing, and give the same oriented counts and homotopies whenever their small-weight compactifications have been*

established. This assertion does not place an arbitrary slowly decaying nonsolution in a faster weighted space.

Proof. All comparisons use closed intervals strictly inside the gaps. At spatially completed torus ends no extra spatial weight is introduced. The longitudinal weights are constant on surface fibres, have zero normal derivative at physical faces and agree on matching traces. Write each longitudinal exponent as $r_e W_e$. Choose $|dW_e| \leq 1$, with cutoff derivatives supported in disjoint longitudinal transitions. The complete stationary mixed operator is self-adjoint on its physical Green-form domain. On a completed compression, its compact-remainder estimate and the acyclic torus model give a normal gap; compact resolvent is not needed. The following argument uses its spectral projections.

Start with the positive decay furnished by the mixed end estimates of Paper I and [1, Section 5.3]. Choose a far end and put the solution in the stationary connection's small Coulomb slice, with Dirichlet condition on an artificial initial slice and the actual Neumann condition at physical faces. The small-weight scalar inverse provides this gauge, preserving the selected stationary arrow. Subtract the exact stationary primary limit and use its translation-invariant matching chart. The equation is

$$(\partial_s - \mathcal{A}_e)v = f(v), \quad f(0) = df(0) = 0, \quad v(s) \longrightarrow 0.$$

The chart and holonomy estimates give, on nested unit windows,

$$\|f(v)\|_{H^m([s, s+1])} \leq C_m \|v\|_{H^{m+\ell_m}([s-1, s+2])}^2$$

for a fixed finite loss ℓ_m . Initial decay holds at every order, so rate α makes the right side have rate 2α . Writing P_-, P_0, P_+ for the spectral projections, variation of constants gives

$$\begin{aligned} P_- v(s) &= e^{(s-S)\mathcal{A}_e} P_- v(S) + \int_S^s e^{(s-\sigma)\mathcal{A}_e} P_- f(\sigma) d\sigma, \\ P_+ v(s) &= - \int_s^\infty e^{(s-\sigma)\mathcal{A}_e} P_+ f(\sigma) d\sigma, \\ P_0 v(s) &= - \int_s^\infty P_0 f(\sigma) d\sigma. \end{aligned}$$

The first integral improves to every rate below $\min(2\alpha, \gamma_e)$; the second retains rate 2α with denominator $\gamma_e^{\text{opp}} + 2\alpha$. The zero-mode integral uses the prescribed limiting primary coordinate. Local elliptic estimates on nested windows improve all derivatives. Iterating with strict margins gives every rate $b < \gamma_e$, without restricting b by the opposite spectral gap. A rate $b > r_e$ gives membership in the desired weighted space. Compact families with common tail-entry windows and spectral margins have uniform bounds.

Allowed arrows between these representatives have their prescribed stationary limit g_∞ . The temporal comparison equation $\partial_s g = g a_s - a'_s g$, integrated from infinity, gives the same rates for $g - g_\infty$. The spatial comparison equation and differentiation give all higher derivatives, after subtracting the stationary intertwining relation. The lifting classes and central transition data are unchanged. This concerns arrows between the controlled representatives, not arbitrary gauges applied to a stationary solution.

Parameter derivatives require differentiating the stationary subtraction. In a family the identity $\mathcal{F}_b(0, q) = 0$ cancels the nondecaying parameter column. Its remainder and every finite stationary column then have all the above decay rates. The linearized equation improves the first parameter derivative; induction improves higher ones. Polynomial factors caused by parameter differentiation are absorbed in a strict rate margin. A common data parameter contributes only its one original column.

Write the full realized operator as

$$D_r : X_r^{\text{dec}} \oplus P \longrightarrow Y_r, \quad D_r(w, p) = D^{\text{dec}}w + Cp.$$

On an end weight conjugation gives $\partial_s - \mathcal{A}_e - r_e$. Its inverse and adjoint inverse are uniformly bounded on a closed weight interval by the reciprocal spectral distance. Small coefficient and domain-transport errors are absorbed beyond a common cutoff. On an infinite spatial torus region the unweighted flat operator has mass $\mu_T > 0$. The conjugation term has norm at most the largest relevant mixed-end exponent r_T ; its inverse bound is at most $(\mu_T - r_T)^{-1}$. The permitted completed spatial gap is bounded by this essential-spectrum threshold. The original closed incoming weights do not vary on that added torus region. Compact-core primal and actual weak-adjoint estimates give Fredholmness over the entire weight rectangle. Smooth stationary and parameter columns remain in every target. If a scalar inverse at a larger weight is needed, scalar Fredholmness throughout its indicial gap, index zero from the small weight and the positive kernel energy identity give it. Scalar gauge modes occur in the full end operator, so the full permissible gap is no larger than that scalar threshold. This avoids extending the sufficient small-weight coercivity bound beyond its range.

For $r_- \leq r_+$ the inclusions of physical fields give a commuting square of source and target inclusions, with identity on P . A kernel element is a stationary column plus a remainder. The coefficient errors and the images of stationary columns have every permitted rate. The same spectral formulas improve the remainder to every weight. This applies also to kernels with compactly supported stabilization columns and to universal kernels.

For the cokernel use distributions paired in the unweighted physical density. A weak cokernel at r_+ lies at dual weight $-r_+$. Window estimates initially bound its growth by $e^{r_+ s}$. Choose β with $r_+ < \beta < \gamma_e$; multiplying by the coefficient error gives a decaying forcing of rate $\beta - r_+$. The adjoint spectral formula excludes growing normal modes, whose rates are at least γ_e . Its zero mode has a finite limit. The resulting bounded field improves its forcing again, and its remainder has every rate below $\min(\gamma_e, \gamma_e^{\text{opp}})$. It therefore belongs to the dual space at $-r_-$ as well. At an ordinary end there is no zero mode, so $r_- = 0$ is included. Retain the conditions $C^*y = 0$ and, in a universal problem, the parameter-column annihilator. These are the same convergent physical pairings at every weight. Hence the kernels and cokernel duals, and therefore the finite-dimensional cokernel quotients, are canonically identified. For unordered weight vectors insert their componentwise minimum and maximum.

Choose finitely many smooth compactly supported target columns $\rho : V \rightarrow Y$ spanning the cokernel on a compact parameter patch. They work at every weight. With $E_r = \ker(D_r + \rho)$, the identity on actual fields identifies

$$\det D_r = \det E_r \otimes (\det V)^*.$$

Enlarging the stabilization gives the same map by its exact sequence. These identifications are continuous, transitive and preserve canonical orientations at invertible operators. They commute with the allowed gauge arrows and shared parameter factor. The weight rectangle is contractible, so its transport has no additional choice of path or component sign.

The reference-orientation paths can initially have slower tails. Prepare them by cutting off their decaying coordinates on a common far end, retaining the exact stationary coordinates and carrying the actual transition data through the preparation transport. The preparation homotopy is made at the initial small weight. A stationary endpoint is fixed; at a solution the endpoint segment has all permitted decay rates by the preceding argument. Thus the prepared path and its endpoint segments admit the full weight rectangle, even when the original path does not. Determinant transport on this rectangle and on the preparation homotopy

proves compatibility with the original reference normalization. Affine-center comparisons are made first at their common small weight, as in Proposition 7.3. The physical solution comparison then uses this proposition's full gap; no old spectrum is assigned to an affine operator.

Finally consider a sequence in an already established small-weight compactification. Fix translations on its retained components. Energy exhaustion gives common tail-entry windows. The estimates above, with $r < b < \gamma_e$, bound a tail beyond R by a constant times $e^{-(b-r)R}$ in the desired weighted norm. Compact convergence therefore improves to convergence in that weight. At ordinary breaks this is a componentwise assertion in the gluing charts; at spatial completion it retains the existing strong spatial tails. The small-weight forward gluing produces the same actual solutions, and the converse and uniqueness give the same local branches in larger weights. Common compactly supported stabilizations make the gluing exact sequences commute with inclusion of physical fields. Cutoffs, evaluation maps, stationary factors, and conversion lines are the same ordered geometric maps. Thus all boundary signs and the actual counts and homotopies are unchanged. $\square$

5.3. Shifting and relative orientations. A constant pair C_x joins the gauge critical point a_x to the symplectic intersection point x . Its operator is invertible. Let $L_S(x) = L_G(a_x)$. More generally, a constant pair C^c may be built from any representative c of the critical point over x (Appendix B); it is gauge equivalent to C_x , its operator is invertible, its positive element corresponds to that of C_x under conjugation, and the almost-complex orientation identifies $L_G(c)$ with $L_G(a_x)$, compatibly with transport (Appendix B). For a mixed configuration c with gauge input a and symplectic output x , shifting and gluing identify it, through a path of configurations, with a configuration of the form $A\#C^{c'}$. Here A is a gauge path from a to a representative c' .

Proposition 5.4. *By Propositions 2.1 and 2.2, there is a relative orientation isomorphism*

$$(5.2) \quad \Gamma_c : L_G(a) \otimes \det D_c \longrightarrow L_S(x),$$

defined up to positive scalar, which agrees with gauge orientation gluing at $A\#C^c$. It is independent of the shifting path, natural under the lifting group, and coherent under ordinary gauge and strip gluing.

Proof. First consider a loop of configurations with fixed limits. The relative family shifting in Proposition 2.2 deforms it to a loop of symplectically constant configurations. A common gluing length works for the compact family. At that end of the deformation, the determinant bundle is the gauge orientation bundle tensored with the determinant of the invertible constant pair. The gauge bundle is oriented by the almost-complex construction, and the second factor has its canonical orientation. The orientation monodromy of the loop is therefore trivial. This proves path independence of the orientation transported from the constant reference.

For a strip attached at the output, first shift the original mixed configuration to $A\#C^c$. Then shift C^c together with the attached strip to another gauge path followed by $C^{c'}$, with c' a representative over y . Associativity of determinant gluing identifies both orders of this construction with gauge gluing at the final reference. Continuity transports that equality back to the original configuration. The gauge-end statement is the same argument with the extra path attached before shifting.

The lifting automorphism preserves gauge orientations, conjugates the constant-pair operator, and commutes with gluing. Conjugation preserves the canonical orientation of an

invertible operator. Hence the construction descends to the framed quotient. If shifting reaches the other lift of a_x , the two end lines are identified by this same action. $\square$

The pure strip orientation is defined by applying (5.2) to $C_x \# u$ and cancelling the constant pair. This is the transported orientation construction of [1, Section 4.7]. For a mixed triangle there are two gauge inputs. Use the outgoing closed cut of Paper I to write its gauge region as an ordinary two-input cobordism followed by a truncated output DFL domain. For the folded triangle the cut is Z and the primary arms are D_0, D_1 . No critical boundary condition is imposed on an arbitrary configuration at this finite cut. On a compact family at positive collapse scales, the compact metric interpolation of Paper I first supplies its product collar. This operation has no asserted uniformity as the scale tends to zero.

The finite-width joined torus domains W_λ^J have the same structure, with no collapse scale. By Paper I, Lemma 10.18, the closed cut $Z_{\text{cut}} \cong Z$ decomposes W_λ^J into the reverse excision cobordism, which is the ordinary two-input cobordism, and the closed-cut piece M_λ , a truncated output DFL region for the pair (A_0, A_1) with arms $D_{0,\lambda}, D_{1,\lambda}$; the cut has a product collar. The construction below applies verbatim with the output region M_λ and the cut Z_{cut} , on compact ranges of widths; no uniformity as $\lambda \rightarrow \infty$ is asserted.

Prepare the output DFL region, and extend it positively by lifting its section to the gauge side as in the proof of [1, Lemma 4.62], keeping a fixed collar of the cut unchanged; this is part of Proposition 2.2. In the coordinate r of Appendix B.5, in which the cut is $\{-c_1\} \times Y_\#$ with $c_1 > 5$, localize the shift by

$$F_\kappa(r) = r + \kappa\chi(r),$$

where χ is nondecreasing, zero near $-c_1$, and one on $[-5, \infty)$. Extend by identity on the ordinary core. Since $F'_\kappa = 1 + \kappa\chi' > 0$, this pulls back the whole connection smoothly; on the section use the usual translation. For a common large finite κ , followed by the straightening and the collar widening of Appendix B.3, the region $r \geq -4$ is a constant mixed pair C^c , with c a representative over the output label y . The remaining connection is an ordinary cobordism configuration A_E ending at c on the parallel copy $\{-7/2\} \times Y_\#$ of the cut, so the reference is $A_E \# C^c$, with the original inputs unchanged. In particular it is not a product of two gauge paths.

Pull back the actual transition gauges as well. If $C_{i,\kappa}$ is the parameter component of this pulled-back connection, solving $\partial_\kappa v_i = -C_{i,\kappa} v_i$ identifies its transitions with the initial ones. This retains the lifting classes and central cocycles. The stationary representatives are the actual primary paths, including the corner pullback, and both operator rows are transported. A common sufficiently small weight gives the family in Proposition 2.1(a); no intermediate operator is assumed invertible.

At the terminal configuration remove compact nonprimary terms whose dependence or output crosses the cut. On the constant mixed factor use its zero-secondary reference datum, retaining the fixed compatible end structures. These are compact Fredholm data homotopies, not assertions about unchanged solution counts. Ordinary determinant gluing then gives the ordinary cobordism line tensored with the canonically oriented constant-pair line. The relative construction preserves this terminal subspace, so the square argument above proves independence and coherence with ordinary gluing. Input lines stay in cyclic triangle order; any conversion line stays in its prescribed position. Associativity is applied before any factor is reordered.

5.4. Parameters and boundary orientations. A parameterized family is oriented by placing the oriented parameter space before the determinant line of the vertical operator.

At a regular point this induces the usual orientation of the finite-dimensional zero set. To see continuity when the vertical kernel dimension changes, choose one stabilization on a neighborhood in the parameter space. The orientation is then the exterior-power orientation of the stabilized kernel with the fixed stabilization factor cancelled. Transition maps between such neighborhoods are the determinant-bundle transition maps; no choice of a basis of the moving kernel is required.

For a one-parameter regular family, the boundary orientation is outward normal first. At a strip break the translation vector is the positive vector $\partial_s u$. Moving its odd line past the preceding generator factors gives exactly the signs in (3.3). At a parameter endpoint the two outward normals are opposite. It follows that a compactified one-dimensional family yields an identity of the form

$$(5.3) \quad F_1 - F_0 = \delta K,$$

with the Koszul signs already included in δ . If an orientation for the whole family has been chosen only up to sign, the corresponding endpoint identity has one overall sign. It never has independently chosen signs for its separate matrix entries.

The compatibility of a geometric boundary collar with this determinant-line orientation rests on ordinary linear gluing at the end where the break occurs (Lemma 3.3) and on the direction of the translation vector along the gluing collar; the computation is carried out in Lemma 8.14. Merely orienting the interior of a moduli space would not prove (5.3).

6. THE MIXED ISOMORPHISM OVER THE INTEGERS

6.1. Differentials and the vertex map. For an index-one gauge trajectory or holomorphic strip, its sign is positive when the relative orientation agrees with its translation vector. For a regular index-zero mixed solution, its sign is the comparison of (5.2) with the canonical element of its invertible determinant line. Set

$$(6.1) \quad \begin{aligned} d_G o_a &= \sum_b \sum_{[A] \in \mathcal{M}_G^0(a,b)} \text{sign}(A) o_b, \\ d_S o_x &= \sum_y \sum_{[u] \in \mathcal{M}_S^0(x,y)} \text{sign}(u) o_y, \\ N_p o_a &= \sum_x \sum_{[c] \in \mathcal{M}_N^0(a,x)} \text{sign}(c) o_x. \end{aligned}$$

The superscript 0 denotes dimension after quotienting translation when that action is present. All sums are finite by the compactness and energy-index bounds of Paper I.

Proposition 6.1. *The transported orientations give $d_S^2 = 0$ and $d_S N_p = N_p d_G$ over $\mathbb{Z}$.*

Proof. The compactified one-dimensional strip space has two-level broken strips as its boundary. Coherence in Proposition 5.4 identifies their boundary orientation with the product of the two zero-dimensional signs, with the sign from the translation factor. The oriented boundary count is zero, yielding $d_S^2 = 0$. For the mixed vertex, the two types of boundary are a gauge trajectory followed by a mixed solution and a mixed solution followed by a strip. The outward-normal orientations of the two ends give opposite signs. Their sum is $d_S N_p - N_p d_G$. The index and energy restrictions exclude additional boundary strata in this dimension. $\square$

6.2. The unit diagonal. The filtration used in Paper I orders the finite generator set so that every index-zero mixed solution goes from a generator to a generator of no larger filtration. At equal filtration the only solutions are the unique small deformations of the constant pairs. This is a statement about the set of solutions, before orienting it.

Lemma 6.2. *Let F be an integral endomorphism of a finite filtered free module. Suppose its associated graded map is diagonal with each diagonal entry in $\{1, -1\}$. Then F is an integral isomorphism. Its inverse preserves the filtration.*

Proof. Write $F = D(1+U)$, with D the diagonal sign matrix and U strictly filtration-lowering. If there are r filtration steps then $U^r = 0$, and

$$F^{-1} = (1 - U + U^2 - \dots + (-1)^{r-1}U^{r-1})D^{-1}.$$

All coefficients are integral. □

Each diagonal solution contributes 1 or -1 , since it is one regular point. Thus the invertibility of N_p already follows from Lemma 6.2; it does not require first proving that every diagonal entry is positive. With the normalization by constant pairs, the entries are in fact $+1$.

Proposition 6.3. *The map N_p is a chain isomorphism. Its diagonal entries are $+1$ in the constant-pair normalization. Its homotopy class is independent of regularizing data, after the corresponding gauge continuation is inserted.*

Proof. Take the segment from zero regularizing data to the chosen small data. At a constant pair the linearization is invertible. The chart implicit-function theorem therefore supplies a nearby branch of regular solutions. The uniqueness statement in the triangularity argument identifies this branch with the diagonal solution. The charts and the continuous Fredholm families of Proposition 2.1(a) make the branch, which is continuous in the chart norm, continuous for determinant-line transport; the constant pair is invertible by Lemma A.2. Its sign is locally constant and begins at the canonical sign $+1$.

The filtration now gives an inverse integral matrix. Since N_p commutes with the differentials, its inverse does too. At fixed ordinary ends, a regular compact-data homotopy gives (5.3). A change of the gauge-end function is treated by the inserted-cylinder comparison in Proposition 6.4, which retains the actual ordinary continuation map. This proves the stated independence. □

The secondary perturbations are supported away from the strip end, so their variation does not change the strip operator. The segment is chosen in a star-shaped smallness neighborhood where all the reference constant-pair operators remain invertible. This is why an arbitrary path of perturbations cannot be substituted in the normalization argument.

6.3. Changing ordinary end data. A change of a far-end equation is compared by inserting its ordinary continuation domain. It is not a compactly supported change on the original mixed domain.

Proposition 6.4. *Fix primary data and a mixed vertex, including the closed- Z vertex at a fixed positive width. With Proposition 2.1, Proposition 2.2 and the compactifications of Paper I, its mixed maps satisfy the following integral chain homotopies.*

- (i) *If f_0, f_1 are two sufficiently small invariant gauge trajectory perturbations vanishing near the whole primary critical set, with the same symplectic end datum, then*

$$N_1 \text{cont}_G(f_0, f_1) - N_0 = d_S H_G + H_G d_{G,0}.$$

- (ii) If the gauge end is fixed and J_0, J_1 are regular ordinary strip structures on the same ordered primary pair, then

$$\text{cont}_S(J_0, J_1)N_0 - N_1 = d_{S,1}H_S + H_S d_G.$$

In the closed- Z case of part (ii), the output pair is the product pair (A_0, A_1) and the structures and continuation are split.

Both homotopies count actual isolated universal solutions of vertical index -1 . The assertion concerns these fixed-primary end changes; it is not a mixed continuation theorem for arbitrary primary paths or arbitrary correspondence-output pairs.

Proof. First record the finite compact-data comparison required below. Gauge sources may vary with fixed compact dependence and output, and section structures may vary on a compact set, keeping all far ends, protected genus matching collars and collapse germs fixed. Coefficients at an ordinary physical Lagrangian boundary may vary through a compact compatible family; its totally real labels remain fixed. Keep split structures on a product target. The mixed local primal and weak-adjoint estimates vary continuously with these coefficients. A compact principal-coefficient change in the section is not called a relatively compact operator: local elliptic estimates, a finite parameter cover and the fixed end parametrices instead give Fredholm estimates with a compact remainder. Both full operator rows and all stationary columns are transported.

For zero Hamiltonian the section energy is twice its symplectic area, even for domain-dependent J . There is no derivative-of- J term. The gauge modified-energy formula and the relative index-period formula give a uniform energy and lower-index bound. Compactness retains the coefficient family on the core; the usual nonnegative four-index loss and positive indices of ordinary breaks remain. Invariant compact gauge variations still detect the entire mixed cokernel with this prescribed J : the scalar inverse removes the scalar component; character-zero holonomy tests, including every nonlocal output in the detecting region, remove the gauge adjoint; and the matching Green form followed by zero extension and unique continuation removes the section adjoint on each component. This uses the genus matching arc and connected gauge region of the stated vertices.

Finite selection is performed in increasing index from a common lower bound. Absence of lower-index cores makes the next-index space compact before its regularity is invoked. Freeze a finite set of detecting variations at that stage; small coefficients preserve the earlier absence and endpoint collars. Relative Sard transversality gives the required universal spaces. In their operator $(v, \tau) \mapsto D_b v + \tau C_b$, the single column C_b includes $(\partial_b J_b)u_t$, target transport and gauge-source variation. Ordinary gluing copies this column once on the retained core. The compactified one-dimensional space has only its data endpoints and ordinary index-one breaks. Strong convergence and the ordinary bordered inverse give both directions of its collars. Parameter-first orientation gives the Hom identity (5.3). This proves the finite comparison even when the two data differ in compact section structure.

For part (i), let β rise from zero to one on $[-1, 1]$ and let χ be the mixed vertex switch, equal to one for $s \leq -3$ and zero for $s \geq -2$. On its incoming gauge end insert the slice functional

$$q_T(s) = \chi(s)(f_0 + \beta(s+T)(f_1 - f_0)), \quad T \geq 5.$$

It retains f_0 on the far end and f_1 on the long intervening cylinder. Translate a fixed regular ordinary continuation perturbation with the β band and retain the N_1 secondary on the mixed core. Their full dependence and output are disjoint for large T . Ordinary continuation is pinned by these data; a rigid continuation has full index zero, without a translation quotient.

The smallness neighbourhood for f_i is chosen so that no additional critical points occur: compactness of the perturbed critical equation would otherwise give a primary critical limit outside the neighbourhood where both functions vanish. Their original critical points and Hessians thus agree.

The modified-curvature energy identity is

$$E_T = 8\pi^2\kappa + 2\|\eta_T(A)\|^2 + 2 \int \partial_s q_T(s)(A_s) ds, \quad \partial_s q_T = \beta'(s+T)(f_1 - f_0) + \chi'(s)f_1.$$

The last integral is bounded in absolute value by $\|f_1 - f_0\|_\infty + \|f_1\|_\infty$, and the translated compact secondary supports have uniformly bounded norm. The autonomous f_1 gradient on the growing middle cylinder is part of modified curvature, not a compact source whose L^2 norm is assumed bounded. Reference configurations with the same intermediate critical lift have cancelling cut actions. Ordinary determinant gluing gives index additivity, so the period formula yields a uniform energy and lower-index bound before extraction.

For part (ii), choose a continuation structure $J_{01}(\sigma, t)$ from J_0 to J_1 , with zero Hamiltonian and constant end collars. It admits regular relative data within this restricted class. Indeed at a nonconstant solution, the set where $u_t \neq 0$ is dense in any free transition region: vanishing on an open set would make u_s zero there and unique continuation would make the solution constant. If a universal adjoint field is nonzero, choose such a point where it is also nonzero. Domain-dependent compatible variations $Y \mapsto Yu_t$ span the target fibre, contradicting the cokernel pairing. Domain localization removes any requirement that the whole strip image be injective. At a constant solution the linear energy identity on the fixed tangent Lagrangians makes the decaying kernel zero. Its index is zero by deformation to an autonomous constant strip; transversality keeps the asymptotic operators invertible during this index comparison. For product outputs use independent variations on each nonconstant factor; every constant factor has that inverse. Relative Sard–Smale, monotonicity and the four-index bubble exclusion give regular continuation spaces through index one, with the parameter counted once.

On the outgoing ordinary strip of N_0 , keep its original datum until $s = T - 1$ and then use $J_{01}(s - T, t)$. The external equation is J_1 for every finite T . The gauge and stationary mixed ends remain fixed. The transition adds no Hamiltonian curvature or boundary work. Consequently the section area-index relation, fixed gauge-source bounds and cancellation of intermediate actions give the same length-independent energy and lower-index bounds. In both parts the finite- T operator has fixed far-end equations; its length derivative is supported in the translated transition band. At $T = \infty$ use separate retained-core charts, rather than declaring that support fixed on the untranslated domain.

We verify exhaustiveness in both constructions. In part (i) the retained factors are the pinned gauge continuation and the N_1 core; in part (ii) they are the N_0 core and the pinned strip continuation. The fixed equations govern the intervening autonomous neck. Extraction gives these two cores and ordinary trajectories W_j , with

$$d - I_1 - I_2 - \sum_j \text{ind } D_{W_j} = 4n, \quad E_{\text{lost}} = 4\pi^2 n, \quad n \geq 0.$$

The continuation contributes its full index. All retained cores are regular with nonnegative index; every nonconstant autonomous trajectory has positive full index. Hence negative-index solutions are absent for sufficiently large T , and at $d = 0$ both cores have index zero, all ordinary breaks are absent and $n = 0$. The common lower bound makes the negative-index

conclusion simultaneous. On the gauge continuation core the source switches are retained; on the section continuation core no derivative-of- J energy is missing.

Zero loss and the original end gaps give strong convergence on the components. Leaving an intermediate critical neighbourhood costs positive neck energy; inside it the gapped cylinder or strip inverse and the small nonlinear remainder control the neck by the component tails. The convergence therefore enters the strong pregluing charts. Conversely each pair of invertible zero-index core operators gives an ordinary bordered neck inverse and an exponentially small gluing error. Contraction gives one solution. There are finitely many rigid pairs, so one threshold works for all of them; strong convergence and uniqueness rule out any other solution or duplicate chart.

The actual lifting quotient is retained in the gauge construction. The connected cylinder contributes no extra central matching class, and its arrows identify the two end lifting groups; the fibre product over the common critical orbit gives precisely matrix-composition multiplicity. A pure strip adds no gauge region or lifting factor. In both cases ordinary determinant gluing compares the product before cancelling the intermediate generator line. For a nonautonomous continuation keep its full stabilized operator on the compact core, its two prescribed end models, and the autonomous inverse only on the matched tail. The bordered exact sequence and the reference and cutoff squares therefore apply with both far ends fixed. They do not replace the compact continuation core by an autonomous trajectory. Relative mixed shifting (Proposition 2.2 and Proposition 5.4) and associativity preserve the constant-pair normalization; the constant strip has its canonical orientation. All maps here are even. Thus the exact rigid counts are

$$F_T = N_1 \text{cont}_G(f_0, f_1) \quad \text{in (i)}, \quad F_T = \text{cont}_S(J_0, J_1)N_0 \quad \text{in (ii)}.$$

No independent sign is chosen for a component or matrix entry.

Choose one such finite T_* . If necessary make an arbitrarily small compact endpoint perturbation to regularize index one, preserving the already regular rigid count by compactness and local uniqueness. In part (i), interpolate the inserted data to N_0 by replacing β with $b\beta$ and interpolating the compact sources; the far input remains f_0 . In part (ii), interpolate to N_1 through compact compatible section changes and compact gauge sources, with far output J_1 . Both paths fit the finite comparison proved at the start. Its actual vertical-index-minus-one count gives H_G or H_S and the displayed identities. Finally Proposition 5.3 extends these physical counts from their common small construction weight to the full permitted intervals. Slow reference paths are prepared before comparing their weights. $\square$

6.4. Parametric orientations. The same reasoning applies to compact parameter families of primary and geometric data whenever their asymptotic intersections remain transverse. Locally choose fixed-center charts and stabilizations. A nearby regular solution has small coordinates in such a chart. The line orientation pulled back from Proposition 5.4 varies continuously, by Proposition 2.2 for families with varying data, while the canonical orientation of an invertible operator is continuous on the invertible locus. Their ratio is consequently locally constant. This fact will be used in the collapse and closed-cut comparisons. It does not require identifying the topology of every Sobolev quotient with the smooth quotient globally.

7. TRIANGLES AND ORIENTATIONS UNDER COLLAPSE

7.1. The triangle product. Order three Lagrangians L_0, L_1, L_2 by the boundary orientation of the triangle. Write $C_{ij} = CF(L_i, L_j)$. Use the end gradings induced by its chosen almost-complex gauge triangle. Orient the positive-scale mixed triangle by the ordinary

cobordism reference above. The inverse collapse construction below, followed by ordered cancellation of its completing vertices, defines the pure triangle orientation. Put all generator lines before its determinant line. The resulting product

$$\mu : C_{01} \otimes_R C_{12} \longrightarrow C_{02}$$

satisfies

$$(7.1) \quad d\mu(a, b) = (-1)^{|b|} \mu(da, b) + \mu(a, db).$$

Indeed the first strip break requires its translation line to pass the second input generator; the second break does not. This is the boundary calculation for the one-dimensional triangle space.

If $\theta \in C_{12}$ is a homogeneous cycle, then $R_\theta(a) = \mu(a, \theta)$ is a closed map of degree $|\theta|$. The product itself is even in this convention. For a change of gauge base at any of the three vertices, the corresponding symplectic conversion is diagonal on the same generator lines. Applying the conversions with (3.4) gives the product in the new bases. If the three conversions have parities t_{01}, t_{12}, t_{02} , the converted product has degree $m = t_{01} + t_{12} + t_{02}$ and satisfies

$$d\mu(a, b) = (-1)^m ((-1)^{|b|} \mu(da, b) + \mu(a, db)).$$

Inserting a cycle θ then gives a closed map of degree $m + |\theta|$. All subsequent changes of bases use this converted formula, with the right tensor convention.

7.2. The determinant comparison at the collapsed limit. The comparison must be defined on compact families of limiting configurations, where kernels and cokernels may occur. We first give that construction. Stationary mixed-end coordinates are included as finite-dimensional domain factors throughout. The fast factor is the determinant contribution from the complement of the harmonic modes on the collapsing compression.

Lemma 7.1 (Collapse of stabilized families). *Let $\{V_b\}_{b \in B}$ be a compact smooth family of limiting configurations with the specified autonomous end models and separated stationary coordinates. Assume that their remainders have a common exponential rate β at every required derivative order, and choose the comparison weights in common gaps strictly below that rate. Retain Paper I's unperturbed collapse datum: the primary perturbation vanishes on the inserted compression, and both the dependence and output of every other perturbation avoid the prescribed collapse neighborhood. Primary data on the remaining pieces may vary in a compact regular family. For sufficiently small $\epsilon > 0$, pregluing gives a continuous isomorphism of orientation lines*

$$(7.2) \quad \det D_{V_b} \longrightarrow \det L_{W_\epsilon(V_b)}.$$

It is independent, up to positive homotopy, of the stabilization and cutoff choices. It commutes with gauge and strip gluing and with the lifting actions. At a constant reference pair its fast factor has the canonical invertible-operator orientation. At a regular solution it is the comparison used by the unsigned collapse construction of Paper I.

Proof. Stabilization. Choose a finite-dimensional real space F and smooth compactly supported target columns $j_b : F \rightarrow Y_b$ so that $\widehat{D}_b = (D_{V_b}, j_b) : X_b \oplus F \rightarrow Y_b$ is onto. Such columns exist by smoothness of the cokernel on the limit domain (Proposition 2.1(a),(b), which includes the zero-scale limit domains) and density of compactly supported tests; a finite cover of B and parameter cutoffs give one F . Write $K_b = \ker \widehat{D}_b$. These kernels form a vector bundle. Choose projections $\Gamma_b : X_b \oplus F \rightarrow K_b$ whose restriction to K_b is the identity. They can be formed from compactly supported pairings, together with the stationary

coordinate projections at original ends and the coordinate tests on F : a nonzero kernel element cannot vanish in all three sets of tests. The F tests also detect redundant or inactive stabilization columns. A finite selection works over each compact parameter patch. Any free stationary coordinate at an artificial exterior end is also detected by compact pairings on K_b , after enlarging the compact set. Choose those pairings instead of an artificial-end limit projection: the artificial end will disappear on attachment. Then $(\widehat{D}_b, \Gamma_b)$ is an isomorphism, with uniformly bounded inverse over B .

Transport the columns to the preglued domain, away from the transition annuli, and denote them by $j_{\epsilon,b}$. Patch kernel fields by the harmonic lift on the collapsed compression and by the usual cutoffs elsewhere. Call this map $J_{\epsilon,b}$. Transport the compact pairings in Γ_b in the same manner, retaining their finite-dimensional factors in the original order. This defines

$$A_{\epsilon,b} = (L_{W_\epsilon(V_b)}, j_{\epsilon,b}, \Gamma_{\epsilon,b}) : X_{\epsilon,b} \oplus F \longrightarrow Y_{\epsilon,b} \oplus K_b.$$

The uniform linear estimate. The linear estimates of Paper I, Section 7, apply here before assuming that V_b solves an equation. On the collapsed strip their inputs are a smooth path of flat compression connections and a smooth section with the prescribed boundary values. The Hodge, trace and energy identities use these conditions and bounded derivatives of their coefficients. The zero-order term of the section linearization is retained. On the unchanged pieces the operator is the full D_{V_b} . The compactness of B bounds all these coefficients. The end and transition estimates use only the fixed end operators, the separated stationary coordinates, and the common gaps. Choose the auxiliary exterior weight $\delta_* < \beta$. Paper I's exterior truncation gives the stationary-column error

$$O(\epsilon^{1/4}) + O(\epsilon^{(\beta-\delta_*)/4}),$$

uniformly over B . Its exponentially convergent coefficient estimates and their weak-adjoint counterparts therefore apply to the original configurations. Exact constant tails are not required. Put $A = 1 + |\log \epsilon|$. First work on the auxiliary exterior of Paper I, Section 7.7, with its free stationary limits and fixed positive end weight. On this exterior the localized estimate holds with a compact remainder on the unchanged pieces and the limiting section. It also holds for the adjoint. If the augmented exterior estimate failed, normalized fields with image tending to zero would have a nonzero limiting field in $\ker(D_{V_b}, j_b, \Gamma_b)$ after passing to a subsequence of b . The compression Hodge estimate eliminates the fast complementary component, and harmonic matching identifies the remaining component with the limiting section. Compact convergence and the finite stationary and F tests detect the limiting field. The adjoint argument gives a field in $\ker(D_{V_b}, j_b, \Gamma_b)^*$. Both are impossible. Thus the augmented exterior inverse is uniformly bounded. Its finite columns and projections are supported off the new necks. Before attachment change the exterior weight from the fixed gap weight to $\delta_L \asymp A^{-1}$, using the small-weight exterior argument of Paper I. The stationary part is integrated explicitly; the normal part retains its fixed spectral gap. The finite compact columns and tests are unaffected. This gives the same polynomial loss at the small weight and avoids exponential weight costs for sources whose supports move out along an overlap.

The actual domain requires an additional attachment argument. Remove the sources in the middle of each overlap using the stationary line inverse of Paper I, the transition inverse, with its small weight of order A^{-1} . This costs $O(A)$; cutoff derivatives of order A^{-1} leave bounded sources. Solve those sources on the augmented exterior and the transitions, setting each transition's stationary value equal to the corresponding free exterior value. The two stationary values are independent. Patch in the source-free bands. The normal components decay exponentially across a band of length comparable to $|\log \epsilon|$, so the patching error is

bounded by $C_B A^N e^{-c|\log \epsilon|}$ for a fixed finite N . A Neumann series gives a right inverse. Cutting a homogeneous field in the same bands and applying the component inverses proves injectivity. The stabilization columns and projections occur only in the exterior equation and are unchanged by these operations. Consequently

$$\|A_{\epsilon,b}^{-1}\| \leq C_B A^N.$$

This is the source-removal proof of the attachment inverse in Paper I, with the exterior operator augmented by F and K_b . A bound independent of ϵ on the whole domain would be false: stationary fields supported on a long new neck have norm ratio of order $|\log \epsilon|$. The polynomial loss suffices below. No equation for V_b was used; its nonlinear residual is not a term in the linear component estimates.

For $k \in K_b$, cutoff differentiation and the horizontal velocity of the harmonic lift give

$$\|A_{\epsilon,b} J_{\epsilon,b} k - (0, k)\| \leq C_B \epsilon^\sigma (1 + |\log \epsilon|)^N \|k\|,$$

for some $\sigma > 0$ and finite N . These are the endpoint and patching estimates of Paper I, Sections 7.5–7.7: the fast lift has weighted size $O(\sqrt{\mathfrak{s}})$, where $\mathfrak{s}$ is the local compression scale ($\mathfrak{s} = \epsilon$ on the middle compression and $\mathfrak{s} = \epsilon e^s$ in the endpoint neck coordinate s of Paper I); the stationary factors are unchanged, and the fixed compact family has uniformly bounded coefficients. Take σ no larger than the positive exponents in these endpoint errors. The truncation estimates are uniform on the compact exponentially convergent family. Multiplication by the polynomial inverse bound still gives an error tending to zero. Hence $k \mapsto A_{\epsilon,b}^{-1}(0, k)$ identifies K_b with $\ker(L_{W_\epsilon(V_b)}, j_{\epsilon,b})$ and is asymptotic to the patched kernel map. Cancel $\det F$ in the two identities $\det D = \det \ker(D, j) \otimes (\det F)^*$ to obtain (7.2).

Compatibility. Two choices of stabilization are compared by their direct sum. The exact kernel sequence for adding the extra columns identifies the two cancellation maps. Cutoffs and compact pairings can be interpolated; the same estimates on the compact interpolation give a homotopy of the resulting maps. Lifting actions preserve the operators and the harmonic lift, so applying a lift to this construction gives the same determinant map after stabilization independence.

For ordinary end gluing choose the stabilizations off both transition regions. If T is the ordinary gluing length, choose a cofinal regime in which $e^{-cT} A^{N'} \rightarrow 0$ for each of the finitely many polynomial losses occurring in the construction. For example $T \geq C \log A + T_0$ suffices with C, T_0 large. Correcting the common patched kernel at once, or first at one neck and then at the other, now gives maps uniformly close to that patched map. Their straight interpolation on the finite-dimensional kernels is invertible. This compares the ordered stabilization cancellations without assuming an estimate on a full rectangle of independent parameters. Any two sufficiently large gluing choices connect through this cofinal regime. The ordinary gluing determinant maps are unchanged along that connection by continuity. We obtain the asserted compatibility, including its graded permutations.

At a constant pair the complementary fast operator is invertible and contributes its canonical positive factor. At a regular solution this is exactly the kernel patching and correction used in Paper I, with no additional stabilization required. This proves all the assertions. $\square$

We use the lemma to *define* the limiting triangle orientation. Complete its incoming ordinary strip ends with the already oriented one-input mixed vertices; the constant vertices at its generators suffice for this definition. Orient $L_{W_\epsilon(V)}$ by the positive-scale several-input construction above, pull its orientation back through (7.2), and cancel the completing-vertex lines in cyclic order. A limiting triangle need not admit a shift to a constant triangle: the

triple intersection of its boundary labels can be empty. The positive-scale shift need not preserve the image of collapse pregluing either. Neither assertion enters this definition.

Continuity for $0 < \epsilon < \epsilon_0$ on a compact configuration family compares sufficiently small scales. On overlaps choose a common smaller scale. Stabilization independence and ordinary gluing compare the auxiliary columns, completing vertices and necks. For a symplectic strip at an incoming end, apply the one-input comparison $C_x \# u \sim A \# C_y$ in its completing vertex, away from the central collapse neighbourhood, and absorb A into the ordinary cobordism factor. Thus the full end-gluing diagram commutes before the vertex lines are cancelled. This proves coherence of the defined triangle orientation without an independent component sign.

For a nonsolution reference family supplied only in a strong weighted topology, use a smaller comparison weight below its remainder rate. This does not put a slow nonsolution into a faster realization. Actual solutions instead use their proved exponential decay in the lemma and are not replaced by truncated solutions.

We may now specialize to a regular index-zero limiting solution V . Its complete realized operator, and the operator at its preglue, are invertible. Denote the actual nearby solution by $\mathcal{G}_\epsilon(V)$.

It remains to compare the preglued operator with the operator at the actual solution. A potential difficulty is that the chart used for the contraction estimate need not be the chart used to define the determinant bundle. The second derivative of an arbitrary fixed mixed chart need not have a uniform bound in the collapse norm. We use the following argument instead.

Lemma 7.2 (Transport around a square). *Let $D_{s,t}$ be a continuous family of Fredholm operators on a square. Suppose the operators on three edges are invertible. Parallel transport of orientations on the fourth edge carries the canonical orientation at its first endpoint to the canonical orientation at its second endpoint.*

Proof. The orientation double cover of the determinant bundle is trivial over the square. Transport around its boundary is therefore the identity. On each of the three invertible edges the canonical orientation is a continuous section, so transport is positive. The remaining edge must have the same property at its endpoints. $\square$

Proposition 7.3 (Affine centers at positive collapse scale). *Fix the collapse datum of Lemma 7.1 and a sufficiently small positive scale ϵ . Let $c_s = \Phi(sz_*) = (A_s, u_s)$, $0 \leq s \leq 1$, be the exact matching-chart segment from the preglue $c_0 = W_\epsilon(V)$ to the solution $c_1 = \mathcal{G}_\epsilon(V)$ supplied by Paper I. At a common sufficiently small original-end weight there is a continuous Fredholm family $D_{s,t}$, $0 \leq s, t \leq 1$, whose $t = 0$ edge is the full contraction derivative and whose $t = 1$ edge is the physical orientation operator at c_s . It retains the finite stationary coordinates, both operator rows, and their physical weak-adjoint domains. If V is regular of index zero, the other three edges are invertible.*

The Fredholm construction also holds in compact smooth families of these small matching-chart neighbourhoods, including nonsolutions. Its bounds may depend on the positive scale.

Proof. Write $g = g_\epsilon$, $\lambda = \lambda_\epsilon$ and $\hat{g} = \lambda^{-2}g$ for the metrics and rescaling in Paper I, Sections 7–8. Put

$$(7.3) \quad C_{s,t} = (1-t)A_0 + tA_s.$$

The center need not solve the primary equation at a mixed end. The exact chart fixes the normal connection trace, so the covariant Neumann condition is common to all these centers. A single normal gauge for A_0 , applied to the whole family, makes this trace zero without

changing the affine relation. For each fixed ϵ , the changed compression region is compact and its minimum scale is positive. Paper I's strong norms are then equivalent to ordinary weighted Sobolev norms, with the stationary coordinates adjoined; these equivalence constants need not be uniform as $\epsilon \rightarrow 0$.

At an original mixed end e , choose the primary representatives $B_e(p)$ in the spatial Neumann Coulomb slice of $B_e(0)$:

$$(7.4) \quad d_{B_e(0)}^*(B_e(p) - B_e(0)) = 0, \quad (B_e(p) - B_e(0))(n) = 0.$$

The spatial scalar inverse and implicit function theorem give this local section of the actual primary family. Extend it into the matching chart after subtracting its stationary value. This ensures that the affine scalar equation has weighted target. If an initial local primary section is used instead, its small spatial gauge gives this normalization; extend that gauge with a fixed longitudinal cutoff. The correction is quadratic in the stationary parameters when the initial tangent is Coulomb. It occurs on the fixed original-end geometry and preserves the initial collapse operator and the quadratic estimates of Paper I. The chart segment is smooth in all stationary-subtracted norms.

We first compare scalar operators without assuming compact support of $C_{s,t} - A_s$. With $d\mu = \lambda^2 dV_{\hat{g}} = \lambda^{-2} dV_g$, the operator $P_{C,A} = \lambda^2 d_C^* d_A$ has weak form

$$(7.5) \quad \langle \xi, P_{C,A} \eta \rangle_\mu = \int \langle d_C \xi, d_A \eta \rangle_{\hat{g}} d\mu.$$

The two factors on the right are generally different. Let C_P be the Poincare constant for A , and let a bound the bracket with $C - A$ in the rescaled metric. Then

$$\operatorname{Re} \langle \xi, P_{C,A} \xi \rangle_\mu \geq (1 - aC_P) \|d_A \xi\|_\mu^2.$$

If $aC_P < 1$, the bounded coercive form gives a weak inverse. Conjugating by an original-end weight $w = e^\omega$ adds form terms controlled by $\|d\omega\|_\infty$ and its square. For a common sufficiently small weight they are absorbed. Neumann elliptic regularity gives the higher Sobolev inverses at this fixed scale. The identical spatial argument makes $d_C^* d_B$ invertible when B is a primary limit and C is sufficiently close to it, whether or not C is primary flat.

We next specify all stationary columns. Let $h_{s,0,e}(p)$ be the connection value of the contraction-chart column at the limit $B_e(s)$, and let $C = C_{s,t,e}$ be the corresponding limit of the center. Define

$$(7.6) \quad \psi_{s,t,e}(p) = (d_C^* d_{B_e(s)})^{-1} d_C^* h_{s,0,e}(p), \quad h_{s,t,e} = h_{s,0,e} - d_{B_e(s)} \psi_{s,t,e}.$$

The new value is C -Coulomb and has the same primary boundary class. At $t = 0$ the correction vanishes by (7.4). Extend ψ by a fixed end cutoff to $\Xi_{s,t}(p)$ and put

$$H_{s,t}(p) = H_{s,0}(p) - G_s \Xi_{s,t}(p), \quad G_s \xi = (d_{A_s} \xi, 0), \quad J_{s,t}(w, p) = J_s^{\text{dec}} w + H_{s,t}(p).$$

Here J_s^{dec} is the decaying derivative of the exact chart. Reading the primary classes at infinity and subtracting $H_{s,t}$ proves that $J_{s,t}$ is a source isomorphism. It is the derivative of an actual matching chart: solve the nonlinear spatial C -Coulomb equation based at $B_e(s)$ and extend its small gauge with the same cutoff. This leaves the base configuration fixed and differentiates to the displayed formula. The scalar stationary image vanishes, and all remaining scalar columns belong to the weighted target. Equivariance gives

$$(7.7) \quad \nabla \mathcal{F}_{c_s} G_s \Xi = \rho_*(\Xi) \mathcal{F}(c_s).$$

This weighted equation column is retained; it need not vanish at a nonsolution.

For Fredholmness, first work in physical decaying spaces. The local primal estimates are [1, Theorem 4.17 and Remark 4.19]; the weak-adjoint estimates are [1, Remark 4.33] and [2, Theorem 5]. Their boundary-domain transport is [2, Proposition 5.27 and Lemma 5.67]. The changed compact core is smooth at this fixed scale. At a mixed stationary limit the current-center cylinder has the form $\partial_\theta - \mathcal{A}_B$, with self-adjoint spatial operator and a positive normal gap γ . Its weighted whole-cylinder inverse satisfies

$$M_\delta \leq \frac{1}{\min(\delta, \gamma - \delta)}, \quad 0 < \delta < \gamma,$$

as well as the spatial first-order estimate. Stationary coordinates are adjoined separately. Changing the scalar center to C adds a zeroth-order row of norm at most $c\|C - B\|_{C^0}$. If $cM_\delta\|C - B\|_{C^0} < 1/2$, a Neumann series supplies the affine cylinder inverse and the inverse of its actual adjoint. No self-adjoint spectral decomposition for the affine operator is required. The true end coefficients and matching domains converge to these models, so their errors are absorbed after a common far cutoff. Ordinary end models are unchanged. Compact-core center changes are relatively compact by Rellich. Hence, after weight conjugation,

$$\|v\|_{H^1} \leq C(\|D_{s,t}^{\text{dec}}v\|_2 + \|v\|_{L^2(K)}), \quad \|y\|_{H^1} \leq C(\|(D_{s,t}^{\text{dec}})^\dagger y\|_2 + \|y\|_{L^2(K)}).$$

Adding the finite stationary columns preserves Fredholmness. Higher estimates and smooth finite stabilizations identify the resulting determinant lines with the strong realizations.

For clarity, the adjoint here is taken after weight conjugation. A zeroth-order physical center change leaves the principal Green form and its matching annihilator unchanged. Write $K_{s,t}(p) = D_{s,t}(0, p)$ for the full target column corresponding to the source lift $H_{s,t}$. A cokernel element satisfies both $(D_{s,t}^{\text{dec}})^\dagger y = 0$ and $K_{s,t}^* y = 0$. A target map S pulls the weak domain back by S^* ; it does not leave the old adjoint boundary label fixed. The source chart, which includes a boundary trace extension, need not extend to a bounded map on bulk L^2 . We use it as a bounded isomorphism of Sobolev domains, which preserves the cokernel annihilator. Equivalently, complete its decaying source in the pulled-back norm $\|J_s^{\text{dec}}w\|_{L^2}$ before forming a coordinate adjoint. Keep the finite stationary space as a separate summand.

There is also a target derivative. On the section factor choose S_c by pointwise parallel transport from $u(z)$ to $u_0(z)$ along the short geodesic in the target. The gauge target is fixed. Shrinking the chart gives uniform normal neighbourhoods; $S_{c_0} = 1$ and $\nabla S|_{c_0} = 0$. Pointwise multiplication estimates preserve Paper I's quadratic bound and initial operator. Differentiating gives

$$(7.8) \quad d(S\mathcal{F})_c[v] = S_c \nabla_v \mathcal{F} + R_c v, \quad R_c v = (\nabla_v S)_c \mathcal{F}(c).$$

The coefficient of R_c tends to zero on the original ends, where the residual tends to zero. It is compact from the weighted decaying H^1 domain to L^2 , by compact truncation and Rellich, and its stationary columns are weighted finite rank. The anchoring gives $R_{c_0} = 0$ even at the nonsolution preglue; also $R_{c_1} = 0$ since c_1 is a solution. The target map itself may have a nonidentity stationary limit.

The required square is

$$(7.9) \quad D_{s,t} = \left(S_{c_s} \nabla \mathcal{F}_{c_s} + (1-t) R_{c_s}, \lambda^2 d_{C_{s,t}}^{*,g} \pi_A \right) J_{s,t}.$$

The preceding estimates prove its Fredholmness and continuity everywhere. Its $t = 0$ edge is exactly the derivative of the contraction equation, including the affine scalar row and (7.8). Paper I's inverse and quadratic bounds make this entire edge invertible. Strong invertibility implies first-order invertibility: kernels regularize into the strong domain, and a weak cokernel

would annihilate every smooth target reached by the strong inverse. Closed range and density exclude that cokernel. At $s = 0$, the centers and source maps are constant and $R = 0$, giving the invertible preglued operator used in (7.2).

At $s = 1$, equivariance gives $\nabla \mathcal{F}_{c_1} G_1 = 0$. The scalar inverse splits the source into horizontal vectors and decaying gauge directions. The resulting blocks are the intrinsic quotient linearization and P_{C_1, t, A_1} . Changing (7.6) changes only the representative of that quotient tangent; (7.7) now vanishes. Regularity and index zero make the quotient block invertible, and scalar coercivity makes the second block invertible. This proves invertibility of the third edge. At $t = 1$ the residual term is absent and the center is A_s , so the remaining edge is precisely the physical orientation operator path. No invertibility assertion is made along its interior.

Finally the required smallness follows from the actual contraction bounds, not an unspecified fixed-scale neighbourhood. On the unchanged original ends the spatial gaps and cylinder constants are fixed. The stationary correction is bounded by $C(1 + |\log \epsilon|)^{m+1} e_\epsilon \rightarrow 0$, with e_ϵ the pregluing error of Paper I. Its strong-control estimate also gives $\|A_s - A_0\|_{C^0(\widehat{g})} \rightarrow 0$. The temporal component in this norm is the rescaled component, with no additional ϵ^{-1} factor. The preglue's uniform scalar Poincare estimate therefore persists, and both scalar and end smallness conditions hold for sufficiently small ϵ . These facts do not give a uniform full mixed inverse at zero scale. Compact parameter families use finite local covers; actual gauge transitions transport both rows and all stationary columns. Ordinary gluing at fixed positive scale commutes with these identifications by the bordered determinant construction, with corrections placed on the mixed factors and fixed ordinary end models. No simultaneous independent neck-and-collapse estimate is needed for this square. $\square$

Proposition 7.4. *The map (7.2), followed by transport from $W_\epsilon(V)$ to $\mathcal{G}_\epsilon(V)$, preserves the sign of the zero-dimensional solution.*

Proof. Use the family (7.9) of Proposition 7.3. Its contraction edge is invertible by the nonlinear estimate, its preglue edge is the initial invertible collapse operator, and its solution edge is invertible by the quotient and scalar block decomposition. Lemma 7.2 identifies the canonical orientations along the remaining physical operator path. Composing this identification with (7.2) proves the claim. $\square$

If the conformal metric in the collapse estimate is changed within the unperturbed collar class, the zero set of the mixed equation is unchanged. At a solution Lemma 5.2 compares the gauge rows by invertible paths, so the sign is unchanged. This permits use of the metric of Paper I without introducing a new orientation convention.

The construction is compatible with compact parameter families because its stabilizations and operator bounds may be chosen locally uniformly. Associativity of ordinary gluing identifies the determinant maps at broken ends. This proves Theorem B.

7.3. Vertex identifications and two-handles. In the triangle degeneration a completing vertex is identified with a product of a gauge path and a constant mixed pair. Normalize this identification using the corresponding end frames, with the outgoing coordinate translated in the positive direction on every arm. The constant pair has sign +1 and the gauge path uses the fixed object line. The resulting vertex identification has sign +1 at the reference configuration. Shifting and gluing show that it has that sign throughout its configuration component.

For the standard surgery triple let θ_G denote the distinguished gauge cycle in the auxiliary connected sum of $S^1 \times S^2$'s. Normalize its symplectic counterpart by

$$\theta_S = N_{12}\theta_G.$$

The unit diagonal and the usual filtration of that auxiliary complex identify this with the distinguished top cycle, with the chosen sign. The mixed triangle family has one endpoint equal to a gauge triangle followed by the mixed map, and the other endpoint is a symplectic triangle with the two mixed inputs. Proposition 7.4 compares their orientations. The broken boundary terms give a homotopy, hence

$$(7.10) \quad N_{02}\mu_G \simeq \pm\mu_S(N_{01} \otimes_R N_{12}).$$

Evaluating the second input on θ_G gives

$$(7.11) \quad N_{p_1}I_J^\#(W_2) \simeq \pm S_{W_2}N_{p_0}.$$

Changing bases conjugates this identity by the right-placed conversions and uses (3.2). Twisting the bundle on the two-handle changes the gauge triangle class but not the orientation argument; its relative bundle is retained throughout the mixed family.

For simultaneous gluing at two triangle ends, use one stabilization off both ends. The determinant comparison is independent of the order of gluing, up to the graded reordering already present in (3.4). Thus the triangle systems are coherent at corners, which is needed when the same product is later folded into a compression cup.

8. TORUS EXCISION OVER THE INTEGERS

8.1. The closed cut and its orientation data. Let $W_k : M \rightarrow M^+$ be the cobordism obtained by adding k one-handles, where $M^+ = M \#^k(S^1 \times S^2)$. The folded triangle for this compression has a closed gauge cut

$$Z \cong (-M) \# M^+ \# T^3.$$

The odd tori $T_a, T_b \subset Z$ are homologous, and the lifting group at this cut is generated by $s_Z = \text{PD}[T_a] \pmod{2}$. Excision along these tori gives

$$Z' = (-M)^\# \sqcup (M^+)^\#.$$

Write $E : C_G(Z) \rightarrow C_G(Z')$ for the gauge excision map and $\bar{E}$ for its reverse. The tensor complex at Z' uses the right convention.

The almost-complex structure induced by the folded triangle at Z need not equal the one induced by the mixed torus domain. A conversion at this interior cut is therefore required. We choose the structures so that

$$(8.1) \quad c_1(J)[T_a] \equiv 0 \pmod{4}.$$

For two such structures, the character (4.1) on s_Z is trivial. Consequently the conversion at Z is defined on the quotient complex.

The choice uses the topology of the particular fillings. In the handle filling of Paper I, Section 11.1, one has

$$B_k = (W_k \setminus \text{int } \nu(\gamma)) \natural (T^2 \times D^2),$$

and T_a, T_b are fibres in the boundary of the displayed torus cap. Choose the product almost-complex structure on the cap. Its first Chern class evaluates to zero on both fibres. The boundary sum is along a three-ball, so Mayer–Vietoris makes $H^2(B_k; \mathbb{Z}) \rightarrow H^2(T^2 \times D^2; \mathbb{Z})$ surjective. The relative obstruction argument of Lemma 4.1, with the whole cap as the prescribed region, extends this product structure: the primary obstruction has zero connecting

class and the secondary relative group in degree four vanishes. The other end restrictions remain free.

On the excision side choose the same product germ on the T^2 times polygon region of Paper I, Section 10. The attachments are away from the fibres at T_a, T_b . Write $A = T^2 \times U$, where U is the disk polygon of that construction. Its complement is the disjoint union of $[0, 1] \times Q_-$ and $[0, 1] \times Q_+$, with $Q_\pm = (T^2 \times I_\pm) \# M_\pm$. Each $Q_\pm$ has an integral cohomology class restricting to the torus generator on both its boundary tori. These classes match the generator on A along all four attaching faces. Mayer–Vietoris therefore gives $H^2(\mathcal{E}; \mathbb{Z}) \rightarrow H^2(A; \mathbb{Z})$ onto. Also $H^3(A; \mathbb{Z}) = 0$ and $H^4(\mathcal{E}; \mathbb{Z}) = 0$, so the secondary relative obstruction vanishes. The same argument applies with the cobordism reversed. Make both structures cylindrical at the closed cut, preserving their product germ near these two fibres. Thus their Chern evaluations are actually zero. This is a choice on the stated fillings and product regions; it does not require a divisibility assertion for arbitrary tori in four-manifolds.

Convention 8.1. The determinant line of the conversion at an interior cut is placed after the complete input generator line and before the complete configuration determinant line. Its position is fixed relative to the entire configuration, independently of where the geometric cut is drawn.

Suppose the configuration is written as a sequence of glued pieces. Moving an odd conversion line with the cut through that sequence would move it past the determinant line of the intervening piece. The sign would then depend on its index. Convention 8.1 avoids this dependence: moving the line to a fixed reference position costs a sign determined by the entire configuration, which is unchanged by moving the cut. At index zero the alternative fixed placement after the complete configuration line agrees with this one.

8.2. Lifting groups and gluing multiplicities. It is useful to perform the gauge excision calculation before passing to a quotient. Let $\mathcal{G}_0(X)$ be the subgroup of bundle automorphisms with trivial lifting class. If a finite group ϕ_X acts freely and orientation preservingly on the relevant zero-dimensional solution set, its quotient count is

$$(8.2) \quad I(X)^{\phi_X}(\bar{a}_0, \bar{a}_1) = \frac{1}{|\phi_X|} \sum_{a_i \in \bar{a}_i} \# \mathcal{M}(X; a_0, a_1).$$

The numerator is divisible by $|\phi_X|$: it is a sum of free orbits, each with a constant sign.

Lemma 8.2. *Let X be connected and let Y be a closed hypersurface with cut manifold X_Y . Suppose the critical limits on Y have trivial stabilizer in the adjoint determinant-one gauge group. For each matched configuration on X_Y that glues, the resulting $\mathcal{G}_0(X)$ -classes form a torsor for*

$$K_Y = \ker(H^1(X; \mathbb{Z}/2) \rightarrow H^1(X_Y; \mathbb{Z}/2)).$$

In particular their multiplicity is $|K_Y|$.

Proof. Choose representatives whose two limits agree under the fixed gluing identification. If two choices glue to the same determinant-one class on the cut manifold, their difference is an automorphism whose lifting class restricts to zero on X_Y . Conversely every such class changes the gluing while leaving the cut class fixed. The stabilizer condition makes this action free. Its transitivity follows by comparing the two gluing identifications on the two copies of Y . The Mayer–Vietoris sequence identifies K_Y with the image of the connecting map from $H^0(Y; \mathbb{Z}/2)$. The matching and analytic assertions needed for the excision cobordisms are proved below in Proposition 2.3. $\square$

A finite lifting group acts freely on the solutions in the present calculation. Indeed, a fixed solution would give a parallel automorphism. Its restriction to an irreducible end must be the identity once its lifting class there vanishes. A parallel automorphism which is the identity on an end is the identity everywhere on a connected cobordism. The almost-complex orientation construction preserves signs under the whole group. Thus (8.2) applies.

8.3. Ordinary self-gluing with regular cut data. We prove Proposition 2.3 using ordinary instanton theory. Write $\mathcal{E}$ for the cobordism defining $\mathbf{E}$ and put $V = \mathcal{E} \cup_{Z'} \bar{\mathcal{E}}$. The local polygon description gives a nonseparating cut $\Sigma = T^2 \times \gamma$; its complement V° is connected and retains the original Z ends. The bundle is odd on the torus fiber. The other composite has a separating torus cut. All components of the cobordisms that we regularize have an original cylindrical end; the standard caps are kept fixed. The geometric cap comparison and the larger lifting groups will be described in the next subsection.

Proof of Proposition 2.3. Perturbations and relative regularity. Keep the entire ordinary end functional fixed, including the trajectory perturbation in $\text{CS}_h + f$ from Paper I. Only the secondary cobordism perturbations vary. On each new three-torus use a fixed flat product metric and the unperturbed equation, throughout all the comparison families below. The odd bundle has two determinant-one flat critical points, both acyclic. The class $\text{PD}[T^2 \times \{\text{point}\}]$ interchanges them; their relative grading is four modulo eight [8, Proposition 5.2 and Lemma 5.3]. Thus the critical points have the same parity and the required lifting action is free.

Every nonconstant unperturbed torus trajectory has full index at least four. Indeed, the two end Hessians are conjugate by an adjoint gauge transformation, so their spectral correction terms agree. The cylinder index and energy are 8κ and $8\pi^2\kappa$, respectively, with $\kappa \in \frac{1}{2}\mathbb{Z}$. Positive energy therefore implies index at least four. The required compactifications have total dimension at most two. After joint regularization of their cut pieces, a torus trajectory of index at least four would leave a negative augmented index for the remaining pieces, counting the data parameter once. Such a stratum is empty. Consequently no torus trajectory perturbation is needed in this dimension range. We do not assert regularity of the higher-index unperturbed torus trajectories. The caps use these same unperturbed end data and have no secondary perturbation.

Choose a product collar joining each component to an autonomous end, and a compact interval J in that collar. The metric is fixed there, including in parameter families. If F_Y is the image on the collar cross-section Y of the allowed lifting group, use secondary functionals

$$(8.3) \quad q_s(B) = \sum_{\ell} c_{\ell} \psi_{\ell}(s) h_{\ell}(B), \quad \text{supp } \psi_{\ell} \subset J,$$

where the h_{ℓ} are F_Y -invariant holonomy cylinder functions on Y . Their holonomy loops lie in a single slice. Elements of the kernel of the restriction to F_Y impose no further condition. Use a countable dense collection of profiles and cylinder functions, with a weighted Banach completion controlling all required smooth norms. The holonomy estimates of [7, Propositions 3.7 and 3.15] persist: time derivatives multiply the corresponding profile bounds, and the derivative in B is bounded on L^2 on bounded sets of data.

This allows sums of compact time profiles in the secondary perturbation construction of [8, Section 3.7]. For the compactness estimate, include the fixed primary cutoff terms in q_s . In a consistent gradient normalization,

$$(8.4) \quad \frac{d}{ds}(\text{CS} + q_s)(B(s)) = -\|\dot{B}(s)\|^2 + \partial_s q_s(B(s)).$$

The additional integral is bounded by the L^1 norm of the time derivatives of the coefficients, uniformly under stretching. Its support remains compact. The spatial holonomy estimates, local regularity and autonomous middle-neck equation are the same as in the ordinary broken-metric construction. No perturbation depends on connections on both sides of a cut.

We give the transversality argument because averaging a generic perturbation does not by itself prove invariant regularity. First, slice continuation holds for these time-dependent collar equations. On a compact slab write the equation for a difference or an adjoint field as

$$(\partial_s + H(s) + K(s))u = 0.$$

Here H is formally self-adjoint on a common smooth domain, H' is L^2 -bounded and K is L^2 -bounded. For the one-form equation, $H = \pm * d_{B(s)}$; the connection derivative is order zero. The difference of two nonlinear holonomy terms has this bounded form by integration of their derivatives along the segment between the two connections. For a compactly supported smooth v and a concave weight φ with $-\varphi'' \geq c > 0$,

$$\|(\partial_s + H + \lambda\varphi')v\|^2 = \|v'\|^2 + \|(H + \lambda\varphi')v\|^2 - \int \langle v, H'v \rangle - \lambda \int \varphi'' \|v\|^2.$$

Absorbing K gives, for large λ ,

$$(8.5) \quad (c\lambda - C)\|e^{-\lambda\varphi}u\|^2 \leq 2\|e^{-\lambda\varphi}(\partial_s + H + K)u\|^2.$$

Choose φ increasing on a short slab. If a solution vanishes before a slice, cut it off farther into the slab; the cutoff error has strictly larger φ than a neighborhood of the slice. Letting λ tend to infinity proves vanishing in that neighborhood. Reversing s proves the opposite continuation. Vanishing on a whole slice suffices, since the equation then makes all time derivatives vanish there and extension by zero is smooth. This estimate does not require ellipticity of $*d_B$ on all one-forms.

Suppose an allowed automorphism fixes a slice of a solution. Extend it constantly in temporal gauge. Invariance and slice continuation make it fix the entire end and its limiting critical point. Freeness of F_Y at that point forces its lifting class to vanish; irreducibility then makes its even lift central. Thus each collar slice has a free F_Y orbit and trivial effective even stabilizer. The cylinder-function density in the proof of [7, Proposition 3.10] detects every nonzero smooth direction transverse to gauge there. It also does so invariantly: take a small finite-dimensional slice through the direction and its finitely many disjoint translates, prescribe equivariant first jets on their union, approximate by cylinder functions, and average. Freeness ensures that the prescribed derivative survives.

Let L_A be the linearized perturbed ASD equation. A cokernel element for (d_A^*, L_A) is a pair (ξ, b) , with b self-dual. Gauge equivariance at a solution gives $L_A d_A = 0$, hence $d_A^* d_A \xi = 0$. The L^2 end condition and integration by parts give $\xi = 0$. On the collar write b using a one-form $y(s)$. The remaining adjoint equations give

$$d_{B(s)}^* y = 0, \quad -\partial_s y + *d_{B(s)} y + K(s)y = 0.$$

If this cokernel annihilates every secondary variation, then

$$\int_J \psi(s) \langle y(s), \nabla h(B(s)) \rangle ds = 0$$

for every time profile and invariant cylinder function. Time localization, invariant density and the Coulomb equation imply $y = 0$ on J . Equation (8.5) propagates this to the unperturbed core. Local unique continuation for the ordinary ASD adjoint gives vanishing on the connected core, and slice continuation handles its other collars. This proves surjectivity of

the universal linearization. All nonlocal terms in this argument are supported on product end collars; spatial unique continuation through nonlocal interior terms is not used.

Sard–Smale now supplies invariant regular data. For a broken configuration with pieces indexed by i and one parameter $b \in B$, the joint linearization is

$$(8.6) \quad ((\xi_i)_i, \dot{b}) \longmapsto (D_i \xi_i + \partial_b \mathcal{F}_i \dot{b})_i.$$

Independent secondary variations on the pieces kill every component of a joint cokernel by the preceding argument. There is only one $T_b B$ summand. Choose regular data on fixed faces first, extend over their collars, and use parameter-dependent variations supported away from them. Intersect the residual sets for the finitely many cut patterns and countably many relative classes. Fixed regular caps require no variation. This gives the relative regular families of [8, Proposition 3.10 and Section 3.9] in the invariant class just constructed.

Matching on the prescribed bundle and finiteness. An even gauge transformation is a section of the bundle with fiber $\mathrm{SU}(2)$ and conjugation structure action; no lift of the underlying $\mathrm{SO}(3)$ bundle is required. The fiber is 2-connected, and conjugation acts trivially on π_3 . If gauge values are prescribed on boundary components B_0 of a compact cobordism W , the sole extension obstruction is in

$$H^4(W, B_0; \mathbb{Z}) \cong H_0(W, \partial W \setminus B_0; \mathbb{Z}).$$

It vanishes when every component has an unprescribed boundary end. Apply this to the two new boundaries of V° , leaving its original Z ends free. Equal critical classes can therefore be represented by equal connections under the prescribed gluing identification, with allowed even changes at the original ends. They glue on the specified global bundle. For capping, keep the cap representatives fixed and make the changes on V° .

Components of the even gauge group on a connected closed three-manifold are classified by $H^3(Y; \mathbb{Z}) = \mathbb{Z}$. On a connected W the obstruction for prescriptions on its whole boundary is the signed sum

$$\bigoplus_{Y \subset \partial W} H^3(Y; \mathbb{Z}) \longrightarrow H^4(W, \partial W; \mathbb{Z}) \cong \mathbb{Z}.$$

It is onto, with kernel the extendible prescriptions. Thus relative classes for a fixed tuple of critical end classes form a $\mathbb{Z}$ -torsor, whose generator changes the ASD index by eight. Every such class can be represented on the fixed bundle in this nonsingular setting [8, Section 3.7]. Central stabilizers introduce no additional degree. A fixed index and tuple therefore allow at most one relative class. The critical sets are finite. In a regular family, each component index is bounded below by minus the base dimension; at fixed total dimension this leaves finitely many index allocations. The shared parameter is counted once in the total dimension.

The final secondary data may be finite sums in (8.3). On a bounded neighborhood of data, (8.4) and the ordinary index-energy relation bound the index below. The preceding class calculation makes the bounded index ranges finite. The regular low-dimensional compactifications have regular gluing neighborhoods at their broken strata. Surjectivity is open on their finite union: a sequence of nonregular solutions for convergent data would limit either to a regular point or to a regular broken configuration, contradicting the local estimate in either case. Finite sums are dense, also for smooth coefficient families on a compact base. Approximate within this open set, relative to the prescribed face collars. This preserves the original end equations and all supports.

Local gluing and exhaustion. At an irreducible nondegenerate critical connection β , the extended scalar-plus-one-form Hessian B_β is invertible. Its spectral gap gives a uniform inverse for $\partial_s + B_\beta$ on the full cylinder. A regular index-zero cut solution A has an invertible

full operator D_A . For self-gluing, patch its inverse on the connected cut piece, including both stubs, with the cylinder inverse. Cutoffs with derivatives $O(T^{-1})$ give an approximate right inverse R_T and, separately, the estimate

$$\|D_{A_T}R_T - 1\| \leq C(T^{-1} + \eta_T), \quad \|u\|_{H^1} \leq C\|D_{A_T}u\|_{L^2} + C(T^{-1} + \eta_T)\|u\|_{H^1}, \quad \eta_T \rightarrow 0.$$

There is one copy of D_A , rather than one for each stub. Compactly supported secondary terms and slice-local primary terms commute with these cutoffs. The full inverse is uniformly bounded for large T . Exponential pregluing error and the uniform quadratic estimate in a sufficiently high Sobolev norm give a unique nearby solution in the Coulomb slice. Stabilization gives the same charts and determinant maps for regular universal families.

Use the broken-metric compactification argument of [8, Section 3.9], with empty singular locus. Its definition permits a nonseparating cut, and the odd torus satisfies the required non-integral condition. On the new torus ends we use the index exclusion above in place of regularity of all trajectories: any nonconstant torus trajectory removes at least four from the index available to the regular joint cut operator. This remains impossible in the two-parameter corner families. The retained strata have regular original-end trajectories, regular joint cut operators and acyclic matching limits, exactly the inputs to the local compactification and gluing arguments. In total dimension at most one, the cut solutions form the broken-metric boundary; further neck breaking has higher codimension, and bubbling is excluded by low-dimensional compactness. The local gluing charts give its collars. Finiteness of the union over relative classes then gives one neck threshold: any additional solutions at lengths tending to infinity would converge to a boundary point and hence belong to its gluing collar.

For disjoint cuts use the multi-parameter corner model; the two-parameter index-zero family has dimension two, still below the source's bubbling threshold. A simultaneous stretching of two components is the diagonal in the two gluing parameters; the product gluing collars restrict to its collar. Equivalently the two-edge composition cycle is obtained by one separating gluing followed by one self-gluing. The cuts Σ and Z' intersect, so their comparison instead joins finite stretching parameters by a relative regular path. It does not stretch intersecting cuts simultaneously.

The lifting fiber. Compare the even lifts on the components of a cut manifold. Their values at paired ends agree up to central signs. Changing a lift by the center on one component changes the incident signs by the graph coboundary. The remaining choices form the first cohomology of the cut graph over $\mathbb{F}_2$, identified with K_Y by Mayer–Vietoris. These classes have actual gauge representatives. For a connected cut, choose a unit section n of the adjoint bundle on Y and, on its normal collar, use the adjoint of

$$\cos(\pi t) + \sin(\pi t)n, \quad 0 \leq t \leq 1,$$

with t constant near the endpoints. The section exists because the Euler class of an oriented rank-three bundle is 2-torsion and $H^3(Y; \mathbb{Z})$ is torsion-free. The adjoint transformation is identity at both ends, while its lift runs from $+1$ to -1 ; its lifting class is $\text{PD}[Y]$.

The action is transitive by comparison of the lifts. It is free: a fixed solution would give a parallel automorphism, even on an original irreducible end, hence identity there and everywhere. The almost-complex determinant orientations preserve this action. Thus the multiplicities are two for the nonseparating Σ cut and the two-edge Z' composition cycle, and one for a separating cut or a cap tree.

Trace signs and homotopies. First use the induced almost-complex structures at the ends, with the orientation torsors of [8, Section 5.1]. Apply its Proposition 5.5 with the enlarged

group exactly $K_\Sigma = \langle \text{PD}[\Sigma] \rangle$. This group restricts trivially to V° , to Σ and to the original ends. The trace therefore uses the full determinant-one complex on Σ . The free orientation-preserving action just proved recovers the raw count as twice that trace. The larger quotient is taken only after this step.

The alternating sign here is an analytic determinant sign, as in the proof of [8, Proposition 5.5], which refers to [9, Lemma 2.4]. In the periodic model $\partial_s + B$, a simple real eigenvalue crossing changes the real zero Fourier mode once; the nonzero modes occur in complex pairs. The sign is spectral-flow parity, hence the parity of the critical generator, with one fixed reference normalization. Excision localizes the comparison to the neck. The cut operator and any stabilization occur on both sides; continuous changes of cutoffs introduce no further sign. Internal basis changes cancel, and the index-eight change of relative class preserves the complex normalization. This explains why there is no independent sign choice for each generator or relative class.

With the traced factor last, write F_{jj} for the diagonal block of a degree- f map F on a tensor end with torus basis g_j . The right tensor convention gives

$$(8.7) \quad \text{Tr}_R(F) = \sum_j (-1)^{(f+1)|g_j|} F_{jj}.$$

For $f = 0$ this is the alternating trace just normalized. Moving the traced line past the degree- f map supplies the additional $(-1)^{f|g_j|}$. Conjugation by the tensor isomorphism T of Section 3 gives the same formula algebraically. The trace commutes with the Hom differential and satisfies

$$\text{Tr}_R((U \otimes_R 1)F(V \otimes_R 1)) = U \text{Tr}_R(F)V.$$

Consequently the external conversion maps can be inserted with their actual degrees. Both torus generators have one parity p , so comparison with unit cap coefficients $u, v \in \{1, -1\}$ has factor $(-1)^{(f+1)p}/(uv)$, up to the fixed overall orientation. Interior conversions retain Convention 8.1.

Finally use the parameter-first orientation of Section 5, with outward-normal-first boundary orientations. The source convention in [8, Section 3.9] is fiber-first. For an r -parameter family the permutation of its determinant factors contributes

$$(-1)^{r \text{ ind } D}.$$

For a zero-dimensional one-parameter count the vertical index is -1 , so this changes the sign of the count. For the universal one-dimensional space the vertical index is zero, so its orientation and its endpoint maps are unchanged. For two parameters the permutation sign is one; boundary faces use the same one-parameter rule.

We include the separate generator-order conversion. At the induced end gradings the unconverted cobordism maps F_i are even. Write d_0 and H_0 for the source differential and its fiber-first homotopy count, so its oriented boundary formula is $F_1 - F_0 = d_0 H_0 + H_0 d_0$. Placing an odd trajectory determinant after its input generator changes d_0 to $d = d_0 \varepsilon$ and H_0 to $H_0 \varepsilon$. The parameter-first change contributes the additional minus sign, so the homotopy operator in our convention is

$$K = -H_0 \varepsilon, \quad dK + Kd = d_0 H_0 + H_0 d_0 = F_1 - F_0.$$

Here ε is the parity operator on the appropriate source complex; the calculation uses that d_0 and H_0 are odd. On disconnected ends the whole ordered input generator line gives the tensor rule (3.3). Thus the parameterized boundary formula agrees with (5.3). Arbitrary external conversions, including odd ones, are then governed by the Hom Leibniz rule. All

these choices are uniform across matrix entries. This proves all three assertions in the stated orientation convention. $\square$

8.4. The capped comparison. Let $V = \mathcal{E} \cup_{Z'} \bar{\mathcal{E}}$ be the composite of excision and reverse excision. The standard local model is T^2 times the doubled excision polygon. It contains a nonseparating hypersurface $\Sigma = T^2 \times \gamma$. Cut along Σ to obtain V° , and cap the two new boundary components by $T^2 \times D^2$. The capped cobordism V' is a product on Z ; this is the geometric comparison in torus excision [8, Theorem 5.6].

The determinant-one complex on Σ has two critical points g_1, g_2 . The class $a = \text{PD}_\Sigma[T^2]$ interchanges them. They must be kept separate when the caps are attached: the caps carry the trivial lifting group, since a does not extend across their meridional disks.

Write $s = \text{PD}_Z[T_a] \pmod{2}$. In $H^1(V^\circ; \mathbb{Z}/2)$ let p be its pullback and h_1, h_2 the meridional classes of the two removed cap neighborhoods. The restrictions are

	Z_{in}	Z_{out}	Σ_{in}	Σ_{out}
p	s	s	0	0
h_1	s	0	a	0
h_2	s	0	0	a
$h_1 + h_2$	0	0	a	a

The group $H^\circ = \langle p, h_1, h_2 \rangle$ has order 8. It contains both groups needed on the cut cobordism, but is not itself the pullback of either of the groups on V and V' .

More precisely, ϕ_V has order 8, its image on V° is $\langle p, h_1 + h_2 \rangle$ of order 4, and its kernel is K_Σ of order 2. The capped product has group $\phi_{V'} = \langle p \rangle$ of order 2. To check these assertions, the two sides of the excision polygon give classes P_+, P_- and the disconnected intermediate cut gives a class c . Their restrictions are $p, p, h_1 + h_2$, respectively. They are independent on V , as seen by loops crossing each of the two polygon sides and the intermediate cut. Their sum $P_+ + P_-$ is dual to Σ , generates the restriction kernel, and is nonzero because Σ is nonseparating.

Choose invariant regularizing data on V° . Let

$$D(\alpha, \sigma; \beta, \tau) = \#\mathcal{M}(V^\circ; (\alpha, \sigma), (\beta, \tau)),$$

where $\sigma, \tau \in \{g_1, g_2\}$. Put $\Delta(\alpha, \beta) = D(\alpha, g_1; \beta, g_1)$. The action of $h_1 + h_2$ makes the two diagonal blocks equal; the action of h_2 identifies the off-diagonal block with $\Delta(s\alpha, \beta)$. Thus, after summing over endpoint lifts, diagonal and off-diagonal blocks have the same sum.

We verify the cap count for the unperturbed torus data chosen in Proposition 2.3. Complete $T^2 \times D^2$ by a product cylindrical end, giving $T^2 \times S$, where S is a complete disk with a cylindrical end. Give T^2 a flat metric and use a product metric on the cap. Let A_0 be the pullback of the odd flat connection on T^2 . Its adjoint local system on T^2 is acyclic: it splits into the three nontrivial real sign local systems. The torus Hodge Laplacian therefore has a positive lower bound in every degree. The product Hodge Laplacian on $T^2 \times S$ has the same lower bound, since its two summands are nonnegative.

For the full flat operator $D_0 = d_{A_0}^* \oplus \sqrt{2}d_{A_0}^+$, the identities

$$D_0^* D_0 = \Delta_{A_0, 1}, \quad D_0 D_0^* = \Delta_{A_0, 0} \oplus \Delta_{A_0, 2}^+$$

show that its kernel and cokernel vanish. Elliptic estimates and the positive gap give an inverse on the cylindrical Sobolev spaces. Thus the flat cap has index zero and is regular.

Call its boundary lift g_1 . The other lift g_2 has relative grading four modulo eight, whereas changing a relative class at a fixed lift changes the cap index by eight. Hence an index-zero cap can approach only g_1 , in the zero-charge class of A_0 . Its unperturbed ASD energy is

$8\pi^2\kappa = 0$, so it is flat. The flat odd torus connection has a unique flat extension across the disk in the prescribed bundle: the disk meridian is contractible and $\pi_1(T^2 \times S) = \pi_1(T^2)$. The two boundary lifts distinguish whether the meridional determinant-one lift extends; only g_1 does. Consequently there is exactly one determinant-one cap class. Its signed coefficient is a unit, and the reversed cap gives the corresponding dual unit. This calculation is at the full two-generator torus complex, before any lifting-group quotient.

Proposition 8.3. *For the regular data of Proposition 2.3, the quotient counts for V and V' have the same absolute multiplicity. In each fixed degree they agree up to the determinant-gluing sign.*

Proof. For V , self-gluing has multiplicity 2 and there are two equal diagonal blocks. Quotienting by the group of order 8 gives

$$\frac{2 \cdot 2}{8} \sum_{\alpha \in \bar{a}, \beta \in \bar{b}} \Delta(\alpha, \beta).$$

For V' , the caps have unit coefficients, their separating gluings have multiplicity 1, and the group has order 2. The result is

$$\frac{1}{2} \sum_{\alpha \in \bar{a}, \beta \in \bar{b}} \Delta(\alpha, \beta).$$

If the second cap selects g_2 , the symmetry of the off-diagonal block gives the same sum. The orientation signs are those in Proposition 2.3. All group actions preserve the orientations, so no cancellation occurs inside an orbit. The two numerical factors agree. $\square$

In particular no division by two is performed on a Floer homology group. The fractions above express integer orbit counts. Computing the caps only in the rank-one quotient would lose the distinction between matching two lifts and matching two orbits.

The one-dimensional gluing family identifies these counts with the composite excision map. The same argument for the other composite gives the reverse comparison. Consequently $H(\mathbb{E})$ and $H(\bar{\mathbb{E}})$ are isomorphisms. On a homogeneous class their composite is that class up to a sign depending on its degree. This is the only inverse-excision statement needed for the handle state below. By Lemma 3.1, the closed excision maps are homotopy equivalences (after shifting degrees if necessary). We do not replace a possibly degree-dependent sign by a single scalar on an ungraded complex.

8.5. The torus domains and their orientation lines. The remainder of this section proves the integral mixed excision theorem. Its unsigned content is Paper I, Section 10: the separate collared domains $W_{a,R}$ ($a = \pm$, $0 \leq R < \infty$), their completions $W_{a,\infty}$, the joined domains W_λ^J ($\lambda > 0$), the loss formula, the torus-completion bijections at large length, the collar and width families, and the closed-cut decomposition. We use those results as they are proved there. The domain list of Section 2 includes the finite-width torus domains, their completed limits and the closed-cut mixed domain, so Proposition 2.1 and Proposition 2.2 apply to them and to the compact families below. What we add is an orientation for every count and every one-dimensional family, and the resulting chain homotopy over $\mathbb{Z}$. No further analytic assumption is made.

At $R = 0$ the separate domain is the original special quintuple, and its count is the vertex map N_a . For every $\lambda > 0$ the closed cut of Paper I, Lemma 10.18, decomposes the joined domain as

$$W_\lambda^J = \bar{\mathcal{E}}_\lambda \cup_{\mathbb{Z}} M_\lambda.$$

Here $\bar{\mathcal{E}}_\lambda$ is the reverse excision cobordism of Kronheimer and Mrowka, with inputs $(-M)^\#$ and $(M^+)^\#$, and M_λ is a mixed domain of DFL type for the pair (A_0, A_1) , with compression bodies $D_{0,\lambda}, D_{1,\lambda}$ and single ordinary input Z . The lifting group is the fibre product $\Phi_J = \phi_{\bar{\mathcal{E}}} \times_{\phi_Z} \phi_M$ over $\phi_Z = \langle s_Z \rangle$.

Labels and lines. An input label is an ordered pair $a = (a_-, a_+)$ of critical points on $(-M)^\#$ and $(M^+)^\#$, with generator line $I_a = L_G(a_-) \otimes L_G(a_+)$. An output label is a point

$$x = (x_-, x_+) \in A_0 \cap A_1 = (\tilde{P}_0 \cap \tilde{P}_1) \times (R_0 \cap R_1).$$

Let $C_{S,-}$ and $C_{S,+}$ be the target complexes of N_- and N_+ , with their transported orientation systems, and let $L_\pm(x_\pm)$ be their generator lines. Put

$$V_x = L_-(x_-) \otimes L_+(x_+), \quad C_I = C_G((-M)^\#) \otimes_R C_G((M^+)^\#), \quad C_V = C_{S,-} \otimes_R C_{S,+}.$$

In Section 9 these are $CF(\bar{p})$ and $CF(q)$.

Choose the invariant ordinary datum f_Z on Z to vanish on a neighbourhood of the entire critical set. The determinant-one critical classes on Z over x are exactly two, as in [1, Lemmas 2.43 and 2.45]: the two genus matching components permit two central signs, and a constant central gauge on either compression piece changes both, which leaves $\{1, -1\}^2$ modulo its diagonal. An $\mathrm{SO}(3)$ arrow of lifting class $s_Z = \mathrm{PD}[T_a] \pmod{2}$, the projection of $\cos(\pi\rho) + \sin(\pi\rho)n$ on a genus collar, exchanges them, and it acts freely (Paper I, Lemma 10.18(iv), for the lifting classes). Both are irreducible and nondegenerate [1, Lemma 2.47], so the quotient critical set is $A_0 \cap A_1$. Let β_x be either lift. The complex $C_Z = CF_X(A_0, A_1)$ is the target of N_Z ; its generator line over x is

$$Z_x = L_G^M(\beta_x),$$

the gauge line of β_x in the almost-complex structure of the mixed piece. The two lifts have lines identified by the action of s_Z , as in Proposition 5.4.

Finally, let Q_x be the conversion line at β_x from the structure induced by $\bar{\mathcal{E}}_\lambda$ to that of M_λ , and let t be its parity. By Section 4, t does not depend on x . Choose one orientation of Q at a base connection and transport it along paths of connections on Z . Take the structure of M_λ with the product germ near the fibres T_a, T_b on its torus region, as in Section 8, so that (8.1) holds for both structures at Z . By Proposition 4.2, the transport therefore gives coherent orientations q_x of all Q_x , invariant under ϕ_Z . The composite $N_Z \bar{\mathcal{E}}$ always means the composite through the right-placed conversion Θ^r at Z , with its line in the position of Convention 8.1; it has degree t .

Lemma 8.4. *Let $C_{M,x}$ be the primary constant pair on M_λ with limit β_x on its ordinary input and with no secondary or inhomogeneous data. Its physical operator, with stationary columns and scalar row, is invertible for every finite $\lambda > 0$ and along the family of Paper I, Lemma 9.7, which changes the finite torus-neck lengths and the compact metric of M_{λ_0} to those of the special quintuple of (D_0, D_1) . The inverse bounds are uniform on compact subfamilies.*

Proof. This is part of Lemma A.2 in Appendix A: the kernel vanishes by the quadratic identity of [1, Lemma 4.40] adapted to these domains, and the index is zero by [1, Proposition 3.19] at the special quintuple and continuity along the family. The proof of that lemma uses the local theory of Section A.1, Lemma A.1, Paper I, Theorem 10.13, Lemma 8.6(iii) and Lemma 8.7; none of the inputs of that lemma uses the present lemma. $\square$

The determinant line of an invertible operator has the canonical positive element 1, and conjugation by a gauge transformation preserves it.

8.6. Orientation maps on the torus domains. All orientation maps below are defined up to positive scalars and take values in the lines just described.

Separate maps. For a configuration c on $W_{a,R}$, $0 \leq R \leq \infty$, Proposition 5.4 gives

$$\Gamma_c^a : L_G(a_a) \otimes \det D_c \longrightarrow L_a(x_a).$$

For a configuration c on the closed-cut piece M_λ , with ordinary input β , the same proposition, with the positive orientation of Lemma 8.4 at the terminal constant pair, gives

$$\Gamma_c^M : L_G^M(\beta) \otimes \det D_c \longrightarrow Z_x.$$

Strip actions. For $b \in \{-, +, M\}$ and a regular strip u from x to y at the output of the corresponding domain, glue u to the constant pair at x and evaluate Γ^b on the positive constant element. This defines an even map

$$R_b(u) : L_b(x) \otimes \det D_u \longrightarrow L_b(y),$$

with $L_M(x) = Z_x$. Proposition 5.4 gives the coherence

$$(8.8) \quad \Gamma_{c\#u}^b(i \otimes \kappa(d \otimes e)) = R_b(u)(\Gamma_c^b(i \otimes d) \otimes e),$$

where κ is ordered determinant gluing. The differentials of $C_{S,\pm}$ and C_Z count index-one strips with these actions. For a split product strip $u = (u_-, u_+)$ put $n_- = \text{ind } D_{u_-}$ and

$$(8.9) \quad R_V(u)(v_- \otimes v_+ \otimes e_- \otimes e_+) = (-1)^{|v_+|n_-} R_-(u_-)(v_- \otimes e_-) \otimes R_+(u_+)(v_+ \otimes e_+).$$

The sign is the move of $\det D_{u_-}$ past $L_+(x_+)$. A regular index-one product strip for split data has exactly one nonconstant factor, of index one, since each nonconstant factor has a translation vector in its kernel and a constant factor at a transverse intersection is invertible. Thus (8.9) gives the right tensor differential (3.3) on C_V .

The decorated closed-cut map. Call a configuration on W_λ^J *terminal* if it agrees with a constant pair $C_{M,x}^c$, for some representative c of the critical point over x , on the part of M_λ beyond a parallel copy of Z in the product collar of the cut, and on a two-sided collar of that copy. Its remaining factor e , the reverse excision cobordism enlarged by the collar, is an arbitrary configuration with the given inputs. Ordinary gluing at the copy of Z identifies its determinant line with $\det D_e \otimes \det D_{C_{M,x}^c}$, and the lines at c are identified with those at β_x as in Section 5. Let $\gamma_e : I_a \otimes \det D_e \rightarrow L_G^{\bar{\mathcal{E}}}(\beta_x)$ be the almost-complex orientation of the gauge cobordism, with the ordered inputs, and let $\Theta_x : L_G^{\bar{\mathcal{E}}}(\beta_x) \otimes Q_x \rightarrow Z_x$ be the conversion. Define

$$(8.10) \quad G_c(i \otimes q \otimes d_e \otimes 1) = (-1)^{t \text{ind } D_e} \Theta_x(\gamma_e(i \otimes d_e) \otimes q).$$

The sign moves Q_x from its fixed position after the input line to the position adjacent to the cut. For a general configuration c on W_λ^J , the localized several-input shift of Section 5 and Proposition 2.2 retains the ordinary cobordism and reaches a terminal configuration. Transport (8.10) back along it. This defines an even map

$$(8.11) \quad G_c : I_a \otimes Q_x \otimes \det D_c \longrightarrow Z_x.$$

Lemma 8.5. *The map (8.11) is independent of the shifting path, natural under Φ_J , and continuous on compact families of finite widths. It has the following compatibilities.*

(i) *If A is an ordinary path at the input, then*

$$G_{A\#c}(i \otimes q \otimes \kappa(d_A \otimes d)) = (-1)^{t \text{ind } D_A} G_c(\gamma_A(i \otimes d_A) \otimes q \otimes d).$$

(ii) If u is a regular strip at the output, from x to y , then, with the coherent orientations q_x, q_y ,

$$G_{c\#u}(i \otimes q_y \otimes \kappa(d \otimes e)) = R_M(u)(G_c(i \otimes q_x \otimes d) \otimes e).$$

(iii) If $c = A\#_Z c'$ is an ordinary gluing at the closed cut of a configuration A on $\bar{\mathcal{E}}_\lambda$ from a to β and a configuration c' on M_λ from β to x , then, with the coherent orientations $q = q_x$ and q_β ,

$$G_c(i \otimes q \otimes \kappa(d_A \otimes d')) = (-1)^{t \operatorname{ind} D_A} \Gamma_{c'}^M(\Theta_\beta(\gamma_A(i \otimes d_A) \otimes q_\beta) \otimes d').$$

Proof. Path independence is the relative argument in the proof of Proposition 5.4. Apply the relative localized shift to a path whose endpoints are terminal. The endpoint tracks and the final edge remain terminal, where (8.10) is continuous. The orientation double cover over the parameter square then shows that the original path preserves (8.10). Applying this to the concatenation of two shifting paths from c proves independence, also when they reach different terminal components. The lifting arrows preserve the almost-complex orientations, the conversion character on ϕ_Z and the positive constant element, and they map terminal configurations to terminal ones. The pullback of a shifting path by an arrow is a shifting path, so path independence shows that they preserve (8.11).

For (i), the path A is attached at an original input, away from the localized shift. At a terminal configuration, ordinary associativity gives $\gamma_{A\#e}(i \otimes \kappa(d_A \otimes d_e)) = \gamma_e(\gamma_A(i \otimes d_A) \otimes d_e)$, and the two sides of (i) carry the exponents $t(\operatorname{ind} D_A + \operatorname{ind} D_e)$ and $t \operatorname{ind} D_A + t \operatorname{ind} D_e$. Continuity along the shift gives (i).

For (ii), glue u to the whole shifting family of c with one long neck. At the terminal end the configuration is $e\#C_{M,x}^c\#u$; shift $C_{M,x}^c\#u$ to $B\#C_{M,y}^{c'}$ with B an ordinary path on Z of index $n = \operatorname{ind} D_u$, and insert a long product neck at Z so that B lies on the $\bar{\mathcal{E}}_\lambda$ side beyond a two-sided collar of Z ; then $e\#B\#C_{M,y}^{c'}$ is terminal. The ordinary action of B in the two structures satisfies the line form of the right conversion convention (4.2):

$$\gamma_B^M(\Theta_x(v \otimes q_x) \otimes d_B) = (-1)^{tn} \Theta_y(\gamma_B^{\bar{\mathcal{E}}}(v \otimes d_B) \otimes q_y).$$

The right side of (ii) at the terminal configuration therefore carries the exponent $t \operatorname{ind} D_e + tn$. The left side places Q before the determinant of $e\#B$, with exponent $t(\operatorname{ind} D_e + n)$. The other factors agree by associativity and the definition of $R_M(u)$. Continuity gives (ii).

For (iii), shift c' on M_λ to $B\#C_{M,x}^c$ and glue A to the whole family at Z with one long neck. Both sides of (iii) are continuous orientations of the same line along this family. At its end the configuration $A\#B\#C_{M,x}^c$ is terminal, and by (8.10) its left side carries $(-1)^{t(\operatorname{ind} D_A + \operatorname{ind} D_B)}$. On the right, $\Gamma_{B\#C_{M,x}^c}^M$ is the action of B in the structure of M_λ , which by the display above contributes $(-1)^{t \operatorname{ind} D_B}$; the prefactor contributes $(-1)^{t \operatorname{ind} D_A}$. The two sides agree at the end, hence along the family. $\square$

The reference comparison. Let $C_{J,x}^{(\lambda)}$ be the joined constant pair on W_λ^J with output x . Its incoming labels are the critical points $a_{x\pm}$, so its input line is $L_G(a_{x-}) \otimes L_G(a_{x+}) = V_x$. Its operator is invertible for every λ (Lemma 8.6(iv) below). Evaluating (8.11) on the positive element gives an even isomorphism $G_{\operatorname{ref}} : V_x \otimes Q_x \rightarrow Z_x$. Define the even map

$$(8.12) \quad K_{0,x} : Z_x \longrightarrow Q_x \otimes V_x, \quad K_{0,x} G_{\operatorname{ref}}(v \otimes q) = (-1)^{t|v|} q \otimes v.$$

The lines are fixed and G_{ref} varies continuously with λ , so $K_{0,x}$ does not depend on λ . Replacing $C_{J,x}^{(\lambda)}$ by the joined constant pair of another representative conjugates the operator by a gauge transformation. It is assembled from the automorphisms relating the representatives

on the two sides; these are central on T , and after multiplying one of them by -1 they agree there and glue across the bridge. The positive elements correspond, and so do the identified lines, by the remark on representatives in Appendix B; hence $K_{0,x}$ is unchanged. Let K_x^q be *literal* contraction of the first factor with q_x : if $K_{0,x}(z) = q_x \otimes v$, then $K_x^q(z) = v$. It has degree t . Put

$$(8.13) \quad \mathcal{K} = K^q, \quad \mathcal{K}_R = K^q \varepsilon_{C_Z}^t.$$

Literal contraction is intended; the right tensor of the odd map q_x^* with the identity would insert an additional Koszul sign.

8.7. The split orientation and completion. On the completed pair $W_{-, \infty} \sqcup W_{+, \infty}$ the product of the separate maps is

$$(8.14) \quad S_{(c_-, c_+)}(i_- \otimes i_+ \otimes d_- \otimes d_+) = (-1)^{|i_+| \text{ind } D_{c_-}} \Gamma_{c_-}^-(i_- \otimes d_-) \otimes \Gamma_{c_+}^+(i_+ \otimes d_+).$$

We extend it to large finite widths through the torus cutoff of Paper I. On the far torus tail of a completed side write a configuration as $\alpha + b$ in the torus-fibre slice of Paper I, Lemma 10.9. A compact family is *in torus normal form beyond r_0* if every member has this form for $r \geq r_0$, with b bounded in every weighted norm and with the weighted norms of b on $r \geq r_1$ tending to zero uniformly as $r_1 \rightarrow \infty$. For a pair $c = (c_-, c_+)$ in torus normal form beyond r_0 and $\lambda \geq 2r_0 + 2$, the preglue $P_\lambda(c)$ replaces b by $\chi_\lambda b$ on both tails, with $\chi_\lambda(r) = \chi(r - \lambda/2)$, and joins them by α on the bridge, as in Paper I, Lemma 10.24. On one side the truncation $P_R(c_a)$ on $W_{a,R}$ is defined in the same way; it equals α near the face and so satisfies the flat-fibre condition.

- Lemma 8.6.** *(i) Every compact family on the completed domains, in the topology of Section 2, is in torus normal form beyond some r_0 after a continuous family of small even gauge transformations which are the identity for $r \leq r_0 - 1$ and on the ordinary ends.*
- (ii) Let $\mathfrak{F}$ be a compact family of pairs in torus normal form beyond r_0 . The residual of $P_\lambda(c)$ relative to c is supported where $\chi'_\lambda \neq 0$ or $\chi_\lambda^2 \neq \chi_\lambda$, and tends to zero in every weighted norm, uniformly on $\mathfrak{F}$, as $\lambda \rightarrow \infty$. The same holds for P_R on one side.*
- (iii) The constant pairs $C_{a,x_a} = C^{c_a}$ whose representatives c_a are α on T , such as those produced in Appendix B, are equal to α on the whole added tail. Then $P_R(C_{a,x_a})$ is the constant pair on $W_{a,R}$, and $P_\lambda(C_{-,x_-}, C_{+,x_+}) = C_{J,x}^{(\lambda)}$ is the joined constant pair.*
- (iv) The operators at these constant pairs are invertible for every R and every λ .*

Proof. (i) Continuity and compactness make the set of tail differences $A - \alpha$ compact in H_δ^m . Their norms on $r \geq r_1$ therefore tend to zero uniformly as $r_1 \rightarrow \infty$. Sobolev embedding on unit boxes, whose geometry is fixed on the flat tail, makes them uniformly small in C^1 beyond some r_1 . Paper I, Lemma 10.9, then gives a unique small even gauge transformation into the torus-fibre slice on $r \geq r_1$, continuous in the family. Cut off its logarithm on $[r_1, r_1 + 1]$, as in Paper I, Lemma 10.11, and take $r_0 = r_1 + 1$. The resulting gauges are connected to the identity by the path of logarithms, so determinant transport along them is canonical, and every orientation map below is natural under them.

(ii) Only the tail is changed. There the residual is $(d\chi_\lambda \wedge b)^+ + (\chi_\lambda^2 - \chi_\lambda)(b \wedge b)^+$, which is bounded by the norms of b on $r \geq \lambda/2 - 1$.

(iii) Choose the stationary representatives equal to α on the whole torus region of the slice, not only on the far collar. For a stationary representative $B = B_T + B_r dr$, flat on the torus band $[b, R]$ and equal to α near $r = R$ (Paper I, Proposition 10.7), solve $\partial_r g = -B_r g$,

$g(R) = 1$, on a slightly larger interval $[c, R]$ in the determinant-one gauge bundle. Then B^g has no radial component there, and flatness gives $B^g = \alpha$ on $T \times [b, R]$. With a cutoff η equal to zero near c and one near $[b, \infty)$, the gauges $q_t(r) = g(R + t\eta(r)(r - R))$, $t \in [0, 1]$, form a path from the identity, supported in a fixed collar, with $B^{q_1} = \alpha$ on $T \times [b, \infty)$. They preserve primary classes and genus germs, and depend smoothly on compact families of primary data. The constant pairs are pullbacks of these representatives, so they equal α on the tail and the truncation and the preglue have no cutoff error; the results are the constant pairs on $W_{a,R}$ and W_λ^J , which are solutions.

(iv) This is Lemma A.2. $\square$

The next lemma is the linear gluing across the torus bridge. The bridge is not an ordinary end, since its cross-section $T \times \mathbb{R}_\theta$ runs into the mixed ends, so Lemma 3.3 does not apply to it.

Lemma 8.7 (Determinant lines across torus completion). *Let $\mathfrak{F}$ be a compact family of pairs in torus normal form beyond r_0 . For $c \in \mathfrak{F}$ let $D_\lambda(c)$ be the physical realization at $P_\lambda(c)$ and $D_\infty(c) = D_{c_-} \oplus D_{c_+}$.*

- (i) *There are a finite-dimensional space E , a linear map $\Lambda : E \rightarrow Y$ into smooth target sections supported in a fixed compact subset of the two cores, and λ_1 , such that $D_\lambda(c) \oplus \Lambda$ is onto, with a uniformly bounded right inverse, for all $c \in \mathfrak{F}$ and $\lambda \in [\lambda_1, \infty]$.*
- (ii) *The kernels of $D_\lambda(c) \oplus \Lambda$ form a continuous vector bundle over $\mathfrak{F} \times [\lambda_1, \infty]$. Near $\lambda = \infty$ its local frames are $\iota_\lambda(v) = \beta_\lambda v - Q_\lambda(D_\lambda \oplus \Lambda)\beta_\lambda v$, where β_λ is multiplication by χ_λ on both tails and Q_λ is the right inverse of (i).*
- (iii) *Consequently the lines $\det D_\lambda(c)$ form a continuous line bundle over $\mathfrak{F} \times [\lambda_1, \infty]$ whose fibre at $\lambda = \infty$ is $\det D_{c_-} \otimes \det D_{c_+}$. This structure does not depend on Λ . Where the operators are invertible, the canonical positive elements form a continuous section.*
- (iv) *Let an ordinary path or strip be glued, with a long neck, at an original gauge input or at the strip output, to every member of $\mathfrak{F}$. The identification of (iii) commutes with the ordinary gluing isomorphism of Lemma 3.3.*

The same statements hold for a compact family in torus normal form on one completed side and its truncations $P_R(c)$ on $W_{a,R}$, $R \rightarrow \infty$. They also hold for any other family of operators of the same form: the chart operators of Paper I, Definition 10.23, and the interpolating operators in the proof of Lemma 8.12, since the proof uses only the properties listed at its beginning.

Proof. Compact-remainder estimates. Steps 1–3 of the proof of Paper I, Proposition 10.26, use only the following: on the torus region beyond a fixed S the operator is the flat odd-torus operator D_α of Paper I, Proposition 10.3, plus a small zeroth-order term, with the normal derivatives at a flat face recovered as in Paper I, Theorem 10.5(b); on the mixed ends it is transported to the stationary models by the maps of [2, Proposition 5.27], with errors tending to zero, and the slice estimates of Paper I, Proposition 10.7 and Corollary 10.4, hold; the ordinary asymptotic operators are invertible; local elliptic estimates hold on a fixed compact region; the coefficients are bounded in C^m on unit patches of finitely many isometry types; and the weight satisfies $\delta < \gamma/2$, $\delta < \mu_T$, with $\tau = \tau_0(\theta)$ on the torus region. The smallness beyond S follows from the uniform tails in the definition of torus normal form and Sobolev embedding; by the local estimates and end models of Section A.1, and compactness of $\mathfrak{F}$, these hold uniformly on $\mathfrak{F} \times [2r_0 + 2, \infty]$. The finite stationary columns and the scalar row of the physical realization are bounded, and they act through the fixed cores and the mixed

ends. The same argument therefore gives

$$\|x\| \leq C(\|D_\lambda(c)x\| + \|x\|_{L^2(K_1)} + |p(x)|),$$

where K_1 is a fixed compact subset of the two cores and $p(x)$ the stationary coordinates. The adjoint of D_α is invertible with the same bound, the stationary models are self-adjoint, and the local estimates for the weak adjoint are recorded in Section A.1. Hence the same steps give

$$\|\eta\| \leq C(\|D_\lambda(c)^*\eta\| + \|\eta\|_{L^2(K_1)})$$

on the dual weighted spaces.

(i) At $\lambda = \infty$ each $D_\infty(c)$ is Fredholm by Proposition 2.1(a). Smooth compactly supported target sections are dense, so finitely many of them span a complement of its range, and by continuity also near c . A finite cover of $\mathfrak{F}$ gives Λ . Suppose there were $\lambda_j \rightarrow \infty$, $c_j \in \mathfrak{F}$ and η_j with $\|\eta_j\| = 1$ and $\|(D_{\lambda_j}(c_j) \oplus \Lambda)^*\eta_j\| \rightarrow 0$. By the adjoint estimate, $\|\eta_j\|_{L^2(K_1)} \geq 1/C$. Pass to $c_j \rightarrow c$. As in Step 5 of that proof, the parts of $W_{\lambda_j}^J$ within distance $\lambda_j/2$ of each core are identified with the corresponding parts of $W_{\pm, \infty}$, and local elliptic bounds give a subsequence converging weakly on compact sets and strongly in $L^2(K_1)$ to some η_∞ on the completed pair. Then $(D_\infty(c) \oplus \Lambda)^*\eta_\infty = 0$ and $\eta_\infty \neq 0$, which contradicts the choice of Λ . Hence $\|\eta\| \leq C\|(D_\lambda(c) \oplus \Lambda)^*\eta\|$ for $\lambda \geq \lambda_1$, and $Q_\lambda = T^*(TT^*)^{-1}$, with $T = D_\lambda(c) \oplus \Lambda$ and T^* its Hilbert adjoint for the weighted inner products, is a uniformly bounded right inverse; the estimate for the formal adjoint on the dual weighted space is equivalent to one for T^* through the Riesz isometry, and it is proved first at the H^1 level and then upgraded by the elliptic estimates of Step 3. The map Q_λ is continuous in (c, λ) for finite λ .

(ii) At $\lambda = \infty$ the kernels form a vector bundle over $\mathfrak{F}$, since the operators are onto. Its unit sphere bundle is compact, so the tails of its elements on $r \geq r_1$ are uniformly small as $r_1 \rightarrow \infty$. Hence $\|(D_\lambda \oplus \Lambda)\beta_\lambda v\| \rightarrow 0$ uniformly for unit v : the error consists of $\sigma(d\chi_\lambda)v$ and of the coefficient difference on the tail, both supported where $r \geq \lambda/2 - 1$. Thus ι_λ is injective for large λ and converges to the identity. It is onto: otherwise unit kernel elements orthogonal to its image would, by the estimate for D_λ and the argument in (i), converge to a nonzero kernel element at $\lambda = \infty$ orthogonal to the whole limiting kernel. For finite λ the kernels of the surjective family form a bundle as usual.

(iii) The exact sequence $0 \rightarrow \ker D \rightarrow \ker(D \oplus \Lambda) \rightarrow E \rightarrow \operatorname{coker} D \rightarrow 0$ identifies $\det D$ with $\det \ker(D \oplus \Lambda) \otimes (\det E)^*$, and (ii) makes these lines continuous. At $\lambda = \infty$ the ordered direct-sum isomorphism gives $\det D_{c_-} \otimes \det D_{c_+}$. Two choices Λ, Λ' are compared through $\Lambda \oplus \Lambda'$ by the standard stabilization maps, which are continuous in (c, λ) because they are built from the same kernel bundles. On an invertible operator the stabilization identification carries the canonical element to the canonical element, so the positive elements are continuous where the operators are invertible.

(iv) Choose Λ supported away from the ordinary gluing neck as well as from the torus bridge; this is possible because the sections spanning the cokernels in (i) have compact support in the cores, which lie at a fixed distance from both necks once they are long. Then the glued operators with both necks long form a family over the product of the two neck-length parameters, including their limits. The kernel transplants of (ii) and of Lemma 3.3 act by cutoffs and projections in disjoint regions, and the estimates of (i) hold uniformly on this product. The resulting determinant bundle is continuous over the product of the two closed length intervals, which is contractible. The two orders of gluing therefore give the same identification.

For one side use the half-plane and strip models of Paper I, Proposition 10.3, with the flat-fibre face, in place of the bridge. $\square$

Let $\mathfrak{F}$ be a compact family of pairs of configurations on the completed domains. By Lemma 8.6(i) we may assume it is in torus normal form. Examples are a pair of index-zero solutions and a pair of constant pairs with strips glued at their outputs. For $c \in \mathfrak{F}$ define

$$(8.15) \quad S_{P_\lambda(c)} : I_a \otimes \det D_{P_\lambda(c)} \longrightarrow V_x, \quad \lambda \geq \lambda_1,$$

as the continuous extension of (8.14) along $\{P_\lambda(c)\}_{\lambda \in [\lambda_1, \infty]}$ given by Lemma 8.7. It is continuous on $\mathfrak{F} \times [\lambda_1, \infty]$ and natural under the joined lifting group $\Phi_J \cong \Phi_- \oplus \Phi_+$.

The separate shifts of Proposition 2.2 give, for $c \in \mathfrak{F}$, a homotopy $h \mapsto H_h(c)$, $h \in [0, 1]$, continuous in the topology of Section 2, from c to $(A_- \# C_{-,x_-}^{c_-}, A_+ \# C_{+,x_+}^{c_+})$ with $A_\pm$ ordinary paths and representatives $c_\pm$ which are α on T . Its image is compact. By Lemma 8.6(i) it can be put in torus normal form beyond a common r_0 , and by Lemma 8.6(iii) its end-points equal α on the tails. Applying (8.15) to this larger compact family gives S on the rectangle $\{P_\lambda(H_h(c))\}$. At $\lambda = \infty$ it is (8.14), which along h is the transport defining $\Gamma^\pm$ in Proposition 5.4. At $h = 1$ the configuration is $(A_- \sqcup A_+) \# C_{J,x}^{(\lambda)}$, and there S is the ordinary orientation of $A_- \sqcup A_+$ followed by the positive element of the invertible constant factor, by Lemma 8.7(iii),(iv). The rectangle is contractible, so S at $P_\lambda(c)$ does not depend on the chosen shifting homotopy.

The same device compares the separate orientations on the finite and completed domains. By construction, Γ^a on the finite domain $W_{a,R}$ is a continuous orientation over its configuration space, defined by the one-input shift on that domain.

Lemma 8.8. *Let $\mathfrak{F}$ be a compact family on $W_{a,\infty}$ in torus normal form. For $R \geq R_1$, the orientation Γ^a of $W_{a,R}$ at $P_R(c)$ is the continuous extension of Γ_c^a along $\{P_R(c)\}_{R \in [R_1, \infty]}$ given by Lemma 8.7.*

Proof. Consider the rectangle $\{P_R(H_h(c))\}$, with H_h the separate shift in torus normal form. The extension from $R = \infty$ is continuous on it by Lemma 8.7. For fixed finite R the orientation Γ^a of $W_{a,R}$ is continuous in h . At $h = 1$ the configuration is $A \# P_R(C_{a,x})$, and $P_R(C_{a,x})$ is the constant pair of $W_{a,R}$ by Lemma 8.6(iii). There Γ^a is the ordinary orientation of A followed by the positive element of the constant factor. The extension from $R = \infty$ is the same: by Lemma 8.7(iv) it commutes with gluing A , and the constant factor is invertible for every R (Lemma A.2) with positive element continuous up to $R = \infty$. The two orientations therefore agree at $h = 1$, and hence on the whole rectangle. $\square$

Corollary 8.9. *The strip actions $R_\pm(u)$ computed on $W_{\pm,R}$ do not depend on $R \in [0, \infty]$, and $R_M(u)$ computed on M_λ does not depend on λ and agrees with the action on the special quintuple of (D_0, D_1) defining N_Z . Thus each of $C_{S,\pm}$ and C_Z carries one orientation system.*

Proof. Each action is an isomorphism between fixed lines, defined through an invertible constant pair with a strip glued at its output. For finite R it varies continuously with R , since the constant pairs are invertible for every R (Lemma A.2); at $R = \infty$ use Lemma 8.8 with $\mathfrak{F} = \{C_{a,x} \# u\}$, which is α on the tail, together with Lemma 8.7(iv) for the strip glued at the output. For R_M use the width interval and the family of Paper I, Lemma 9.7, along which $C_{M,x}$ stays invertible by Lemma 8.4. A continuous family of isomorphisms between fixed lines is constant. $\square$

Proposition 8.10. *On the family $P_\lambda(H_h(c))$, $c \in \mathfrak{F}$, for $\lambda \geq \lambda_1$ and $h \in [0, 1]$,*

$$(8.16) \quad K_{0,x} G_c = (1_{Q_x} \otimes S_c)(\sigma_{I,Q} \otimes 1_{\det}),$$

where $\sigma_{I,Q}$ moves Q_x past the entire input line. For the literal insertion $G_c^r(i \otimes d) = G_c(i \otimes q_x \otimes d)$ and $m = \text{ind } D_c \pmod{2}$,

$$(8.17) \quad \mathcal{K} G_c^r = (-1)^{t|I|} S_c, \quad \mathcal{K}_R G_c^r = (-1)^{t(m+1)} S_c.$$

Proof. Both sides of (8.16) are continuous on the rectangle at finite λ , so it suffices to verify it at $h = 1$. There the configuration is $(A_- \sqcup A_+) \# C_{J,x}^{(\lambda)}$. Let h_A be the total parity of $\det D_{A_-} \otimes \det D_{A_+}$ and let $v = \gamma_A(i \otimes d_A) \in V_x$, so $|v| = |i| + h_A$. By Lemma 8.5(i), $G_c(i \otimes q \otimes \kappa(d_A \otimes 1)) = (-1)^{th_A} G_{\text{ref}}(v \otimes q)$, and by (8.12)

$$K_{0,x} G_c(i \otimes q \otimes \kappa(d_A \otimes 1)) = (-1)^{th_A + t|v|} q \otimes v = (-1)^{t|i|} q \otimes S_c(i \otimes d).$$

This is the right side of (8.16). For the second assertion, the tensor of maps $1_Q \otimes S_c$ with even S_c introduces no sign, so literal contraction gives $\mathcal{K} G_c^r = (-1)^{t|I|} S_c$. Since G_c is even, $|G_c^r(i \otimes d)| = |i| + t + m$, and ε^t contributes $(-1)^{t(|i|+t+m)}$. The product is $(-1)^{t(t+m)} = (-1)^{t(m+1)}$. $\square$

At index zero the coefficient in (8.17) is $(-1)^t$; at the isolated points of a one-parameter family, where m is odd, it is $+1$. The index-zero sign must not be applied to homotopies.

Proposition 8.11. $\mathcal{K}_R : C_Z \rightarrow C_V$ is an isomorphism of graded groups, and

$$d_V \mathcal{K}_R = (-1)^t \mathcal{K}_R d_Z.$$

Thus $\mathcal{K}_R$ is a closed isomorphism of degree t .

Proof. $K_{0,x}$ is an isomorphism of lines, and the generators of C_Z and C_V are both indexed by $A_0 \cap A_1$; hence $\mathcal{K}_R$ is an isomorphism. Let $u = (u_-, u_+)$ be a regular index-one product strip from x to y for the split data. Apply Proposition 8.10 to the pair of separate configurations $(C_{-,x_-} \# u_-, C_{+,x_+} \# u_+)$, which is in torus normal form, and to its preglue $C_{J,x}^{(\lambda)} \# u$. By (8.8) on each side and (8.14), the split orientation at $\lambda = \infty$ is

$$S_{C_J \# u}(v \otimes \kappa(1 \otimes e)) = R_V(u)(v \otimes e).$$

Indeed, $\Gamma_{C_\pm}^\pm(v_\pm \otimes 1) = v_\pm$, the determinant of $C_\pm \# u_\pm$ is $\kappa(1 \otimes e_\pm)$ with $\text{ind } D_{C_- \# u_-} = n_-$, and (8.8) gives $\Gamma^\pm(v_\pm \otimes \kappa(1 \otimes e_\pm)) = R_\pm(u_\pm)(v_\pm \otimes e_\pm)$. Formula (8.14) with $i_\pm = v_\pm$ then contributes $(-1)^{|v_+|+|n_-|}$, which is the sign in (8.9); reordering the even constant elements costs nothing. By continuity in λ , and Lemma 8.7(iv) for the strip glued at the output, the identity holds at large finite widths. By Lemma 8.5(ii) with $c = C_{J,x}^{(\lambda)}$,

$$G_{C_J \# u}(v \otimes q_y \otimes \kappa(1 \otimes e)) = R_M(u)(G_{\text{ref}}(v \otimes q_x) \otimes e).$$

Apply $K_{0,y}$ and use (8.16) for $C_J \# u$ and (8.12). Both sides of the resulting identity, evaluated on $G_{\text{ref}}(v \otimes q_x) \otimes e$, equal $(-1)^{t|v|} q_y \otimes R_V(u)(v \otimes e)$. Let $J_u : Q_x \rightarrow Q_y$ be the conversion transport along the ordinary path obtained by shifting $C_{M,x} \# u$; by coherence $J_u q_x = q_y$. Hence

$$K_{0,y} R_M(u) = (J_u \otimes R_V(u))(K_{0,x} \otimes 1_{\det D_u}),$$

with Q to the left of both V and the strip determinant; since $R_V(u)$ is even, the tensor of maps inserts no sign. Literal contraction with q_x, q_y gives $K_y^q R_M(u)(z \otimes e) = R_V(u)(K_x^q z \otimes e)$. Summing over index-one strips gives $d_V \mathcal{K} = \mathcal{K} d_Z$. Since $d_Z \varepsilon^t = (-1)^t \varepsilon^t d_Z$, the assertion for $\mathcal{K}_R$ follows. $\square$

8.8. Signed counts at large torus length. Fix regular data as in Paper I, Section 10.11, steps (1)–(4). The completed index-zero sets are finite and regular, there are no completed solutions of negative index, and for $R \geq R_0$, $\lambda \geq \lambda_1$ Paper I, Theorem 10.31, gives the bijections $\mathcal{G}_R$ and $\mathcal{G}_\lambda$ onto regular index-zero solutions, with no solutions of negative index. Write $N_{a,\infty}$ for the completed separate counts, oriented by Γ^a , and $N_a^{(R)}$ for the counts on $W_{a,R}$. On W_λ^J let $F_\lambda^S : C_I \rightarrow C_V$ count index-zero solutions oriented by S , and $F_\lambda^Z : C_I \rightarrow C_Z$ the same solutions oriented by G^r ; their degrees are 0 and t . The orientation S at these solutions is defined by the following lemma.

Lemma 8.12. *Let V be a completed index-zero solution, or a pair of them, and for $L \geq L_1$ let $z_L(V)$ be the small zero of Paper I, Proposition 10.29, so that $\mathcal{G}_L(V) = \Psi_L(z_L(V))$. Consider the rectangle of configurations*

$$\mathcal{R}_V = \{P_L(V) : L \in [L_1, \infty]\} \cup \{\Psi_L(sz_L(V)) : s \in [0, 1], L \in [L_1, \infty)\},$$

with $P_\infty(V) = V$. The physical determinant lines form a continuous line bundle over $\mathcal{R}_V$, and determinant transport carries the canonical positive element at V to the canonical positive element at $\mathcal{G}_L(V)$.

Proof. Over the first edge the line bundle is that of Lemma 8.7 with $\mathfrak{F} = \{V\}$, and its extension clause covers the chart operators used below; the second edge is a continuous family on a fixed domain for each L .

Use also the chart operators T of Paper I, Definition 10.23. At the preglues they are invertible with a uniform bound by Paper I, Proposition 10.26; at $L = \infty$ the chart is centered at V and the operator is invertible by Step 4 of its proof and index zero. Along the contraction path the chart operator is $D_L + DN_L(sz_L)$ with $\|DN_L(sz_L)\| \leq 2C_{\text{quad}}r \leq (2C_{\text{inv}})^{-1}$, as in the proof of Paper I, Proposition 10.29, so it is invertible. The chart operator at $\Psi_L(sz)$ is centered at the preglue. Moving its slice center, trace chart and target projection to the configuration itself changes it by at most $C(\|sz\| + \|D\Psi_L(sz) - 1\|)$ in operator norm, which is small, so the straight path between them consists of invertible operators. The resulting operator and the physical realization at the same configuration have the same decaying block; their stationary columns and scalar rows differ by a bounded finite-rank map (Paper I, remark after Definition 10.23). The straight path between these two is therefore a continuous Fredholm path, and these identifications are continuous over $\mathcal{R}_V$. The positive elements of the invertible chart operators thus give a continuous nonvanishing section of the physical determinant bundle.

At the two solutions V and $\mathcal{G}_L(V)$ this section is the canonical positive element. Indeed, there the change of stationary columns is unipotent: the chart lifts and the Coulomb lifts differ by infinitesimal gauge directions $d\psi_e$ and by decaying fields already in the domain (Paper I, Step 4 of the proof of Proposition 10.26), and a unipotent change of source coordinates preserves the positive element. The change of the scalar row is block triangular with invertible gauge block, as in the proof of Lemma 5.2, using Paper I, Lemma 10.25. At these points every operator of the interpolating paths is invertible. No kernel at a nonsolution of $\mathcal{R}_V$ is identified with a tangent space. $\square$

Extend (8.15) from $P_\lambda(V)$ to $\mathcal{G}_\lambda(V)$ along the second edge of $\mathcal{R}_V$. Both sides of (8.16) are continuous along it, so the comparison persists at $\mathcal{G}_\lambda(V)$.

Proposition 8.13. *For R and λ sufficiently large,*

$$N_a^{(R)} = N_{a,\infty} \ (a = \pm), \quad F_\lambda^S = N_{-, \infty} \otimes_R N_{+, \infty}, \quad \mathcal{K}_R F_\lambda^Z = (-1)^t N_{-, \infty} \otimes_R N_{+, \infty}.$$

Proof. Let V be a completed index-zero solution on a separate side. By Lemma 8.8 with $\mathfrak{F} = \{V\}$, the orientation Γ^a of $W_{a,R}$ at $P_R(V)$ is the continuous extension of Γ_V^a ; along the second edge of $\mathcal{R}_V$ it is continuous on the fixed domain. By Lemma 8.12 the canonical positive element is also transported continuously from V to $\mathcal{G}_R(V)$. Hence $\Gamma^a(i \otimes 1)$ has the same sign at $\mathcal{G}_R(V)$ as at V . The bijection $\mathcal{G}_R$ preserves labels and lifting orbits, so the counts agree entry by entry.

In the joined case S at $\mathcal{G}_\lambda(V)$ is, by definition, the continuous extension of (8.14) over $\mathcal{R}_V$, and the same lemma transports the positive element. At index zero $\text{ind } D_{V_-} = 0$, so (8.14) evaluated on the positive element is the product of the two separate signs. The joined lifting group is $\Phi_- \oplus \Phi_+$ and the bridge has no gluing parameter (Paper I, Theorem 10.31), so the orbit counts multiply. The generator factors in (8.14) are those of the right tensor of maps (3.4) of the even maps $N_{\pm, \infty}$. This proves the second identity. The third follows from the second and (8.17) with $m = 0$, applied solution by solution, since (8.16) holds at $\mathcal{G}_\lambda(V)$. $\square$

8.9. Boundary orientation of the comparison families. The comparison families below are those of Paper I: Lemma 10.15 with Propositions 10.16 and 10.17 for collar length and width, Lemma 10.20 for the change of data at fixed width, and Theorem 5.4 for compactly supported changes of secondary data. In each case the data are constant near $b = 0, 1$, the endpoints are regular in index at most one, and a relative invariant regularization gives: finitely many isolated universal solutions of vertical index -1 , each with invertible augmented operator $(v, \tau) \mapsto Dv + \tau C$, and a compact one-manifold of universal index-zero solutions whose boundary consists of the two endpoint fibres and one branch for each pair of an index -1 solution with an index-one trajectory at an ordinary end. Here C is the parameter column; on a varying mixed end it is the weighted column of Paper I, Lemma 10.15, not a compactly supported one.

Let Ω denote the orientation map in use: Γ^a on a separate domain, Γ^M on the closed-cut piece, or G on a joined domain. Put $f = 0$ in the first two cases and $f = t$ in the third. The endpoint counts F_0, F_1 are oriented by Ω on the positive element. At an isolated universal solution c , orient the parameter first: choose $h \in \det D_c$ such that $db \otimes h$ is the positive element of the invertible augmented operator, and count c by $\Omega(i \otimes h)$, or $\Omega(i \otimes q \otimes h)$ when $f = t$. Let H be the resulting count. Its degree is $f + 1$.

Lemma 8.14. *With these conventions,*

$$(8.18) \quad F_1 - F_0 = d_{\text{out}} H + (-1)^f H d_{\text{in}} = \delta H.$$

For a fixed regular datum, the index-one boundary gives $d_{\text{out}} F = (-1)^f F d_{\text{in}}$, so F is closed of degree f .

Proof. Orient the universal index-zero space by $\det(T_b[0, 1]) \otimes \det D$ and Ω , as in Section 5; the boundary is oriented by the outward normal first. At $b = 1$ the outward normal is ∂_b , and at $b = 0$ it is $-\partial_b$; there the vertical operators are invertible, so these contribute $F_1 - F_0$.

At an outgoing strip break (c, u) the glued solutions are $U_T = \Psi(p_T + z_T)$, where the preglue p_T has the strip translated to $u(s - T, \cdot)$ and z_T is the small correction of Paper I, Theorem 5.1. Differentiate the defining identity of z_T in T , using local identifications of the domains which fix the mixed core and translate the strip core. The T -derivative of p_T is the cut-off translation vector $-\partial_s u$ plus an error which is exponentially small in T . By the uniform right inverse, the T -derivative of z_T is bounded by the T -derivative of the residual of p_T and by $\|z_T\|$ times a bounded quantity, and both tend to zero. For pseudoholomorphic curves this T -derivative estimate, with exponential decay, is [6, Theorem 6.4]; the same

argument applies on the fixed ordinary gauge end. Hence $\partial_T U_T = -\partial_s u + o(1)$ on the strip core after projection to the slice. Linear gluing (Lemma 3.3) identifies $\det A_c \otimes \det D_u$ with the determinant along the branch, and the positive translation vector $\partial_s u$ orients $\det D_u$. The outward direction at $T = \infty$ is ∂_T , which is negative in this order. With the parameter already first in $\det A_c = \det(T_b[0, 1]) \otimes \det D_c$ and (8.8) or Lemma 8.5(ii), this branch contributes $-d_{\text{out}} H$.

At an incoming break (A, c) , with A an index-one trajectory, the trajectory is translated to $A(s + T, \cdot)$, and the same argument gives $\partial_T U_T = +\partial_s A + o(1)$, positive in $\det D_A \otimes \det A_c$. The outward direction is again ∂_T . Moving the one-dimensional parameter line to the front past the odd line $\det D_A$ contributes -1 . The input compatibility of Ω (Proposition 5.4 for Γ , Lemma 8.5(i) for G) contributes $(-1)^f$, because $\text{ind } D_A = 1$. This branch contributes $-(-1)^f H d_{\text{in}}$. For a joined domain with two inputs, the other-arm permutations are those of the right tensor differential, by Lemma 3.2. The total signed boundary vanishes, which is (8.18). Every branch is counted once with its matched lifting orbit: Paper I, Theorem 5.4(ii), gives one branch per pair, and Theorem 5.1(ii),(iv) with hypothesis (H5) there gives the orbit matching.

For a fixed datum the same computation without the parameter line gives $-d_{\text{out}} F + (-1)^f F d_{\text{in}} = 0$. The sign determinations use only the direction of the translation vector, the ordinary linear gluing of Lemma 3.3 at the ordinary end where the break occurs, and the invertibility of the bordered augmented operator along the gluing paths; no orientation is attached to a physical kernel at a nonsolution. By Corollary 8.9 the output differential is the same at every point of the family. $\square$

For a simple check of the incoming sign, take $D_A : \mathbb{R}v \rightarrow 0$, $D_c : 0 \rightarrow \mathbb{R}y$ and $C : \mathbb{R}\mu \rightarrow \mathbb{R}y$ the identity. In the order $\det D_A \otimes \det A_c$ the kernel is oriented by v . In the parameter-first order $\mu \otimes v \otimes y^*$ cancelling μ against y leaves $-v$.

8.10. The finite families and the closed neck.

Proposition 8.15. *Let $N_{\pm}$ be the regular vertex maps at $R = 0$, and fix a regular joined datum at a width $\lambda_0 > 0$. Choose R and Λ as in Proposition 8.13, and at these lengths add small invariant secondary perturbations supported in the cores which make index one regular and preserve the index-zero counts. Then there are maps $P_{\pm}$ of degree one and B_J of degree $t + 1$ with*

$$N_{a,\infty} - N_a = \delta P_a, \quad F_{\Lambda}^Z - F_{\lambda_0}^Z = \delta B_J.$$

Proof. The added perturbations exist by Paper I, Section 10.11, step (5): the index-zero sets are finite and regular, so openness and compactness preserve them, their signs and the absence of negative indices. For the separate side, join the data at $R = 0$ to those at the large length by one family which first changes the collar length and then makes a compactly supported change of secondary data. Its endpoints are regular through index one. Paper I, Lemma 10.15, Proposition 10.16 and Theorem 5.4, give its relative regularization and compactification, applied to the concatenated family, whose two segments each satisfy the hypotheses of Lemma 10.15, and Lemma 8.14 with $\Omega = \Gamma^a$ gives $N_a^{(R)} - N_a = \delta P_a$. Proposition 8.13 replaces $N_a^{(R)}$ by $N_{a,\infty}$. For the joined side use one family consisting of the width change of Paper I, Proposition 10.17, from λ_0 to Λ , followed by the change of secondary data at width Λ , regularize it relatively as a single family, and apply Lemma 8.14 with $\Omega = G$. The joined families lie at finite widths throughout, so G is defined along them. $\square$

With the right tensor convention, $\delta(F \otimes_R G) = (-1)^{|G|} \delta F \otimes_R G + F \otimes_R \delta G$. Since $N_{\pm}$ and $N_{\pm, \infty}$ are even and closed,

$$(8.19) \quad P = P_- \otimes_R N_+ + N_{-, \infty} \otimes_R P_+, \quad \delta P = N_{-, \infty} \otimes_R N_{+, \infty} - N_- \otimes_R N_+.$$

By Proposition 8.11 and (3.2), $\delta(\mathcal{K}_R B_J) = (-1)^t \mathcal{K}_R \delta B_J$. Together with Proposition 8.13 this gives

$$(8.20) \quad (-1)^t \mathcal{K}_R F_{\lambda_0}^Z - N_- \otimes_R N_+ = \delta(P - \mathcal{K}_R B_J).$$

Proposition 8.16. *Let N_M be the count on the closed-cut piece M_{λ_0} , with the data of N_Z at width λ_0 , oriented by Γ^M . There are maps B_{cut} of degree $t+1$ and H_M of degree one with*

$$N_M \bar{E} - F_{\lambda_0}^Z = \delta B_{\text{cut}}, \quad N_Z - N_M = \delta H_M.$$

Proof. Paper I, Proposition 10.21, inserts a closed neck $[-\ell, \ell] \times Z$ carrying the cut-off term f_Z , and shows that for large ℓ , after a small invariant perturbation making index one regular, the joined index-zero solutions are exactly the gluings $A \#_{Z'} c'$ of index-zero solutions on $\bar{E}$ and on M_{λ_0} , one for each matched lifting orbit. The lifting group is the fibre product over ϕ_Z and all three pieces are connected, so by Lemma 8.2 the gluing fibre is trivial and the quotient count is the matrix composite. By Lemma 8.5(iii), at index zero the orientation G of $A \#_{Z'} c'$ is $\Gamma_{c'}^M$ applied to the right-placed conversion of γ_A ; the prefactor $(-1)^{t_{\text{ind}} D_A}$ is one. Linear gluing along the long neck is continuous and the operators along the gluing paths are invertible, so the sign is that of the composite. Hence the count at the long-neck data is $N_M \bar{E}$. Paper I, Lemma 10.20, connects the data at width λ_0 to the long-neck data by a family with compactly supported parameter derivative, and Lemma 8.14 with $\Omega = G$ gives B_{cut} .

For the second identity, the mixed-end clause of Paper I, Lemma 9.7, changes the finite torus-neck lengths of D_{i, λ_0} to those of the special quintuple of (D_0, D_1) , together with a compact interpolation of the metric, with the ordinary datum on Z fixed. This is a noncollapsing family with the weighted parameter column of Paper I, Lemma 10.15. Lemma 8.14 with $\Omega = \Gamma^M$ gives H_M . The input complex $C_G(Z)$ is the same at both ends, since f_Z is fixed. $\square$

Since $\bar{E}$, with the conversion at Z , is closed, $\delta(H_M \bar{E}) = (N_Z - N_M) \bar{E}$ by (3.2). With $B = B_{\text{cut}} + H_M \bar{E}$,

$$(8.21) \quad N_Z \bar{E} - F_{\lambda_0}^Z = \delta B.$$

8.11. The mixed excision identity.

Theorem 8.17 (Mixed excision). *Assume the unsigned analytic results of Paper I. With the conversion at the closed cut placed as in Convention 8.1 and $\mathcal{K}_R$ as in (8.13),*

$$(8.22) \quad (-1)^t \mathcal{K}_R N_Z \bar{E} - N_- \otimes_R N_+ = \delta H, \quad H = P - \mathcal{K}_R B_J + \mathcal{K}_R B.$$

Each term of H is a signed count of isolated universal solutions in an actual one-parameter family, or its composite or right tensor with closed counts. Here $\bar{E}$ is the reverse excision map at the closed-cut data of Proposition 8.16; by Proposition 2.3(c) it is chain homotopic to the map at the data of that proposition. In particular

$$(8.23) \quad \mathcal{K}_R N_Z \bar{E} \simeq (-1)^t (N_- \otimes_R N_+).$$

For $z \in C_G(Z)$ in the structure of $\bar{E}$, write $N_Z z$ for $N_Z \Theta^r z$, as in the composite $N_Z \bar{E}$. Then for a homogeneous cycle z ,

$$(8.24) \quad [\mathcal{K}_R N_Z z] = \pm [(N_- \otimes_R N_+) E z].$$

All terms are expressed at the fixed end bases.

Proof. By (3.2) and Proposition 8.11, $\delta(\mathcal{K}_R B) = (-1)^t \mathcal{K}_R \delta B$. Adding this, applied to (8.21), to (8.20) gives (8.22), and (8.23) is its restatement. For (8.24) apply (8.23) to Ez . After the continuation homotopy of Proposition 2.3(c), the capped comparison gives $[\bar{E}Ez] = \pm[z]$ for this homogeneous class, with a sign depending on its degree, and applying $\mathcal{K}_R N_Z \Theta^r$ gives the claim. $\square$

The sign $(-1)^t$ in (8.23) refers to the fixed choices of ordinary orientations, of the base orientation of Q and of the reference comparison (8.12). Changing one of these choices changes the identity by one overall sign; no choice changes individual matrix entries.

Equality (8.24) is equality of homology classes of tensors. If the tensors differ by a boundary, applying the chain isomorphism (3.6) produces a boundary in the Hom complex. This is the mechanism that will turn the handle-state calculation into a chain homotopy. No inference from equality of induced maps on homology is involved.

9. THE COMPRESSION CUP BEFORE REFLECTION

9.1. The tensor supplied by the folded triangle. We spell out the geometry from Paper I, Section 10.1. Write a presentation p of M as a pair of torus compression bodies U_0, U_1 of genus $h \geq 2$, with perturbed Lagrangians $\tilde{P}_0, \tilde{P}_1 \subset M_h$. Let $C = C_{h,g}$, $g = h+k$, add k handles and put $V_i = C \circ U_i$. The resulting three-manifold is $M^+ = M \#^k (S^1 \times S^2)$. Restriction of flat connections on C gives the embedded Lagrangian correspondence $L_C \subset M_h^- \times M_g$.

Choose product Morse height functions on the clean auxiliary intersections $L(V_i) \cap (L(U_i) \circ L_C) \cong \mathrm{SU}(2)^{g-1}$. Each height function has 2^{g-1} critical points. Perturb the primary data of V_i by t times this function, for $t > 0$ small, and write $R_i \subset M_g$ for the resulting Lagrangian. An *upper height presentation* is the presentation with compression bodies V_0, V_1 , Lagrangians R_0, R_1 , and these height and auxiliary regular data. For this fixed upper height choose an open neighbourhood $\mathcal{N}(t)$ of zero in the product of the lower primary parameter spaces for U_0, U_1 . Require the two auxiliary height intersections to remain transverse, with their critical points continuing uniquely from the product Morse model. Such a neighbourhood exists by openness of transversality for these compact intersections. Shrink it inside the chosen primary ball and choose the lower data there; the additional generic conditions on the lower pair are imposed below. This is the lower-data neighbourhood used in Section 13. In the product symplectic manifold

$$X = M_h^- \times M_g, \quad A_i = \tilde{P}_i^- \times R_i,$$

require the three pairs among (A_0, L_C, A_1) to be transverse and their triple intersection to be empty. These are the generic conditions used for the folded triangle in Paper I. The two auxiliary complexes are $CF_X(A_0, L_C)$ and $CF_X(L_C, A_1)$. Their distinguished top cycles are q_0^* and q_1^* , respectively, with top degree defined by the auxiliary integer grading m of Paper I, Section 11.3. Its differences satisfy $m(a) - m(x) = \mathrm{ind}(u) - 8\kappa_{\mathrm{aux}}(u)$. Here κ_{aux} is the relative class of Paper I's auxiliary pair. Let $\mathcal{A}_i$ denote its critical action, the values of the actual local exact-graph primitive, with the sign fixed by the ordered pair. Writing ω_X for the symplectic form of X , the relation is

$$4\pi^2 \kappa_{\mathrm{aux}}(u) = \int u^* \omega_X - (\mathcal{A}_i(a) - \mathcal{A}_i(x)).$$

At the flat height data $\mathcal{A}_i$ is the signed height function th_i ; under small primary changes its actual values are retained. Thus the symplectic area includes an endpoint correction; its quotient by $4\pi^2$ is the total topological energy. Attaching a positive generator sphere

increases κ_{aux} by $1/2$ and the index by 4, leaving $m(a) - m(x)$ unchanged. The normalization of m agrees with the Morse grading. The parity grading alone does not define a top class. Depending on the order of the pair, the corresponding critical point is the maximum or minimum of the height function.

Denote the chosen upper presentation by $q = C \circ p$; this notation includes its height perturbation. With a split almost-complex structure on X , folding identifies the output complex with $CF(\bar{p}) \otimes_R CF(q)$. Let $\mathcal{K}_R$ be the closed identification (8.13) of Section 8, with its transported orientation lines. It uses the orientation system transported independently at $\bar{p}$; the reflection pairing with $CF(p)^\vee$ will be constructed later. The triangle product for the ordered triple (A_0, L_C, A_1) , including the conversion Θ at its closed gauge cut Z in the position of Convention 8.1, is denoted by μ_X^Θ . It gives the cycle

$$(9.1) \quad z_C = \mathcal{K}_R \mu_X^\Theta(q_0^* \otimes_R q_1^*) \in CF(\bar{p}) \otimes_R CF(q).$$

Define

$$(9.2) \quad \check{U}_C = B_{CF(\bar{p})}(z_C) : CF(\bar{p})^\vee \longrightarrow CF(q).$$

For $k = 0$, L_C is the diagonal and we denote the upper height presentation by r . An *admissible fold pair* (p, r) means such a lower and upper presentation in one common admissible ball, with a height t , lower data in $\mathcal{N}(t)$, and the transversality and auxiliary regularity conditions just stated. Put $B(p, r) = \check{U}_C$ for this pair. Thus a fold pair is a particular pair of nearby presentations with a fixed diagonal triangle and its two height caps.

The top cycles are normalized using the integral mixed maps on the auxiliary vertices: $q_i^* = \pm N_i \theta_i$. The sign is one sign of each distinguished cycle. The auxiliary complexes have the standard filtration, and the unit-diagonal calculation in Proposition 6.3 makes this an integral normalization, not merely a mod-two one.

9.2. The gauge handle state. Let B_k be the gauge filling whose closed boundary is Z , obtained from the folded triangle and its two auxiliary fillings. Its relative invariant is a homogeneous cycle $\Theta_{B_k} \in C_G(Z)$. Choose a single almost-complex structure on this filling and use its restrictions on the pieces. Convert at Z to the structure used for mixed torus excision. After excision the resulting state is a cycle on $C_G(-M) \otimes_R C_G(M^+)$.

Bending the source end of the one-handle cobordism W_k gives the identity

$$(9.3) \quad (\Phi_{G,M} \otimes_R 1)[E\Theta_{B_k}] = \pm[\text{ten}(I_J^\#(W_k))].$$

To check the sign convention, start with the product cylinder: its bent state is the coevaluation tensor. Glue W_k to its second end and use the ordered determinant gluing. This gives the tensor of the right-placed map. An odd endpoint conversion is already part of that map through (4.3); moving it to the other side would change the tensor by a grading operator. The residual signs in (9.3) are therefore scalar. The same argument holds for $k = 0$, when the cobordism map is gauge continuation.

9.3. The cup formula. Define the transposed reflected mixed map

$$(9.4) \quad n_p = \text{ev}_{C_G(M)}^{-1}(N_{\bar{p}} \Phi_{G,M}^{-1})^\vee : CF(\bar{p})^\vee \longrightarrow C_G(M).$$

It is an even chain isomorphism, using the paired choices of gauge base. This definition uses gauge duality and the mixed map at the reflected presentation; it makes no assumption about symplectic duality.

The integral version of Paper I, Theorem 11.8, is the folded form of the triangle comparison (7.10).

Lemma 9.1 (Folded triangle comparison). *Let N_0, N_1 be the integral mixed maps of the product-target pairs (A_0, L_C) and (L_C, A_1) , with gauge inputs $Z_0 = D_0 \cup (-C)$ and $Z_1 = C \cup (-D_1)$, and let $\mu_{\mathcal{W}} : C_G(Z_0) \otimes_R C_G(Z_1) \rightarrow C_G(Z)$ be the integral gauge map of the cobordism formed by the pieces D_0, C, D_1 . With the conversion at Z placed as in Convention 8.1,*

$$\mu_X^\Theta(N_0 \otimes_R N_1) \simeq \pm N_Z \Theta^r \mu_{\mathcal{W}}.$$

Proof. The unsigned argument is the proof of Paper I, Theorem 11.8: the family of data of Paper I, Lemma 9.12, collapses the middle compression body C , which is an odd compression body with Lagrangian L_C , stretches the two ordinary necks cutting off the vertex domains of (A_0, L_C) and (L_C, A_1) , and stretches the closed cut Z . We orient every configuration in the family by the several-input construction of Section 5, with the conversion line at Z in its fixed position. At the collapsed end, Theorem B identifies the orientations of the index-zero solutions with the orientations of the symplectic triangle glued to the two mixed vertices. It applies to this collapse: Section 7 rests on the collapse analysis of Paper I, Sections 6–8, and the proof of Paper I, Theorem 11.8, supplies its inputs for the collapse of $C = C_{h,g}$, with product target, both boundary surfaces matched and the lower sheet reversed, using the two-trace slice space $H^1(\Sigma_h; \text{ad}) \oplus H^1(\Sigma_g; \text{ad})$; ordinary gluing along the two stretched necks (Lemma 3.3) gives the count $\mu_X^\Theta(N_0 \otimes_R N_1)$, with the input lines in cyclic order. At the stretched cut, gluing at the connected closed cut Z has no extra gluing fibre (Lemma 8.2). The stretched cut decomposes the domain into the triangle cobordism of D_0, C, D_1 , glued at Z to the closed-cut vertex. Lemma 8.5 is stated for W_λ^J , but its proof uses only a decomposition at a closed cut, the localized shift, which Proposition 2.2(b) supplies on folded triangles, and Convention 8.1. The same proof gives parts (i)–(iii) for this decomposition, and part (iii) identifies the orientation with that of the gauge cobordism followed by the conversion and the mixed map; this gives $N_Z \Theta^r \mu_{\mathcal{W}}$. Parts (i) and (ii) give the compatibility at the input and output breaks. The family between the two ends is a compact one-parameter family with compactly supported parameter derivative, and Lemma 8.14, with $f = t$, turns its compactified index-zero space into the stated homotopy. The overall sign reflects the fixed orientation choices at the two ends. $\square$

Proposition 9.2 (Cup formula).

$$(9.5) \quad \check{U}_C \simeq \pm N_q I_J^\#(W_k) n_p.$$

In particular, for a lower datum p and a compatible upper height datum r ,

$$(9.6) \quad B(p, r) \simeq \pm N_r \text{cont}_G(p, r) n_p.$$

Proof. Apply Lemma 9.1 to the two distinguished input cycles $q_i^* = \pm N_i \theta_i$. Since the gauge filling B_k is the triangle cobordism glued to the two auxiliary fillings, Θ_{B_k} is homologous to $\pm \mu_{\mathcal{W}}(\theta_0 \otimes \theta_1)$ by gauge composition. Hence the folded triangle output is, up to a boundary and one sign, $N_Z \Theta^r \Theta_{B_k}$. Apply (8.24) to the homogeneous cycle Θ_{B_k} . The result is

$$[z_C] = \pm [(N_{\bar{p}} \otimes_R N_q) \Theta_{B_k}].$$

Equation (9.3) identifies this with the tensor of $N_q I_J^\#(W_k) n_p$. The identification follows directly from (3.6) and the graded double-dual map in (9.4). Since the difference of the tensors is a boundary, their images under $B_{CF(\bar{p})}$ differ by δK . This proves (9.5). The same calculation with $k = 0$ gives (9.6). $\square$

This proves a formula for the cup tensor without selecting the reflection pairing needed for the source of Paper I’s map. The next sections construct that pairing and show that it gives the ordinary symplectic continuation when $k = 0$.

10. SPHERE CHARACTERS AND VARIATION OF PRIMARY DATA

10.1. The character of an orientation system. For a transverse pair $P = (P_0, P_1)$ let $\mathcal{P}(P_0, P_1)$ be the path space of paths in M_h starting on P_0 and ending on P_1 . For $h \geq 2$, the compression-body Lagrangians are simply connected and have vanishing second homotopy group. The relative homotopy classes of strip configurations with fixed ends are consequently a torsor for $\pi_2(M_h) \cong \mathbb{Z}$ in the lift convention of [1, Section 2]. Fix its positive generator S_h ; attaching it changes the index by four. For $h = 1$ the moduli space is a point and the sphere-character assertions below are empty.

The determinant line of a sphere operator, together with the inverse determinant of the matching tangent space at its node, has its complex orientation. Compare node gluing with the coherent strip orientation Γ^P . The ratio is a sign

$$(10.1) \quad \chi^P : \pi_2(M_h) \longrightarrow \{1, -1\}.$$

Associativity of node gluing makes it a character. Multiplying generator lines by signs does not change it: the two endpoint factors occur in both sides of the comparison. Multiplying them by their grading signs has the same property. Thus the character is information about the configuration-level coherent system, beyond the signs of its differential.

At a transverse point x , a second description is available. Attach a fixed representative sphere to the constant mixed pair C_x and compare its transported mixed orientation with the product of the canonical orientation of C_x and the complex node orientation. Denote the ratio by $\chi(h, x; d)$, where h is the primary datum and d contains the regularizing and almost-complex data.

Lemma 10.1. *At regular data, $\chi(h, x; d) = \chi^P(S_h)$ for every transverse intersection point x . The ratio is constant along any compact smooth family of primary data carrying a persistent transverse intersection point. It is also independent of small regularizing data at fixed primary data and fixed gauge base.*

Proof. Glue a strip to the constant pair and attach the sphere away from the ordinary gluing collar. The two possible orders of these operations induce the same determinant-line isomorphism. The sphere and node factors are complex, hence even as real graded lines, so commuting them with the strip factor gives no sign. Cancelling the strip orientation gives equality of the pointed and strip definitions. A strip configuration exists between any two intersection points, in some relative homotopy class; it need not solve the holomorphic equation. Applying the same argument to that configuration shows independence of x .

For a family with a persistent point, the constant-pair operators and the node-gluing map vary continuously by Proposition 2.1, and the transported orientation of Proposition 5.4 by Proposition 2.2. The ratio is a continuous function to $\{1, -1\}$ and is therefore constant. To vary regularizing data, use the segment to zero data. The perturbations are supported off the strip end, and the constant-pair operators stay invertible. Transporting their canonical orientations along the segment proves the last assertion. Compatible almost-complex structures can be varied through the same argument, with the gauge base held fixed. $\square$

The persistent point need only remain transverse at that point. Other intersections of the two Lagrangians may become degenerate. The lemma is about determinant lines over configurations and therefore does not require a regular moduli space along the family.

10.2. A common ball of primary perturbations. Let $\mathcal{C}_i$ be the finite real span of the sign-sensitive holonomy functions on C_i° , modulo constants. Use the norm

$$(10.2) \quad \begin{aligned} \|h_i\|_i &= \sup_B \|\nabla_B h_i\|_{L^\infty} + \sup_{B \neq B'} \frac{\|\nabla_B h_i - \nabla_{B'} h_i\|_{L^2}}{\|B - B'\|_{L^2}}, \\ \|h\| &= \max(\|h_0\|_0, \|h_1\|_1). \end{aligned}$$

The suprema are taken over smooth connections in a fixed determinant bundle. Cylinder functions have finite norm by the holonomy estimates of [1, Proposition 6.1]. If the norm vanishes, the gradient vanishes on the affine connection space, so the function is constant; thus this is a norm modulo constants.

Choose a finite trace family F_0 whose restrictions to the compact flat-connection moduli spaces give embedding coordinates. In the model $L(C_i) \cong \mathrm{SU}(2)^{h-1}$, quaternion coordinates can be obtained from sign-sensitive traces using the fixed holonomies on the odd torus. Such traces distinguish central sign changes; passing to unsigned $\mathrm{SO}(3)$ traces would lose that property. For a finite family F containing F_0 , put

$$\mathcal{U}(F) = \{h \in \mathrm{span} F : \|h\| < r_*\}, \quad \mathfrak{B} = \bigcup_F \mathcal{U}(F).$$

The radius is independent of the additional functions in F , for the fixed splitting and metric data. These sets are nested under inclusion of finite families and are convex.

Proposition 10.2. *The radius $r_* > 0$ can be chosen so that every h in this ball has the following properties.*

- (i) *The perturbed flat moduli space of each compression body is a smooth compact manifold identified with its unperturbed moduli space; its boundary restriction is an embedded Lagrangian close to the unperturbed one.*
- (ii) *The connection representatives and their first derivatives in the moduli parameters are close in L_1^2 and L^2 , respectively, uniformly as $\|h\| \rightarrow 0$.*
- (iii) *The functions from F_0 remain embedding coordinates on these perturbed moduli spaces.*
- (iv) *If the two boundary Lagrangians intersect transversely, the corresponding closed gauge critical points are nondegenerate and are exactly the lifts of those intersections.*

Proof. The norm gives $\|F_B\|_{L^\infty} \leq \|h_i\|_i$ for an h_i -flat connection. If a sequence with norm tending to zero escaped every gauge neighborhood of the flat moduli space, Uhlenbeck compactness on the compact compression body would give a subsequence weakly converging in $W^{1,4}$, after gauge, to a flat connection. The $W^{1,4}$ embedding in C^0 is compact in three dimensions. A Coulomb–Neumann slice relative to the flat limit and the elliptic estimate for (d_B^*, d_B) then improve this to convergence in L_1^2 , a contradiction. The boundary slice and compactness statements are the versions for a manifold with boundary in [15].

In a slice at a flat connection split into the finite-dimensional harmonic space and its complement. The linearized perturbed-flat operator on the complement has a bounded right inverse. The second term of (10.2) makes the holonomy part a small Lipschitz perturbation in L^2 , so the contraction gives a graph over the harmonic space. Differentiating its fixed-point equation in the finite-dimensional parameter gives the asserted first-derivative control. Hodge decomposition and boundary elliptic regularity identify these solutions with the smooth perturbed-flat solutions; the exact regularity inputs are recorded in Appendix A. Compactness of the flat moduli space makes the thresholds uniform over finitely many slice charts.

For a trace function τ , its bounded Lipschitz gradient extends the identity

$$\tau(B') - \tau(B) = \int_0^1 \langle \nabla_{B+s(B'-B)} \tau, B' - B \rangle ds$$

to L^2 connections. Differentiating on the graph family shows that the trace-coordinate map remains C^1 -close to its unperturbed embedding. Openness of embeddings of a compact manifold proves (iii). The matching exact sequence for the two compression-body boundary problems identifies the kernel of the closed Hessian with the tangent intersection of the Lagrangians; hence transversality gives (iv), as in [1, Section 2.3]. $\square$

No radius controlling all higher derivatives of every cylinder function is asserted here. The norm controls the two estimates actually used in this proposition. On a fixed finite-dimensional perturbation family all higher smooth seminorms are finite and are handled separately in compact-family regularity statements.

10.3. Paths with only birth–death singularities. Enlarge F to include the monomials of degree at most five in the trace coordinates F_0 . These remain cylinder functions: a monomial is a smooth invariant function of the joint finite holonomy family.

Lemma 10.3. *Two regular primary data in $\mathfrak{B}$ can be joined within a common $\mathcal{U}(F)$ by a smooth path with finitely many singular parameters. At each singular parameter there is one quadratic tangency, with versal normal form*

$$(10.3) \quad S_t(x, y) = x^3 + \sigma(t - t_*)x + Q(y), \quad \sigma \in \{1, -1\},$$

where Q is nondegenerate. All other intersections are transverse.

Proof. Start with the segment from h_a to h_b and perturb it to

$$h_t = (1 - t)h_a + th_b + \beta(t)c,$$

where β vanishes near the endpoints and is positive on the interior where singularities may occur. The segment is compact in the open convex ball, so small c keeps the path in the ball.

Variation of the holonomy function induces an exact Lagrangian deformation; its Hamiltonian on the perturbed Lagrangian is the restriction of that function to the connection moduli space. The trace coordinates remain embedding coordinates by Proposition 10.2. Polynomials of degree at most three in embedding coordinates realize arbitrary third jets at one point. Polynomials of degree at most five realize arbitrary second jets at two distinct points. For the latter assertion, choose an affine function ℓ with $\ell(a) = 0$ and $\ell(b) = 1$. Given a second jet at b , multiply ℓ^3 by the quadratic Taylor polynomial at b of that jet divided by ℓ^3 . The product has degree at most five, realizes the prescribed jet at b , and vanishes to order three at a . Use $(1 - \ell)^3$ in the same way for the jet at a , then add the two polynomials. Restriction of ambient jets to an embedded submanifold is surjective.

In a Weinstein chart, varying one Lagrangian changes the relative generating function by this Hamiltonian to first order. The map on its jets is triangular with identity diagonal. Parametric transversality therefore excludes higher singularities, two simultaneous tangencies, and tangencies with zero crossing speed: these strata have codimension at least two. A generic one-parameter path meets only the codimension-one birth–death stratum, transversely. Its parameter-dependent splitting normal form is (10.3).

The singular-parameter set is closed by compactness of the Lagrangians. Each singular parameter is isolated by the normal form, and the endpoint neighborhoods are regular. It is consequently finite. No uniform separation of the two points in the two-point jet argument is required; transversality is applied on the open configuration space of distinct pairs. $\square$

Lemma 10.4. *Suppose $H(C_G(M)) \neq 0$. The path in Lemma 10.3 has a finite subdivision with regular subdivision points such that each subinterval carries a continuous transverse intersection point throughout the subinterval.*

Proof. Every regular primary datum has at least one intersection. Otherwise its gauge complex would have no generators by Proposition 10.2(iv), contradicting invariance and nonvanishing of the gauge homology.

At a birth–death time the local pair in (10.3) is absent on one side. If it were the only intersection at the singular time, compactness would force the nearby regular intersection set on that side to be empty. Thus another intersection exists at that time and is transverse. The implicit function theorem continues it across the wall.

On a closed interval with no walls, the incidence set of intersections is a compact one-manifold projecting locally diffeomorphically to the interval. It is a finite covering, hence a disjoint union of sections. Extend the chosen transverse point from each wall to neighboring regular subdivision times. This gives the required finite collection of persistent sections. $\square$

Theorem 10.5 (Constancy of the sphere character). *Fix a splitting, metric data and gauge base, and suppose $H(C_G(M)) \neq 0$. Then $\chi^P(S_h)$ is independent of the regular datum P with primary data in $\mathfrak{B}$.*

Proof. First remove the regularizing data at both endpoints using Lemma 10.1. Join the primary data by the subdivided path of Lemma 10.4. On each subinterval the pointed character is constant along its persistent transverse point. At a regular subdivision time the pointed character equals the global character for any choice of point, so one may switch to the persistent point for the next subinterval. Chaining these equalities gives constancy along the whole path. Restore the endpoint regularizing data. The same argument allows variation of the compatible symplectic almost-complex structure at a fixed gauge base. $\square$

The argument uses nonvanishing only to find a point persisting through each birth–death. It does not assume that individual generators persist along an arbitrary path, or that the regular locus of the perturbation ball is connected.

11. REFLECTION OF THE TRANSPORTED ORIENTATIONS

11.1. Reflection and sphere gluing. The reflected presentation $\bar{p}$ interchanges the two compression bodies. A strip is reflected by

$$\bar{u}(s, t) = u(-s, 1 - t).$$

This map preserves the orientation of the strip domain and keeps the surface almost-complex structure. It reverses the translation direction. The target symplectic manifold retains the original orientation; the reversed order of the Lagrangians supplies the dual complex. Mixed solutions themselves are not identified by this formula. The orientation at $\bar{p}$ is transported through its own mixed domain.

For a strip from x to y , the reflection of its coherent orientation has differential sign

$$(11.1) \quad \text{sign}_{\text{dual}}(\bar{u}) = -(-1)^{|y|} \text{sign}(u).$$

The minus sign is the reversed translation vector; the other sign is the graded transpose. Thus this reflected system is the one for which the paired generator bases give the graded dual differential.

Attaching S_h changes the index by four. The determinant factor at a sphere node is complex and even. The additional transpose sign is $(-1)^{4|y|} = 1$. Consequently reflection

of a coherent strip system preserves its sphere character. What remains is to compare this reflected system with the system transported through the reflected mixed domain.

11.2. Capping, unfolding, and end orientations. We record the end-gluing comparison before using it to compare sphere characters. For a capped folded configuration c with outputs x on $\bar{p}$ and y on r , write F_c for its determinant line after contracting the two normalized auxiliary cap factors. The conversion line at Z is retained in its fixed position. Transport from the mixed configuration gives $\Gamma_c : F_c \rightarrow L_{\bar{p}}(x) \otimes_R L_r(y)$. Thus this is a tensor with two outgoing ends. Bending the first end gives a map from $L_{\bar{p}}(x)^\vee$, not from $L_{\bar{p}}(x)$.

Lemma 11.1. *Let u be any smooth outgoing strip from x to x' on $\bar{p}$, and w any smooth outgoing strip from y to y' on r . Capping commutes with their determinant gluing. For $k = 0$, unfolding along the diagonal commutes with the same gluing, with the first outgoing strip reflected into an incoming strip by graded transpose. These statements hold for arbitrary strip indices and configuration classes.*

Proof. Put $d_u = \det D_u$ and $d_w = \det D_w$. The assertion about the end systems is the commutative diagram

$$\begin{array}{ccc} F_c \otimes d_u \otimes d_w & \xrightarrow{(\Gamma_u \otimes_R \Gamma_w) \Gamma_c} & L_{\bar{p}}(x') \otimes_R L_r(y') \\ \downarrow G & & \parallel \\ F_{c\#(u,w)} & \xrightarrow{\Gamma_{c\#(u,w)}} & L_{\bar{p}}(x') \otimes_R L_r(y'). \end{array}$$

In the upper arrow the strip determinant factors are inserted next to their corresponding generator factors using the right tensor convention. In particular moving d_u past $L_r(y)$ contributes $(-1)^{\text{ind}(u)|y|}$.

Here G is ordered determinant gluing, including the cap contractions. Before contraction take the order

$$\mathcal{J}, \quad \det \Theta, \quad \det D_c, \quad \det D_{a_0}, \quad \det D_{a_1}, \quad d_u, \quad d_w,$$

where $\mathcal{J}$ is the complete generator input line in cyclic triangle order and a_0, a_1 are the two caps. If a sphere is attached, append its complex node factor last. Both routes use this order. Associativity regroups factors without interchanging them; any permutation needed for an end convention is performed on both routes with its graded sign. In particular the conversion line is never carried through the moving configuration determinant. The unit cap maps contract their paired generator and determinant factors by even evaluation maps. No assumption that the individual cap generator has even parity is needed.

First consider the completed mixed triangle at positive scale. Its localized shift retains the ordinary two-input cobordism and ends at that cobordism followed by a constant mixed pair. There ordinary determinant associativity gives the end-gluing diagram, with the conversion line and the fixed unit cap maps in the order above. The relative family construction transports it over the positive-scale configuration families.

Pull back the entire ordered diagram through Lemma 7.1. Its ordinary-gluing compatibility applies to the strip and cap operators at their ends in the stated cofinal neck-and-scale range. Only after this diagram commutes do we cancel the completing vertices and contract the two cap factors. The fixed-width closed-cut line comparison retains the conversion factor; its completion is expressed through the separate output lines through the reference comparison $K_{0,x}$ of Proposition 8.10, not by completing the fixed closed- Z input itself. This proves the first diagram for the limiting configurations. It does not shift a limiting triangle with empty triple intersection to a constant triangle.

For unfolding with $k = 0$, first choose doubling data retaining the Hodge almost-complex structure and zero inhomogeneous germ in the collapse neighbourhood, the compatible ordinary ends and the fixed cap models. Choose a folded configuration obtained from a smooth configuration on the unfolded strip. The two coefficient systems are then restrictions of one smooth system. At the $W^{1,p} \rightarrow L^p$ realization, $p > 2$, equal traces along the diagonal are exactly the condition for the two sections to join to a $W^{1,p}$ section after reflecting the lower side. Restriction is the inverse, and the same identification holds for L^p target sections. This is an operator identification. At higher Sobolev orders equal traces alone would not suffice. Instead identify the higher-order determinant lines through their common smooth kernels and cokernels, using smooth stabilization columns. Elliptic regularity makes these identifications commute with gluing and parameter variation.

The comparison is first made on this doubling locus. Any finite set of smooth configuration classes has representatives there: smooth the joined maps in a small collar of the diagonal and keep the end data fixed. The smoothing is a relative homotopy and changes no sphere class. Interpolate the auxiliary almost-complex and perturbation data to doubling data within the admitted data space, preserving these germs and end values; Proposition 2.1(a) gives the continuous Fredholm families. Determinant transport extends the resulting comparison back to the original finite configuration families. The first gluing diagram and path independence in Proposition 5.4 make it independent of these homotopies. With stabilizations off the joining regions the operator identification on the doubling locus gives

$$\begin{array}{ccc} F_c \otimes d_u \otimes d_w & \xrightarrow{G} & F_{c\#(u,w)} \\ \downarrow U_{\text{ord}} & & \downarrow U \\ d_{Ru} \otimes \det D_{Uc} \otimes d_w & \xrightarrow{G} & \det D_{Ru\#Uc\#w}. \end{array}$$

Here Ru runs from x' to x on p , so it attaches to the incoming end of Uc , which runs from x to y . The left arrow U_{ord} applies unfolding and reflection and then interchanges F_c and d_u , with sign $(-1)^{|F_c|\text{ind}(u)}$. At the operator level the maps are restriction, reflection and the same ordinary-end cutoff patching, whose supports are disjoint from the diagonal. Their two composites agree. This is a diagram on the zero-scale section problem; no unfolding map for the positive-scale gauge operator is asserted, and no additional commutation of unfolding with collapse is used. The displayed graded interchange expresses that agreement in the chosen tensor order. Cancelling the ordered stabilizations proves the displayed statement. The diagonal normal factor has its complex orientation and is even. Reflection of the strip u of index n moves its index line across the paired endpoint line and contributes $(-1)^{n|x'|}$ with the endpoint order used in the graded transpose. For a strip counted modulo translation the reversed translation vector adds the minus sign in (11.1). To check the total sign, put $a = |x|$, $b = |y|$ and $f = |F_c| = a + b$. The outgoing gluing sign is $(-1)^{nb}$. On bending, the reflected input sign $(-1)^{na}$ and the swap $(-1)^{nf}$ have product $(-1)^{n(a+f)} = (-1)^{nb}$. Thus the diagrams give the same sign at every index. $\square$

11.3. One folded comparison. Fix one admissible fold pair with lower primary datum $\tilde{p}$ and upper height datum r , both in the admissible ball. Choose the height after the radius of that ball, as in Section 13. Double the folded triangle and attach its auxiliary caps. Before bending, the resulting disk has two outgoing strip ends modeled on $\tilde{p}$ and r . By Lemma 11.1, its transported orientation is coherent with the independently transported strip system on $\tilde{p}$ and the transported outgoing system on r . The comparison with the dual system on p is made only after the sphere characters have been identified.

Proposition 11.2.

$$\chi^{\bar{p}}(S_h) = \chi^r(S_h).$$

Proof. Choose a smooth reference configuration c_0 on the doubled disk. Insert a representative sphere on the first outgoing strip end at a fixed small positive plumbing parameter. Move this smooth inserted configuration through the unfolded disk to the second outgoing end, then refold. The middle is a compact smooth configuration homotopy; we do not apply an interior-node theorem at a node on the diagonal. The two resulting configurations lie in the same relative class. Under diagonal unfolding the two sphere classes are identified with the same generator of the relative sphere group. In the product description this moves the class between the two factors modulo the diagonal class. A reversal of its representative replaces the generator by its negative and does not change a character with values in $\{1, -1\}$.

There is a continuous orientation on the determinant bundle along this path. Its construction on the folded domain uses the collapse transport of Section 7, the closed-cut conversion, and the reference comparison of Proposition 8.10. For the endpoint comparisons place the sphere in an ordinary strip factor attached away from the collapse neighbourhood. At the fixed positive plumbing parameter, Lemma 7.1 commutes with gluing that entire strip. Its interior-node determinant comparison is the usual punctured Cauchy–Riemann bordering and complex evaluation map; the gauge rows are absent on this pure strip. Ordinary-gluing associativity thus gives the endpoint comparison with the complex even node factor appended last. The collapse scale and ordinary neck can be chosen after this plumbing parameter. No simultaneous node-and-collapse degeneration estimate is used. At the two ends, the first diagram of Lemma 11.1 identifies the transported orientation with both specified strip-node gluing orientations. No Koszul sign is introduced by moving the complex node factor. Their ratios are therefore equal, giving the assertion.

This comparison uses arbitrary smooth configurations, not only holomorphic solutions. To justify coherence where the fold is attached to prescribed strip configurations, select one reference configuration for each of the finitely many generator pairs and their required sphere classes. Choose the height and neck lengths so that ordinary gluing applies to this finite set. The determinant-line construction then extends over each configuration component by path independence. Proposition 2.1 and Proposition 2.2 are needed on all the domains in this finite construction; a comparison only at the constant solution would not suffice. $\square$

Changing the almost-complex base on the closed cut changes the output generator lines by a fixed diagonal conversion. That conversion cancels in a sphere ratio. Thus the character comparison is unchanged by the conversion required in Section 8, although that conversion remains essential for the cup identity as a chain-level formula.

Corollary 11.3 (Reflection invariance). *For every regular datum p in the admissible ball at an object with nonzero integral gauge homology,*

$$\chi^{\bar{p}}(S_h) = \chi^p(S_h).$$

Proof. Apply Theorem 10.5 on $-M$ to compare $\bar{p}$ with $\tilde{p}$. Apply Proposition 11.2 to compare the latter with r . Apply character constancy on M to compare r with p . The reflected ball is the same ball under the chosen reflection identification. Gauge duality makes nonvanishing for M and $-M$ equivalent. $\square$

11.4. The reflection pairing. Compare on the strips of $\bar{p}$ its own transported system and the dual system of (11.1). Their ratio is constant on smooth configuration components and multiplicative under concatenation. Choose one reference intersection point. A configuration

from it to any other intersection determines a sign $\kappa_p(x)$. Changing that configuration changes its class by a sphere; Corollary 11.3 says that the ratio on this sphere is one. Hence the ratio on a strip from x to y is $\kappa_p(x)\kappa_p(y)$, independent of the strip class.

Proposition 11.4. *The map*

$$(11.2) \quad \Phi_p : CF(\bar{p}) \longrightarrow CF(p)^\vee, \quad o_x^{\bar{p}} \longmapsto \kappa_p(x)(o_x^p)^\vee$$

is an even chain isomorphism. The configuration-level comparison determines it up to one sign. The pairings may be chosen so that

$$(11.3) \quad \Phi_{\bar{p}} = \Phi_p^\vee \text{ ev}.$$

Proof. Equation (11.1) and the ratio just computed give exactly the dual differential under (11.2). The paired gauge bases make the sum of the two endpoint parities zero, so the map is even. Any two comparisons have constant ratio: configurations exist between every pair of intersection points, and the ratio on all those configurations fixes the relative generator signs. Only the initial reference sign is free.

Reflect twice on a strip of index n with endpoints x, y . The two transpose signs multiply to $(-1)^{n(|x|+|y|)} = (-1)^n$. The corresponding change of the paired basis is the sign in the graded evaluation map. Thus the pairing obtained by the second reflection is exactly $\Phi_p^\vee \text{ ev}$, after the common reference sign is fixed. This proves (11.3) at configuration level. $\square$

A chain-map condition alone would not determine κ_p up to one sign. If the differential splits into two subcomplexes, one can change the sign on one summand and still obtain a chain map. The preceding construction uses all strip configurations, including those which are not solutions, to exclude that extra freedom.

Put

$$(11.4) \quad \widehat{\Phi}_p = \Phi_p^\vee \text{ ev} : CF(p) \longrightarrow CF(\bar{p})^\vee.$$

This is the reflection pairing which will be composed with the cup map (9.2).

12. CONTINUATION AND THE DIAGONAL COMPRESSION CUP

12.1. Coherent orientations with prescribed ends. A continuation datum from p to q has a strip-like surface with Lagrangian boundary conditions varying on a compact interval and equal to those of p, q at its ends. The almost-complex and Hamiltonian data are likewise constant outside that interval. Throughout this section the primary paths lie in the common convex ball of Proposition 10.2; after Hamiltonian identification, their exact-graph primitives lie in a fixed convex neighbourhood. Homotopies have fixed endpoints and remain in this neighbourhood. Almost-complex and Hamiltonian continuation data are varied with these fixed boundary identifications. All statements of independence below concern this class of data. For generators x, y , let $\mathcal{B}_{p,q}(x, y)$ be its smooth configuration space, with the prescribed exponential limits. A coherent orientation system assigns relative orientations to the determinant lines over these spaces and commutes with attaching strips at either end.

The space of configurations in a fixed relative homotopy class is path connected. One may first choose a continuous relative homotopy, smooth it relative to the fixed ends and labels, and then make it constant beyond common end cutoffs. Local charts are given by an exponential map for a metric making the boundary labels totally geodesic. The topology in Section 2 makes these paths continuous. The relative classes are a torsor for the same sphere group as fixed-label strips, and the index changes by four under its generator. In particular there is at most one index-zero component for fixed endpoints.

Lemma 12.1 (Rigidity of coherent systems). *Two coherent continuation orientation systems with the same end systems differ by one sign. Such a system exists if and only if the sphere characters at the ends agree.*

Proof. The ratio of two systems is constant on each configuration component. If a strip is attached at either end, the common end orientation cancels, so the ratio does not change. Choose one reference configuration from x_0 to y_0 . For any x, y , choose strips from x to x_0 and from y_0 to y . Attaching these strips transports the ratio to a configuration with ends x, y . Any difference in relative class can be supplied by a sphere class on either strip. Since the end systems are the same, this also leaves the ratio unchanged. Thus there is only one sign for all configurations.

For existence choose a reference orientation and extend it by the same gluing procedure. The determinant bundle on each configuration component is orientable: the boundary Lagrangians are the parallelizable compression- body Lagrangians, and the remaining sphere factors have complex orientation. The only compatibility left when the gluing paths are changed is moving a sphere class from the incoming end to the outgoing end. The two extensions agree precisely when the two sphere characters agree. This also proves necessity by moving the node of one sphere through a continuation configuration. The construction is continuous because the determinant line is a line bundle over the smooth configuration space. $\square$

In an index-zero comparison a finite form of the lemma suffices. Choose one strip for each input generator and one for each output generator, in the classes needed to reach the index-zero component. Coherence on that finite collection fixes the ratios of all zero-dimensional counts. This form permits the height and neck thresholds to be chosen after a finite set of configurations, without a claim of uniform estimates over the whole configuration space.

Theorem 10.5 supplies existence for p, q in the same admissible ball. For a lower datum and an auxiliary upper height datum, the folded configuration itself supplies a coherent system and hence the required character equality. The latter observation applies even when the auxiliary upper datum is outside the character-constancy ball, provided it satisfies the separate primary regularity threshold.

12.2. Parity and continuation maps. For a continuation configuration c from x to y , the number

$$d(p, q) = |y| - |x| - \text{ind } D_c \pmod{2}$$

is independent of the configuration and its end generators. It is additive under composition of continuation data. The configuration description by sphere classes, together with a homotopy in the specified small family, shows its independence of the chosen datum in that family. For the paired gauge bases of this paper, $d(p, q) = 0$ at the data used in the comparison. Here one can use the integral cup formula to check the parity without assuming that integral nonvanishing implies mod-two nonvanishing.

Indeed, at a fold pair (p, r) the map $B(p, r)$ is homogeneous and (9.6) identifies it up to homotopy with the even isomorphism $N_r \text{cont}_G(p, r)n_p$. On nonzero integral homology this map cannot be both even and odd. Thus its degree is zero. The doubled configuration has the same index as the continuation problem, and the gauge reflection pairing is even; hence $d(p, r) = 0$. A common upper height datum and additivity give $d(p, q) = 0$. All uses of the integral cup formula in this argument precede the diagonal-cup identification and therefore do not use the continuation properties being proved.

With a coherent system define

$$\text{cont}_S(p, q) : CF(p) \longrightarrow CF(q)$$

by the signed zero-dimensional continuation count. It is even and commutes with the differential. The compactification of a one-dimensional continuation moduli space proves this exactly as in Proposition 6.1.

Proposition 12.2. *The continuation maps satisfy, up to sign and chain homotopy,*

$$(12.1) \quad \text{cont}_S(q, r) \text{cont}_S(p, q) \simeq \pm \text{cont}_S(p, r),$$

$$(12.2) \quad \text{cont}_S(p, p) \simeq \pm 1.$$

Their homotopy classes are independent of the continuation datum. They are homotopy equivalences.

Proof. For a long strip neck, ordinary gluing identifies the zero-dimensional solutions with pairs of continuation solutions. The determinant gluing is coherent, so the signed matrix is the composite matrix. The glued orientation system has the same end systems as the chosen direct continuation system. Lemma 12.1 compares them by one sign. Varying the neck length gives the chain homotopy in (12.1).

A homotopy of continuation data, constant near the ends, gives the corresponding homotopy by its oriented one-dimensional boundary. The pure symplectic compactness and gluing used here are Proposition 2.1(c). For constant data the zero-dimensional solutions are constant strips, with canonical positive orientation. This normalizes the unit. Apply composition to a datum and its reverse to obtain a homotopy inverse. $\square$

12.3. Reflection of continuation. There is no translation quotient for a continuation configuration. Reflecting it therefore has the graded transpose sign but no additional translation sign. Using the pairings of Proposition 11.4 gives

$$(12.3) \quad \text{cont}_S(\bar{p}, \bar{q}) \simeq \pm \Phi_q^{-1} \text{cont}_S(q, p)^\vee \Phi_p.$$

To verify coherence of the reflected system, glue an input strip, a continuation configuration, and an output strip, and reflect the three pieces in reverse order. If their indices are n_u, n_c, n_w , the reordering sign is $(-1)^{n_u n_c + n_u n_w + n_c n_w}$. Combining it with the three transpose signs leaves $(-1)^{n_u d(p, q)} = 1$. The generator corrections κ_p and κ_q identify the reflected end systems with the own transported systems. Rigidity then gives the one sign in (12.3). This argument also shows why parity zero is needed here.

12.4. Identification of the diagonal cup. For $k = 0$ the compression correspondence is the diagonal. Double the folded triangle, glue the two distinguished caps, and unfold along that diagonal. The resulting surface is a continuation strip. This is the geometric identification used in Paper I. To obtain an integral statement we compare its coherent systems before counting.

Proposition 12.3 (Diagonal cup). *At every admissible fold pair (p, r) ,*

$$(12.4) \quad B(p, r) \hat{\Phi}_p \simeq \pm \text{cont}_S(p, r).$$

Proof. Use both diagrams of Lemma 11.1. After unfolding, the incoming system is the graded dual of the own transported system on $\bar{p}$. Its comparison with the system on p is exactly the dual configuration-level ratio defining κ_p . Indeed for every strip from x to y that ratio is $\kappa_p(x) \kappa_p(y)$; these are precisely the two endpoint corrections in $\hat{\Phi}_p$. On a homogeneous generator its coefficient is $(-1)^{|x|} \kappa_p(x)$, since the graded evaluation map enters

(11.4). Bending by $B_{CF(\bar{p})}$ supplies the second factor $(-1)^{|x|}$; the two evaluation signs cancel. Thus precomposition by $\widehat{\Phi}_p$ changes the unfolded input system into the prescribed system on p . The outgoing system is already the prescribed one on r . This is a statement for every configuration class and every strip index, not only for differential trajectories.

The pulled-back capped system and the chosen continuation system therefore have the same two end systems. Their relative sign is constant under the two gluing operations in the diagrams. Rigidity (Lemma 12.1) applies and leaves one scalar sign. A nonconstant generator-sign automorphism would change one of those configuration-level gluing maps; a grading operator would change an odd-index gluing map. Both have been fixed before invoking rigidity. This proves the required comparison of coherent systems.

For sufficiently long cap necks, the gluing theorem gives a bijection of zero-dimensional solutions and preserves these orientations. A cap of index zero has one regular solution with unit coefficient, so no extra multiplicity occurs. The counts are therefore the same up to that sign. Varying the cap length and the triangle data to the ones defining $B(p, r)$ gives a chain homotopy through the one-dimensional moduli spaces. The preglue-to-solution path is within the pure section chart, and determinant transport on that path preserves the canonical orientation of a regular solution. This is the pure symplectic gluing input of Proposition 2.1(c). $\square$

This completes Theorem C. The statement uses the compression cup of the manuscript. It does not replace continuation by an inverse of a chosen mixed map.

12.5. Objects with vanishing homology. Suppose $H(C_G(M)) = 0$. Lemma 3.1 and Proposition 6.3 make both $C_G(M, p)$ and $CF(M, p)$ contractible for every presentation. Every closed map from or to a contractible complex is a boundary: compose it with a contracting homotopy and use (3.2). Hence there is exactly one morphism between it and any other object in the homotopy category.

At such an object we take the continuation and reflection maps needed for the comparison to be zero representatives of those unique morphisms. In particular a reflection map here is a homotopy equivalence between zero objects, not an invertible matrix. We assert neither character constancy nor existence of the same configuration-level reflection pairing at this object. Every identity involving it is interpreted in the homotopy category.

This convention also handles a word passing through an acyclic intermediate object: its class is zero, as is the corresponding gauge composite. Reduction modulo two preserves a contracting homotopy. Thus Paper I's maps incident to that object are null-homotopic as well.

Nonzero mod-two gauge homology implies nonzero integral gauge homology, since an acyclic finite free integral complex would be contractible. The standard mod-two framed calculation therefore places rational homology spheres and $\#^k(S^1 \times S^2)$ in the nonvanishing case [13, Corollary 1.4 and Section 7.8]. We do not use the converse implication: a nonzero finite odd-torsion integral group need not be detected modulo two.

13. CHOICES, PRIMARY NATURALITY AND HANDLE MAPS

13.1. Order of the auxiliary choices. All choices in a fixed comparison involve finitely many presentations. The following order avoids requiring a perturbation to be simultaneously small relative to a parameter chosen after it.

- (1) Fix the splittings, metric data, paired gauge bases and trace-coordinate families for all objects and their reflections. Choose a common primary regularity threshold r_H for this finite set, as in Proposition 10.2.
- (2) Choose the first auxiliary heights t_*, t'_* so their height perturbations and small generic corrections have norm less than r_H . These data are used as upper auxiliary presentations.
- (3) Choose the character-constancy radius r_* below r_H and inside the small-primary neighborhoods required for those heights. Use the same choice for a presentation and its reflection.
- (4) Choose smaller heights t'' for the direct duality argument, t_f for the reflection fold, and t_C for the compression cup. Their height data lie inside the corresponding balls. Choose the lower datum of a cup after t_C , sufficiently small relative to that height.
- (5) Fix the ordinary gauge trajectory data and the symplectic strip-end data for every required end, including the intermediate closed cuts. Make the simultaneous regular end choices before selecting their weights. Choose a common sufficiently small weight below the relevant primal, adjoint and scalar thresholds. For configuration reference paths it must also lie below their remainder rates; relative end preparation then supplies faster reference paths when needed. The completed-domain estimates at these choices are Proposition 2.1(b).
- (6) Select the finite compact reference families needed for coherence, and regularize the limiting vertex, triangle and completed-core problems with these fixed ends. Their primary transport, finite stabilizations and compact secondary supports are fixed before taking a geometric limit. In particular an auxiliary upper datum is a full presentation with its own vertex-map thresholds. Keep the collapsing compression unperturbed, including the dependence regions of the holonomy terms.
- (7) Choose the ordinary necks after these limiting data have been fixed. Then choose the collapse scales from the bounds for the compact reference families and the finite limiting index-zero solution sets, with their decay rates and the already fixed weights. For simultaneous determinant comparisons use the cofinal neck-and-scale regime in Lemma 7.1. No uniform threshold for all independent neck and collapse parameters is used.
- (8) At the resulting positive geometric parameters choose any further compactly supported secondary regularizers within the openness thresholds for the endpoint counts, preserving every frozen end equation. Extend them relatively over the comparison families. If an end datum or a limiting regularization is changed, reselect the dependent necks and collapse scales and use the corresponding data homotopies. Independence over the full permitted weight range is applied only after this common small-weight construction, as in Proposition 2.1(c).

Every choice has a positive threshold once the preceding finite data have been fixed. Auxiliary upper data may lie outside the final character ball; the folded comparison supplies their coherent continuation system directly. The lower data and the handle target where reflection is used lie inside the relevant ball.

13.2. Small-primary naturality. We now prove the comparison for changes of primary data. All maps in the next two propositions are even.

Proposition 13.1. *For regular p, q in one admissible ball,*

$$(13.1) \quad N_q \text{cont}_G(p, q) \simeq \pm \text{cont}_S(p, q) N_p.$$

Proof. Choose one upper height presentation a of $-M$ generic for $\bar{p}$ and $\bar{q}$. Apply the cup formula and Proposition 12.3 to both fold pairs. They give

$$\begin{aligned}\text{cont}_S(\bar{p}, a) &\simeq \pm N_a \text{cont}_G(\bar{p}, a) n_{\bar{p}} \widehat{\Phi}_{\bar{p}}, \\ \text{cont}_S(\bar{q}, a) &\simeq \pm N_a \text{cont}_G(\bar{q}, a) n_{\bar{q}} \widehat{\Phi}_{\bar{q}}.\end{aligned}$$

All factors being eliminated are homotopy equivalences. Cancel the common upper datum using continuation composition. Thus

$$\begin{aligned}\text{cont}_S(\bar{p}, \bar{q}) &\simeq \pm (n_{\bar{q}} \widehat{\Phi}_{\bar{q}})^{-1} \text{cont}_G(\bar{p}, \bar{q}) \\ &\quad \cdot n_{\bar{p}} \widehat{\Phi}_{\bar{p}}.\end{aligned}$$

Use the reflection formula (12.3) on the symplectic side and its gauge analogue from (4.5). The symmetric conventions give

$$\widehat{\Phi}_{\bar{p}} = \Phi_p, \quad \Phi_{G,M} n_{\bar{p}} = N_p^\vee,$$

and the same equalities at q . Cancelling these pairings transforms the displayed identity into

$$\text{cont}_S(q, p)^\vee \simeq \pm (N_p \text{cont}_G(q, p) N_q^{-1})^\vee.$$

Transpose and rename p, q to obtain (13.1). Every double-dual map occurs together with its inverse. Because the maps here are even, the cancellations introduce no grading-dependent sign. $\square$

This proof uses the same pairing in the diagonal-cup and reflection identities. Choosing an unrelated chain-level duality isomorphism would leave a diagonal automorphism after the cancellation.

13.3. Mixed duality. Put $N_p^\dagger = n_p \widehat{\Phi}_p$. This is the transpose of the reflected mixed map, with the gauge and symplectic pairings inserted.

Proposition 13.2. *For every regular p in the admissible ball,*

$$(13.2) \quad N_p N_p^\dagger \simeq \pm 1.$$

Proof. First take p small enough for a fold at the height t'' , with upper datum r in the same ball. The two cup identities give

$$\text{cont}_S(p, r) \simeq \pm N_r \text{cont}_G(p, r) n_p \widehat{\Phi}_p.$$

By primary naturality the right side is homotopic, up to sign, to $\text{cont}_S(p, r) N_p N_p^\dagger$. Cancel the continuation equivalence.

For a general datum q in the ball, transpose primary naturality for the reflected presentations. The symmetric pairings give

$$N_q^\dagger \text{cont}_S(p, q) \simeq \pm \text{cont}_G(p, q) N_p^\dagger.$$

Combine this with primary naturality and (13.2) at p :

$$N_q N_q^\dagger \text{cont}_S(p, q) \simeq \pm \text{cont}_S(p, q).$$

Cancel the continuation once more. The contractible case has the interpretation of Section 12.5. $\square$

13.4. One- and three-handle squares. At the compatible handle data define the integral compression cup by

$$(13.3) \quad U_C = \check{U}_C \widehat{\Phi}_p : CF(p) \longrightarrow CF(q), \quad q = C \circ p.$$

Define the cap by reflected transposition. If $\bar{C}$ is the reflected compression and $\check{U}_{\bar{C}} : CF(p)^\vee \rightarrow CF(\bar{q})$ is its unpaired cup, put

$$(13.4) \quad \check{V}_C = \text{ev}_{CF(p)}^{-1}(\check{U}_{\bar{C}})^\vee, \quad V_C = \check{V}_C \widehat{\Phi}_q.$$

Then $V_C : CF(q) \rightarrow CF(p)$. Equivalently it is the transpose of the paired reflected cup with the two pairings inserted. These are the orientations of the compression cup and cap counts of Paper I. At other presentations they are conjugated by the geometric continuation maps.

Theorem 13.3. *For the one-handle cobordism W_k and its corresponding three-handle cobordism W_3 , one has*

$$(13.5) \quad U_C N_p \simeq \pm N_q I_J^\#(W_k),$$

$$(13.6) \quad V_C N_q \simeq \pm N_p I_J^\#(W_3).$$

The identities hold at all admissible endpoint presentations.

Proof. The cup formula and mixed duality give

$$U_C N_p \simeq \pm N_q I_J^\#(W_k) n_p \widehat{\Phi}_p N_p \simeq \pm N_q I_J^\#(W_k),$$

which proves (13.5). For the cap, transpose the cup formula at the reflected compression. The graded evaluation maps and $\Phi_{G,-M} = \Phi_{G,M}^\vee \text{ev}$ give the gauge transpose of the reversed handle map. Mixed duality at q cancels the reflected mixed map. In detail, in the homotopy category,

$$\begin{aligned} V_C N_q &= \check{V}_C \widehat{\Phi}_q N_q \\ &\simeq \pm \check{V}_C n_q^{-1} \\ &\simeq \pm N_p \Phi_{G,-M}^{-1} I_J^\#(\overline{W}_3)^\vee \Phi_{G,-M} \\ &\simeq \pm N_p I_J^\#(W_3). \end{aligned}$$

The only potentially odd map in the transpose comparison is the handle map; all pairings and mixed maps are even. Its degree is retained in the graded transpose, so any remaining sign is scalar.

Finally conjugate both handle maps by continuations from arbitrary admissible endpoints to the compatible handle data. Primary naturality moves the mixed maps past these continuations, and gauge functoriality identifies the gauge composite with the same cobordism. Homotopies compose by (3.2). $\square$

14. DESCENT TO COBORDISMS AND PROOF OF THE MAIN THEOREM

14.1. The remaining changes of presentation. The geometric presentation moves are the ones in Paper I. We record how the integral orientations enter them. A change of regularizing data uses the family defining Proposition 6.3. A change of strip-end almost-complex structure uses Proposition 6.4(ii), with its actual outgoing continuation map and the fixed transported end systems. For a change of gauge base, both the gauge and symplectic generator lines are converted by the same right-placed maps. The reference constant pair has no extra line, so its orientation comparison commutes with this conversion.

An isomorphism of decorated data pulls back the equation, its linearization and the shifting construction; the pullback of a shift is a shift, and the orientation maps do not depend on the shifting path (Proposition 5.4). Choose pulled-back stabilizations in the determinant construction. The kernel gluing maps then commute with the isomorphism, and the mixed square commutes exactly on chains at the pulled-back data. A subsequent continuation gives the square at the originally chosen data, up to sign and homotopy. This applies to frame changes and diffeomorphisms preserving the torus decoration.

A Heegaard stabilization is realized by the cancelling one- and two-handle pair of Paper I. Compose (13.5) and (7.11). The decorated gauge cobordism is the identity, so its map is continuation up to sign and homotopy. The reverse stabilization is treated by the cancelling two- and three-handle pair. Passing through a common stabilization treats changes of metric data and isotopies of the splitting. The order of choices in Section 13 allows the new vertices to satisfy the integral height and ball conditions. No orientation conclusion is inferred from cancellation of unsigned counts alone.

Thus every elementary presentation arrow $e : p \rightarrow q$ has a symplectic map S_e constructed from the curves of Paper I and a square

$$(14.1) \quad N_q G_e \simeq \pm S_e N_p.$$

The gauge arrows G_e represent the relevant decorated cobordism or identity continuation. This includes both even presentation moves and possibly odd handle arrows.

14.2. Projective descent.

Lemma 14.1. *Suppose a connected presentation category maps to a cobordism category and every cobordism has an elementary word. Let G_e be a projective gauge functor on these words. Suppose N_p are isomorphisms in a homotopy category and the elementary maps satisfy (14.1). Then S_d depends only on the represented cobordism and the endpoint presentations, up to sign and homotopy.*

Proof. For a word $d = e_m \cdots e_1$, compose the squares to obtain

$$S_d N_{p_0} \simeq \pm N_{p_m} G_{e_m} \cdots G_{e_1}.$$

Every composition of homotopies is valid by (3.2). If d' represents the same cobordism, gauge functoriality identifies the two gauge words up to sign and homotopy. Cancel N_{p_0} to identify S_d and $S_{d'}$.

For words representing the identity cobordism this also gives the transitive system of comparison maps between presentations. The identity word is the identity map, and concatenation of words gives composition up to sign. Hence these systems descend to a projective functor and the classes of N_p give a projective natural isomorphism. $\square$

Proof of Theorem A. Proposition 6.3 gives the integral mixed isomorphisms. Equation (7.11), Theorem 13.3, and the presentation comparisons above supply all elementary squares. Generation and connectivity are the geometric presentation results of Paper I, with the additional smallness conditions arranged as in Section 13. Proposition 4.4 and Lemma 14.1 therefore give (1.1) and the projective functoriality.

The mixed maps preserve the chosen object gradings. On homology, the comparison with the homogeneous gauge map therefore gives the asserted parity of the symplectic map. No separate assertion of homogeneity for every chosen chain representative of a long word is needed.

Modulo two all determinant orientations give the same counts. The conversions become the identity on the generator basis, the signs in the tensor and dual formulas disappear, and the pairing (11.2) becomes the paired-basis reflection pairing. Thus the geometric maps reduce to those of Paper I at the same data. At an acyclic object the zero representatives reduce to maps homotopic to those of Paper I by Section 12.5. $\square$

14.3. Scope and further questions. Theorem A assumes the unsigned analytic results of Paper I. The additional analysis used here, the linear foundations of Proposition 2.1 and the relative preparation and shifting of Proposition 2.2, is proved in Appendices A and B, and the oriented torus comparison is derived in Section 8. The ordinary nonseparating self-gluing is supplied by Proposition 2.3; its regularization uses only collar-supported ordinary gauge perturbations.

Several further extensions ask different questions. Exact composition signs require more than projectivizing the maps; the regrading discussed after (4.6) indicates the extra grading information involved. An identification with integral homology-oriented instanton maps requires a comparison of orientation systems on the framed quotient. An identification with integral quilted Floer maps requires a comparison with relative-spin orientations on the compression correspondences. None of these identifications follows from the integral theorem by reducing modulo two.

APPENDIX A. ANALYTIC FOUNDATIONS

This appendix proves Proposition 2.1. All references to Paper I mean the companion version accompanying this manuscript. The present paper does not supply an independent proof of that manuscript’s unsigned collapse, torus excision, or generation results; it uses them as they are stated there. The arguments below combine the local theory of [1, 2] with explicit end models. They are the local-plus-ends arguments of [1, Lemma 4.34], carried out on the domains of this paper. Theorem, proposition and remark numbers in [2] are those of arXiv:2109.07038v1.

A.1. Local theory and end models. *Local estimates.* On a compact region meeting the matching seam, with a product metric and the fixed almost-complex structure near the seam, the local estimates for the mixed operator are [1, Theorem 4.17]; they are uniform along C^∞ -convergent sequences of configurations by [1, Remark 4.19]. For domain-dependent almost-complex structures, and for the formal adjoint, the local estimates and weak regularity are [2, Theorem 5], with [1, Remark 4.33]; their uniformity on neighbourhoods of the configuration and of the data, for both the operator and its adjoint, is [2, Remarks 5.73 and 5.76]. Away from the seam these are the interior and boundary estimates for the anti-self-duality operator with the normal boundary condition and for real Cauchy–Riemann operators with totally real boundary conditions. Holonomy perturbations contribute zeroth-order terms with the bounds of [1, Proposition 6.1]. The conformal factor of Definition 5.1 does not change the anti-self-duality equation; it rescales the scalar row by f^{-2} and adds zeroth-order terms to it. A compact family in the topology of Section 2 has coefficients converging in every C^k norm on compact sets, so these estimates are uniform on it.

Charts. At a smooth configuration the matching-boundary chart is built from [1, Lemmas 4.1, 4.2 and 4.4] and the construction following Lemma 4.4, with the slice of [1, Proposition 4.6] and the L^q radius of Paper I, Remark 5.2; the L^2 radius stated in [1, Lemma 4.1]

does not suffice. The construction is local, so on a compact family the charts vary continuously with their centers. Charts with a radius uniform as a torus length tends to infinity are used only at the preglued references of Paper I, Definition 10.23 and Lemmas 10.27–10.28.

End models. (O) At an ordinary gauge or strip end the operator is translation invariant with invertible asymptotic operator, the extended Hessian at a nondegenerate irreducible critical point or $J\partial_t$ with transverse boundary conditions. Weighted estimates hold for weights below the gap, by [4, Section 3.3.1] for gauge ends and by the same Fourier argument for strip ends. (E) At a mixed end the stationary operator is self-adjoint with kernel the tangent space of the primary critical family and a spectral gap [1, Section 4.2.1 and Proposition 4.21]. On the torus-extended slices Y_R and the joined slices $D_{i,\lambda}$ the gap is uniform for lengths bounded below (Paper I, Proposition 10.7), and on compact ranges of lengths by continuity (proof of Paper I, Proposition 10.17). The moving boundary domain is transported to a fixed one by [2, Proposition 5.27 and Lemma 5.67], and the weighted estimate for weights in $(0, \gamma)$ follows by Fourier transform in the end coordinate, as in Step 1(E) of the proof of Paper I, Proposition 10.26. (T) On a flat torus region the operator and its adjoint are the invertible operators of Paper I, Proposition 10.3, with the stated weighted bounds; at a flat face the normal derivatives are recovered from the equation as in Paper I, Theorem 10.5(b).

A.2. Proof of Proposition 2.1(a).

Lemma A.1. *Let $\mathfrak{F}$ be a compact family as in Proposition 2.1. There are a compact subset K of the cores and a constant C such that, for every member and every x in the domain of its realized operator D and every η in the domain of its weak adjoint,*

$$\|x\| \leq C(\|Dx\| + \|x\|_{L^2(K)} + |p(x)|), \quad \|\eta\| \leq C(\|D^*\eta\| + \|\eta\|_{L^2(K)}),$$

where $p(x)$ denotes the stationary coordinates.

Proof. Choose cutoffs subordinate to a fixed compact core, the ordinary ends, the mixed ends and, on torus domains, the torus region beyond a fixed radius. Their gradients can be made small on the noncompact overlaps, as in Step 2 of the proof of Paper I, Proposition 10.26. On each end apply the corresponding model of Section A.1; the coefficients differ from the model by terms which are small beyond a common truncation, uniformly on $\mathfrak{F}$. On the core apply the local estimates. Summing and absorbing the small commutator and coefficient terms gives the first estimate for the decaying part; the finite stationary columns and the scalar row on them are bounded, which accounts for $|p(x)|$. The adjoint estimate is the same argument with the adjoint models, whose invertibility is recorded in Section A.1.

Varying geometry is handled as follows. The collar identifications ψ_R of Paper I, Definition 10.1(a), are the identity outside the region N_X , which is compact in the normal direction and extends along both mixed ends; they change the mixed-end slices from Y to Y_R . The width identifications in the proof of Paper I, Proposition 10.17, are the identity outside the bridge. Pulled back by them, the metrics form a compact family of uniformly elliptic metrics, and on the mixed ends model (E) is applied on the slices Y_R and $D_{i,\lambda}$ for lengths in the given compact range. At positive collapse scale, identify X_ε , $\varepsilon \in [\varepsilon_1, \varepsilon_2]$, with X_{ε_1} by the insertion coordinates of Paper I, Definition 6.2; each member is a noncollapsing domain (Paper I, Proposition 6.9). The zero-scale limit domains are obtained, in the proofs of Paper I, Theorems 9.10 and 11.8, by joining DFL-type domains of compression pairs, with the new mixed ends of Paper I, Proposition 6.9, and a polygon domain along ordinary strip ends; the joined domain has a compact core, and the argument above applies to it directly, with model (E) at the new ends. On the completed torus domains a compact family is in torus normal

form after the gauge transformations of Lemma 8.6(i), and model (T) applies beyond a fixed radius. $\square$

Proof of Proposition 2.1(a). Charts are recorded in Section A.1. By Lemma A.1 and Rellich’s theorem on K , each realized operator has finite-dimensional kernel and closed range, and so does its adjoint; hence it is Fredholm with cokernel the kernel of the weak adjoint. Local regularity for the operator and its weak adjoint makes these elements smooth. After transport to fixed spaces by the boundary maps of Section A.1 and the domain identifications above, the coefficients vary continuously in operator norm on the family, so the operators form continuous Fredholm families and their determinant lines form continuous line bundles. For the $W^{1,p} \rightarrow L^p$ realization, $p > 2$, of the Cauchy–Riemann summand used in unfolding in Section 11, the standard L^p theory of real Cauchy–Riemann operators with totally real boundary conditions and nondegenerate ends gives the Fredholm property, and elliptic regularity identifies its kernel and cokernel with those of the Hilbert realization. The comparison of realizations is Lemma 5.2; its paths are compact families of this kind, and its proof uses only the Fredholm and continuity statements just proved. The affine-center comparison is Proposition 7.3. $\square$

A.3. Proof of Proposition 2.1(b). The end critical sets are finite and nondegenerate, and the primary critical families are compact and regular. Hence there is a common gap for the finitely many end models of a compact family, and a common small weight below all gaps and scalar thresholds; at mixed ends the stationary coordinates are split off before the weight is imposed [1, Section 4.2.1].

The scalar operator. The Neumann operator $d_A^* d_A$ is Fredholm by the argument of Lemma A.1: its end models have positive gaps by irreducibility of the end connections, and on the torus region the fibre operator has mass μ_T because $H^0(T; \text{ad } \alpha) = 0$. Integration by parts identifies its kernel at weight zero with parallel sections, which vanish on an irreducible collar; it is self-adjoint of index zero there, and sufficiently small weights preserve the index and add no kernel. This is the argument in the proof of Paper I, Theorem 5.6(ii). Near the preglued references the uniform version is Paper I, Lemma 10.25.

A regular zero-dimensional solution has an invertible realized operator by definition, and this does not depend on the realization by Lemma 5.2. Uniformity for the preglued families as the torus lengths tend to infinity is Lemma 8.7. At positive collapse scale the estimates are those of Lemma A.1 on compact scale intervals; nothing uniform as the scale tends to zero is claimed beyond Lemma 7.1. The constant reference pairs are treated in the next lemma. For small secondary or inhomogeneous data the operators remain invertible by openness, with the bounds of Lemma A.1.

Lemma A.2. *The constant reference pairs, with no secondary or inhomogeneous data, have invertible physical operators on every DFL-type domain of a transverse compression pair in the class of Paper I, Lemma 9.7 (including the vertex domain and the truncated output regions of the triangles), on the separate domains $W_{a,R}$, $R \in [0, \infty]$, on the joined domains W_λ^J , $\lambda > 0$, and on the closed-cut piece M_λ , including along the family used in the proof of Paper I, Lemma 9.7, as applied in Paper I, Proposition 10.21. The inverse bounds are uniform on compact families of lengths, and the positive elements extend continuously through torus completion.*

Proof. The quadratic identity. Write d_A^h for the perturbed exterior derivative of [1, (4.42)], which adds the primary Hessian term on the compression cylinders and equals d_A on the flat

genus and torus regions. For every physical variation (ζ, ν) satisfying the linearized matching and arc conditions, with the stationary behavior of the realization,

$$(A.1) \quad \int_X \operatorname{tr}(d_A^h \zeta \wedge d_A^h \zeta) + \int_U \Omega(d\nu, d\nu) = 0.$$

On the original pieces this is [1, Lemma 4.40]: the interior terms are total derivatives by the symmetry of the primary Hessian and the primary equation, the matching terms cancel because the surface trace is closed with class the section value, and the arc terms vanish on the Lagrangian tangent spaces. A primary tangent vector satisfies the linearized primary equation, so its pure stationary contribution at a truncated mixed end vanishes; the terms involving the decaying remainder tend to zero with the truncation, and the ordinary ends contribute zero. The flat torus region contributes only Stokes terms. At a finite outer torus face the conditions are $\zeta(\partial_r) = 0$ and $d_\alpha(\zeta|_T) = 0$ (Paper I, Definition 10.23). Write the trace as $a + \psi d\theta$ with $d_\alpha a = 0$; since $H^1(T; \operatorname{ad} \alpha) = 0$ we have $a = d_\alpha f$, and $\int_T \operatorname{tr}(a \wedge \partial_\theta a) = \int_T \operatorname{tr}(d_\alpha f \wedge d_\alpha \partial_\theta f) = 0$ by Stokes on T . The cross term $\int_T \operatorname{tr}(a \wedge d_T \psi)$ vanishes by Stokes on T because $d_\alpha a = 0$, as in [1, (4.52)]. On a completed domain use spatial cutoffs equal to one on all primary supports, whose gradients are of size C/R on the flat tail; the resulting errors tend to zero because the tail field is in H^1 . On a joined domain the bridge has no boundary, and its interfaces cancel those of the two cores. On the closed-cut piece the ordinary input Z contributes nothing.

The kernel. Let x be in the kernel and let (ζ, ν) be its physical variation. The mixed equations give $(d_A^h \zeta)^+ = 0$ and the linear Cauchy–Riemann equation for ν , so both integrands in (A.1) are nonnegative with the trace convention of [1]. Hence $d_A^h \zeta = 0$ and $d\nu = 0$. The section variation decays at its strip output, so $\nu = 0$ on each component. The section limit of the stationary coordinates is their image in the tangent space of the primary Lagrangian, and the restriction map is an embedding, so all stationary coordinates vanish. What remains is a decaying ζ with $d_A^h \zeta = 0$, $d_A^* \zeta = 0$ and the normal and face conditions.

A primitive. Construct η with $d_A \eta = \zeta$ as in [1, Proposition 4.54]. On each compression cylinder take the temporal primitive from the gauge input; it converges because the kernel field solves the full gauge-fixed equation on the input cylinder, whose asymptotic operator is nondegenerate, so the field decays exponentially there. The primary equation and Hessian symmetry make it a primitive of the spatial part. On a flat genus product the surface class of ζ is $\nu = 0$, and the surface scalar inverse gives the primitive; the base components agree because a parallel adjoint scalar on an odd surface vanishes. On the flat torus region put $\eta_T = \Delta_\alpha^{-1} d_\alpha^* \zeta_T$ fibrewise. Since $H^1(T; \operatorname{ad} \alpha) = 0$ this is a primitive of the fibre part, and since $H^0(T; \operatorname{ad} \alpha) = 0$ the base components agree. The primitives agree on overlaps for the same reason, also across a joined bridge. The fibre estimate $\|\eta_T\| \leq \mu_T^{-1} \|\zeta_T\|$, the scalar spatial estimates on the mixed ends and the irreducible gauge input put η in the weighted H^1 space. Integration by parts with exhausting cutoffs gives $\|\zeta\|^2 = \langle d_A \eta, \zeta \rangle = \langle \eta, d_A^* \zeta \rangle = 0$; the boundary term vanishes by the normal and face conditions. Thus the kernel is zero.

The index. On the special quintuple [1, Proposition 3.19] gives index zero for the realization of [1]. That realization and the one of Definition 5.1 use the same decaying space and the same stationary parameter space; their stationary columns, including the scalar ones, differ by a bounded finite-rank map, so the indices agree (as in Paper I, the remark after Definition 10.23); a change of weight inside the gap does not change the index (Lemma 5.2). If the almost-complex structure near the matching line is not the one of [1], a homotopy of compatible structures localized in the seam collar keeps the constant pair a solution and the operators Fredholm, by Lemma A.1. For a DFL-type domain in the class of Paper I,

Lemma 9.7, and for the closed-cut piece along the family in the proof of that lemma, the metric changes are conformal on the ends or lie in flat regions, and the flat references do not depend on the end metric, so the constant pair remains a solution and the index is constant along the family. For $W_{a,R}$ with finite R the index is constant along the collar-length family (Paper I, Theorem 10.13(i)). By Lemma 8.6(iii), which does not use the present lemma, $P_R(C_{a,x})$ is the constant pair on $W_{a,R}$ and $P_\lambda(C_{-,x_-}, C_{+,x_+}) = C_{J,x}^{(\lambda)}$. Hence Lemma 8.7(ii) gives $\text{ind } D_\infty = \text{ind } D_R = 0$ for large R , and the index at $C_{J,x}^{(\lambda)}$ is the value 0 at $\lambda = \infty$ for large λ ; it is constant in $\lambda \in (0, \infty)$ by Proposition 2.1(a). In all cases the kernel is zero and the index is zero, so the operator is invertible.

Uniformity. On a compact family of finite lengths the inverses vary continuously. Through torus completion the positive elements are continuous by Lemma 8.7(iii), and uniform inverse bounds for large lengths follow from its part (i) with $E = 0$. $\square$

A.4. Proof of Proposition 2.1(c). The weight comparison is Proposition 5.3.

Compactness and gluing of pure moduli spaces. The compression-body Lagrangians are simply connected with $\pi_2 = 0$, and $\pi_2(M_h) \cong \mathbb{Z}$ is generated by a class of Maslov index four; in the product target the correspondence L_C has π_2 generated by spheres of zero area and zero Chern number (proof of Paper I, Lemma 6.7, and Section 10). The pairs and correspondences are therefore monotone with minimal Maslov number at least three, and sphere and disc bubbling has codimension at least two in the dimensions used. The compactness and transversality of the pure continuation, cap and strip moduli spaces is then [17, Theorem 3.9], and their gluing is the construction in the proof of [17, Theorem 3.13], with [12, Theorems 3.3 and 3.7] and [11] for the quilted disc families. The strip gluing and exponential decay are those of [14]; in the transverse case the full-strip statement is [14, Proposition 2.3.1]. For a finite collection of regular configurations with nondegenerate strip ends, every nearby zero-dimensional solution at a long neck is the unique correction of a preglued configuration, the correction tends to zero with the neck length, and the estimates are uniform over compact families. Along the path from the preglue to the solution the operators are invertible, by the uniform right inverse at the preglue and the smallness of the correction, as in the proof of Lemma 8.12; hence determinant transport preserves the canonical orientation. The unsigned low-dimensional counts agree with those of Paper I, Section 11.

Interior nodes. Linear gluing of Cauchy–Riemann determinant lines along strip ends is Lemma 3.3. For a family of real Cauchy–Riemann operators on surfaces degenerating at an interior node, the determinant lines form a topological line bundle [16, Proposition 2.4.12], and the line isomorphism is given by the exact sequence of [16, Remark 2.4.6 and Corollary 2.4.7]; its construction, in the proof of [16, Theorem 2.4.5], is local to the node disc, and it is associative [16, Proposition 2.4.8]. The node contributes its complex fibre, canonically oriented and even, with no additional sign [16, Theorem 4.3.3(a)]; numbering as in arXiv:1503.07803v6. These statements localize. Choose stabilizations of a mixed or pure operator supported away from a disc containing the node and separated from the matching and boundary collars. Then the stabilized kernel near the node is obtained by the local gluing map of the cited proposition, and the rest of the operator is unchanged. Hence the line bundle structure and the gluing isomorphism hold for the real Cauchy–Riemann part of any such operator. Associativity follows from the cited associativity, with disjoint supports. Commutation with ordinary strip gluing follows because the two stabilizations and cutoffs have disjoint supports, as in the proof of Lemma 8.7(iv). A cohomological composition identity alone is not used as a substitute for these moduli-space gluing statements.

A.5. Boundary Hodge theory and holonomy regularity. For a flat orthogonal bundle on a compact compression body, the input needed in Proposition 10.2 is the Hodge decomposition on $W^{1,2}$ with normal boundary condition. Its harmonic fields are smooth and finite dimensional, and its exact part is the covariant differential of $W^{2,2}$ sections. The Dirichlet Green operator for $d_B^* d_B$ preserves smoothness, so $1 - d_B G_D d_B^*$ is a smoothness-preserving projection onto coclosed forms. These statements follow from the flat covariant de Rham case of [10, Section 1.6 and Theorems 3.10–3.11]; the normal and tangential versions are interchanged by Hodge star.

The boundary elliptic estimate has the form

$$\|b\|_{W^{1,p}} \leq C(\|d_B b\|_{L^p} + \|d_B^* b\|_{L^p} + \|b\|_{L^p})$$

for $*b|_{\partial C} = 0$. For smooth fixed B , adjoint bundle coefficients add only bounded zeroth-order terms in local trivializations. The compactness and Coulomb-slice inputs are Theorems A, D(a) and F of the introductory statements in [15].

Finite holonomy functions have bounded gradients and Lipschitz gradient differences as used in (10.2) [1, Proposition 6.1]. Their higher derivatives obey the local tame estimates obtained by differentiating parallel transport. Combined with the boundary elliptic estimate, these give the bootstrap from the slice fixed point to the smooth perturbed-flat solution. For holonomy families with moving basepoints, restriction and composition are performed in the smooth topology, not asserted bounded on arbitrary low-order Sobolev spaces.

A.6. Shifting and self-gluing. Relative preparation and shifting are proved in Appendix B. Naturality under isomorphisms of the decorated data is automatic: the pullback of a shift is a shift, and the orientation maps are independent of the shifting path (Proposition 5.4).

For ordinary gauge self-gluing, Proposition 2.3 applies the low-dimensional instanton compactification, verifies matching on the prescribed bundle and identifies the lifting fiber. Its orientation comparison first traces over the full determinant-one torus complex and then takes the larger quotient. Together with the finite group calculation in Section 8, this gives integral unit multiplicity.

APPENDIX B. RELATIVE PREPARATION AND SHIFTING

This appendix proves Proposition 2.2. The one-input statement is the family version of [1, Lemmas 4.62 and 4.64], whose proof is sketched there; we give it in the topology of Section 2, jointly continuous in configurations and data, relative to the terminal subspace, and on the domains of this paper. Throughout, a family is a continuous map from a compact metrizable space K to actual configurations on a fixed bundle, together with continuously varying data. No quotient by gauge transformations is taken; the lifting groups enter only through Proposition 5.4.

The terminal subspace. For an output label x , a *representative* of the critical point over x is a connection $c = B \cup \text{pr}^* b \cup B'$ on $Y_\#$, where b is a flat connection on Σ over the point of $\mathcal{M}(\Sigma)$ determined by x and B, B' are primary solutions on Y_0, Y'_0 which are products with value b on the collars $[-5/2, -2] \times \Sigma$ of their boundaries fixed in Section B.2. Every such c is a critical point of the perturbed Chern–Simons functional in the gauge orbit of a_x or of its other lift, and the constant pair C^c built from it is gauge equivalent to C_x . Its operator is therefore invertible, its positive element corresponds to that of C_x under conjugation, and the almost-complex gauge orientations are invariant under the bundle automorphism. The lines at c are identified with those at a_x by this automorphism. The identification agrees with transport along any path of connections from a_x to c : the gauge line is orientable over closed

loops in the configuration space of $Y_\#$ [4, Section 5.4], and the character of the conversion line depends only on the lifting class (Proposition 4.2), so it is trivial for automorphisms of trivial lifting class, of any degree; an automorphism of nontrivial lifting class is the lifting action of Proposition 5.4. The terminal subspace $\mathcal{T}_x$ consists of the configurations which, outside the gauge input end, agree with C^c for some representative c ; they are gauge paths ending at c followed by C^c . Proposition 2.2 is to be read with this terminal subspace. It cannot be replaced by the configurations ending at one fixed C_x : those have a single topological energy, whereas attaching a sphere of nonzero area changes the topological energy of a configuration without changing its labels (a, x) .

B.1. Reduction to a special quintuple. Let (Y, Y') be a pair of odd compression bodies with common boundary Σ , a surface with two components on each of which the bundle is odd (Paper I, Definition 3.2), with primary data whose Lagrangians $L(Y), L(Y') \subset \mathcal{M}(\Sigma)$ are embedded (Paper I, Section 3.2). For the torus domains $\Sigma = \Sigma_h \sqcup T$; for the closed-cut piece and the product-target vertex pairs $\Sigma = \Sigma_h \sqcup \Sigma_g$. Let X_D, U_+ be the gauge region and section domain of the special quintuple [1, Section 3.1]: the gauge region is $\mathbb{R} \times Y_0 \cup U_- \times \Sigma \cup \mathbb{R} \times (-Y'_0)$, the matching seam is $U_\partial \times \Sigma$, the Lagrangian arcs η_+, η'_+ carry $L(Y)$ and $L(Y')$, and the gauge input end is $(-\infty, -3] \times Y_\#$. Every flat connection on Σ is irreducible with central stabilizer.

The configuration space of Section 2 depends only on the decorated smooth manifold (bundle, matching seam, arcs with their Lagrangians, ends with their limits) and on the primary data. It does not depend on the metric, the almost-complex structure or the secondary data: perturbed flatness involves no metric, and the weights are functions of the end coordinates. Each one-input domain of Proposition 2.2(a) with finite torus length is identified with a special quintuple by a diffeomorphism of decorated manifolds, depending continuously on the lengths and widths: for $W_{a,R}$ the map ψ_R of Paper I, Definition 10.1(a), which is independent of θ and is a translation near the flat face; for the closed-cut piece with input Z the identification of Paper I, Lemma 10.18(ii); for the vertex domains of Paper I, Lemma 9.7, the diffeomorphism of that lemma, which moves the seam and the conformal structure. The output regions used in part (b) are identified in Section B.5. It therefore suffices to construct the deformations on a special quintuple, for families in which the primary data vary in a compact set.

B.2. Preparation. Write $C_Y = [-2, 0]_s \times \Sigma$ for the collar of $\Sigma = \partial Y$ in Y which forms part of $U_- \times \Sigma$ on the mixed ends [1, Section 3.1], and $C_{Y'}$ likewise. It extends to a flat collar $[-3, 0] \times \Sigma$ whose part $[-3, -2] \times \Sigma$ in Y_0 is free of the primary perturbation, so that primary solutions are flat on it. The collars of ∂Y_0 used below are $[-9/4, -2] \times \Sigma$, in the normal form, and $[-5/2, -2] \times \Sigma$, in the definition of $\mathcal{T}_x$. We use the following normal form, with constants $\Theta \geq 2$ and $S \geq 3$:

- (N1) u is the constant map to x on $[S, \infty) \times [-1, 1]$;
- (N2) near η_+ and η'_+ , u is independent of the normal coordinate;
- (N3) on the mixed-end regions $|\theta| \geq \Theta$ the configuration is a stationary pair (π^*B, q) , (π'^*B', q') , where B equals pr^*b on $[-9/4, 0] \times \Sigma$, that is, on C_Y and on the collar of width $1/4$ of ∂Y_0 in Y_0 , for a flat connection b on Σ , and similarly for B' ;
- (N4) $u(s, \theta)$ is independent of s for $s \in [0, 1]$;
- (N5) on $[-1, 0] \times \mathbb{R}_\theta \times \Sigma$ the connection is the pullback of a connection $b(\theta) + \psi(\theta) d\theta$ on $\mathbb{R}_\theta \times \Sigma$ whose slices $b(\theta)$ are flat, with $\psi = 0$ for $|\theta| \geq \Theta$.

Lemma B.1. *For a family $f : K \rightarrow \mathcal{A}(a, x)$ on the special quintuple there are Θ, S and a deformation f_t , $t \in [0, 1]$, jointly continuous in (k, t) and in the data, with $f_0 = f$ and f_1 in normal form. It is the identity on the gauge input end $(-\infty, -3] \times Y_\#$. Members that are stationary on the original ends beyond Θ and S remain so, and members of $\mathcal{T}_x$ are fixed.*

Proof. Collar gauge at the mixed ends. Let B_q be the actual limit representative at the + mixed end; it is continuous in the family and flat on the flat collar. Let K be its temporal gauge along the collar, the solution of $\partial_s K = -(B_q)_s K$ on $[-3, 0]$ with $K(0) = 1$. Then $K^* B_q$ has no ds component on $[-3, 0] \times \Sigma$ and, being flat, is a product there. Choose a smooth $\sigma_N : [-3, 0] \rightarrow [-5/2, 0]$ with $\sigma_N(s) = s$ on $[-9/4, 0]$ and $\sigma_N = 0$ near -3 , and put $k_\lambda = K(\lambda \sigma_N(s))$, extended by 1. This is a path of gauge transformations from 1 to $k = k_1$, with $k_\lambda = 1$ at the seam for every λ , and $k^* B_q = \text{pr}^* b_q$ on $[-9/4, 0] \times \Sigma$, because $k = K$ there. Apply the gauge $k_{\lambda(\theta)}$ on the + mixed end, with $\lambda(\theta) = 0$ for $\theta \leq \Theta_0$ and 1 for $\theta \geq \Theta_0 + 1$, and do the same at the - end. Gauge transformations preserve the matching condition, and this one is the identity at the seam. If the limit representative is already a product on $[-5/2, 0] \times \Sigma$, then $K = 1$ on $[-5/2, 0]$, which contains the range of σ_N , so $k_\lambda = 1$; in particular members of $\mathcal{T}_x$ are fixed.

Ends. On the strip output end write $u = \exp_x \xi$ in an exponential chart at x in which both Lagrangians are linear, and put $u_t = \exp_x((1-t\beta)\xi)$ with a cutoff $\beta(s)$ equal to one for $s \geq S$. On a mixed end, beyond some $\Theta_1 \geq \Theta_0 + 1$ the configuration lies in the matching chart Ψ_{R_q} (Section A.1) centered at its stationary limit $R_q = (\pi^* k^* B_q, q)$: $c = \Psi_{R_q}(\zeta)$ with ζ in the linearized boundary conditions. Put $c_t = \Psi_{R_q}((1-t\beta(\theta))\zeta)$ with $\beta = 1$ for $\theta \geq \Theta$. The chart is fibrewise over the base, and the linear conditions are pointwise in θ , so they are preserved; the chart gives the nonlinear matching and Lagrangian conditions. A compact family has uniformly small tails in the topology of Section 2, so common Θ_1, Θ, S exist, and the charts depend continuously on the limits and the primary data. Members already stationary beyond Θ remain so.

Collars. Let $\phi_0 : [0, \infty) \rightarrow [0, \infty)$ be smooth and nondecreasing, with $\phi_0 = 0$ on $[0, 1]$ and $\phi_0(s) = s$ near and beyond 2, and put $\phi_t = (1-t)\text{id} + t\phi_0$. Pull back u by ϕ_t in the normal coordinate of the seam in U_+ , and A by $(s, \theta, y) \mapsto (-\phi_t(-s), \theta, y)$ on $[-2, 0] \times \mathbb{R} \times \Sigma$; since ϕ_t is the identity near 2, the result is smooth where $[-2, 0] \times \Sigma$ meets Y_0 . Since $\phi_t(0) = 0$, the seam values, hence the matching condition, are unchanged. Do the same in the normal coordinates of η_+, η'_+ , which preserves the arc values; along the strips $[0, 2] \times [2, \infty)$ the two pullbacks fix the seam and the arc pointwise and commute. At $t = 1$, (N2), (N4) and (N5) hold; stationarity beyond Θ is preserved, and so is (N3), because $\text{pr}^* b$ is independent of s . Members of $\mathcal{T}_x$ are independent of s on these collars and are fixed.

All three steps are explicit, continuous in the data, and supported away from the gauge input end. $\square$

B.3. Positive extension, shift and straightening. Use the reparametrizations $\Phi_\pm$ of [1, Section 4.4], chosen polar about the corners for $|\theta| \geq \Theta + 1$, preserving the lines of constant θ near the seam for $|\theta| \leq \Theta$, smooth in between, where the prepared configuration is stationary, and carrying the normal collars of η_+, η'_+ onto collars of $t = \pm 1$; DFL prescribe $\Phi_\pm$ only on the ends. In these coordinates the gauge region is $(-\infty, 0] \times Y_\#$ with $\{0\} \times Y_0$ and $\{0\} \times Y'_0$ removed, the section domain is $[0, \infty) \times [-1, 1]$ with the corners $\pm i$ removed, the matching seam is $\{0\} \times [-1, 1] \times \Sigma$ against $\{0\} \times [-1, 1]$, and the collars $C_Y, C_{Y'}$ of the mixed ends collapse to $\{0\} \times \partial Y_0$ and $\{0\} \times \partial Y'_0$, the collar coordinate becoming the angle about these corners. By (N3)–(N5) a prepared configuration is independent of the normal coordinate

near $r = 0$, it is pr^*b with no angular component near the corners, and its seam restriction is $b(\theta(t)) + \tilde{\psi}(t) dt$ with $\tilde{\psi}(t) = \psi(\theta(t))\theta'(t)$, which vanishes near $t = \pm 1$.

Lemma B.2. *Let $\mathcal{P}$ be the space of smooth primary solutions on Y_0 which are products on the collar $[-9/4, -2] \times \Sigma$ of ∂Y_0 , and let $\mathcal{S}$ be the space of smooth flat connections on Σ whose class lies in $L(Y)$; a torus component of Σ contributes only the point of $\mathcal{M}(T)$. Restriction $\rho : \mathcal{P} \rightarrow \mathcal{S}$ is a locally trivial fibre bundle, also over a compact family of primary data. A continuous family $a : K \times [0, \infty) \rightarrow \mathcal{S}$, smooth in r with all r -derivatives continuous, with a lift $B(\cdot, 0)$ of $a(\cdot, 0)$, has a lift B with the same regularity, constant in r where a is.*

Proof. Near $a_0 = \rho(B_0)$ choose a smooth local section $[a] \mapsto \hat{B}$ of $\mathcal{P} \rightarrow L(Y)$ by a primary slice; the primary moduli space is regular and its restriction map is an embedding (Paper I, Section 3.2). Then $\rho(\hat{B})$ and a represent the same class, so $\rho(\hat{B}) = g^*a$ for a gauge g near 1 on Σ , unique up to the center, by the surface Coulomb slice and irreducibility on both components. Extend g over the collar by $\exp(\beta \log g)$, with $\beta = 1$ on $[-9/4, -2] \times \Sigma$ and 0 at depth; this gives a local section with $\rho = a$. Thus ρ is locally trivial; its fibre is the space of gauge transformations of Y_0 which are central constants on the collar, modulo the center. On $K \times [0, \infty)$ patch the local sections with a partition of unity that is smooth in r to a connection on the pulled-back bundle; the horizontal lift solves an ordinary differential equation $g'g^{-1} = \xi$ pointwise on Y_0 , so it has the stated regularity and is constant where a is. $\square$

Positive extension. For a prepared family, let $\alpha(r, t)$ be the horizontal lift of $r \mapsto u(r, t)$ in flat connections on Σ , with $d_\alpha^* \partial_r \alpha = 0$ and initial value $b(\theta(t))$; on a torus component it is constant in r , since $H^1(T; \text{ad}) = 0$. It exists, is unique and depends smoothly on u , since the flat connections are irreducible; near $t = \pm 1$ it is independent of t by (N2). By Lemma B.2 choose $B(r)$ on Y_0 and $B'(r)$ on $-Y'_0$, products with values $\alpha(r, \pm 1)$ on the collars of (N3), with $B(0), B'(0)$ the limit representatives of (N3). Let A_u be the connection on $[0, \infty) \times Y_\#$ equal to $B(r)$ on Y_0 , $B'(r)$ on $-Y'_0$, and $\alpha(r, t) + \tilde{\psi}(t) dt$ on $[-1, 1] \times \Sigma$, with no dr component. By (N3)–(N5) the configuration and A_u are independent of r near $r = 0$ and agree there, so the connection $\tilde{A}$ on $\mathbb{R} \times Y_\#$ equal to A for $r \leq 0$ and to A_u for $r \geq 0$ is smooth. For $r \geq S$ it is independent of r ; write c_S for its restriction there, whose Σ -slices $\alpha(S, t)$ all represent the point determined by x .

Lemma B.3. *For $\tau \in [0, S + 4]$ put*

$$A_\tau(r, y) = \tilde{A}(r + \tau, y) \quad (r \leq 0), \quad u_\tau(r, t) = u(r + \tau, t).$$

*This is a family of configurations with labels (a, x) , jointly continuous in (k, τ) and the data, beginning at the prepared family, and it translates the gauge input end by τ . At $\tau = S + 4$ the region $r \geq -4$ carries pr^*c_S and $u_\tau \equiv x$. Members of $\mathcal{T}_x$ remain in $\mathcal{T}_x$ for all τ .*

Proof. At the seam the slice of A_τ over t is $\alpha(\tau, t)$, which represents $u(\tau, t) = u_\tau(0, t)$; the arcs carry $u(r + \tau, \pm 1) \in L$. Near a removed corner the pair is smooth in the reparametrized coordinates, with limit the stationary pair of $B(\tau)$, whose collar value is $\alpha(\tau, 1)$, a product. In the original end coordinate it converges to this stationary pair at the exponential rate one, which exceeds the weight δ ; the configuration therefore lies in the space of Section 2, with the mixed limit moving along the arc. Translation over a bounded range is continuous in that topology. For $\tau = S + 4$ and $r \geq -4$ we have $r + \tau \geq S$. If a member lies in $\mathcal{T}_x$, then $u \equiv x$ and its seam restriction is pr^*b , so $\alpha \equiv b$, $\tilde{\psi} = 0$, B, B' are constant, and $\tilde{A}$ is pr^*c on $[-3, \infty) \times Y_\#$; the translates keep this form. $\square$

Straightening. At $\tau = S + 4$ contract the t -dependence of c_S . The slices $\alpha(S, t)$ are independent of t for $|t| \geq 1 - \epsilon$, by (N2). Choose a smooth nondecreasing $\rho : [-1, 1] \rightarrow [-1, 1]$ with $\rho(t) = t$ for $|t| \leq 1 - \epsilon$ and $\rho = \pm 1$ near ± 1 . For $\lambda \in [0, 1]$ let c_λ be the connection on $Y_\#$ equal to $\alpha(S, (1 - \lambda)\rho(t)) + (1 - \lambda)\tilde{\psi}(t) dt$ on $[-1, 1] \times \Sigma$ and to the lifts B_λ, B'_λ of Lemma B.2 of the paths $\lambda \mapsto \alpha(S, \pm(1 - \lambda))$, starting at $B(S), B'(S)$. Near $t = \pm 1$ the slices are the constant values of these paths, so the pieces join smoothly, and at $\lambda = 0$ the connection is c_S . With a smooth χ equal to 0 for $r \leq -5$ and to 1 for $r \geq -4$, replace the connection A_{S+4} by $A_{S+4} + \chi(r)(\text{pr}^*c_\lambda - \text{pr}^*c_S)$ and keep $u \equiv x$. This is pr^*c_λ on $[-4, 0] \times Y_\#$ and the unchanged connection for $r \leq -5$; the interpolation lies in the gauge input end, where no boundary condition is imposed, and at $\lambda = 0$ nothing changes. The slices represent the point of x , so matching holds, and the mixed limits are stationary pairs of B_λ, B'_λ . At $\lambda = 1$ the connection $c_1 = B_1 \cup \text{pr}^*b \cup B'_1$ is independent of t on the neck, with B_1, B'_1 products on the collars of (N3). On members of $\mathcal{T}_x$ the slices $\alpha(S, \cdot)$ are constant and $\tilde{\psi} = 0$, so this step is the identity.

Collar widening. Let K_1 be the temporal gauge of $B_1 \cup \text{pr}^*b$ along the flat collar, as in the proof of Lemma B.1; it equals 1 on $[-9/4, 0]$. Choose a smooth $\sigma_W : [-3, 0] \rightarrow [-3, 0]$ with $\sigma_W(s) = s$ on $[-5/2, 0]$ and $\sigma_W = 0$ near -3 , put $g_\mu = K_1(\mu\sigma_W(s))$ on Y_0 , extended by 1, and define g'_μ on Y'_0 likewise. For $\mu \in [0, 1]$ these gauge transformations equal 1 on $[-9/4, 0] \times \Sigma$, hence near the neck and the seam, and $g_1^*B_1$ is a product on $[-5/2, -2] \times \Sigma$. Apply the gauge transformation $g_{\mu\chi(r)} \cup 1 \cup g'_{\mu\chi(r)}$ on the gauge region, with χ as above. It changes the mixed limits to the stationary pairs of $g_\mu^*B_1$ and $g'_\mu^*B'_1$, and at $\mu = 1$ the region $r \geq -4$ carries pr^*c with $c = g_1^*B_1 \cup \text{pr}^*b \cup g_1'^*B'_1$, a representative of the critical point over x in the sense above. The configuration therefore lies in $\mathcal{T}_x$. A member of $\mathcal{T}_x$ is carried to the constant pair of $g_\mu^*B \cup \text{pr}^*b \cup g_\mu'^*B'$, which is again a representative since $g_\mu = 1$ on $[-5/2, 0]$ in that case; so members of $\mathcal{T}_x$ remain in $\mathcal{T}_x$.

Proof of Proposition 2.2(a) for finite torus lengths. Concatenate Lemma B.1, Lemma B.3 and the straightening, and transport by the identifications of Section B.1. Relative preparation of the original tails is the second step of Lemma B.1. Continuity in the data holds because every choice above (the collar gauges, the charts at stationary pairs, the lifts of Lemma B.2, the horizontal lifts, the widening gauges) depends continuously on the primary data, and the identifications depend continuously on the lengths. The deformation is the identity on, or a translation of, the gauge input end away from $[-5, -3] \times Y_\#$. Hence if the family has the form $A_0 \#_{\ell} c$ with an ordinary configuration A_0 glued at the input with a long neck ℓ , the deformation is $A_0 \#_{\ell+\tau} c_\tau$, and the determinant gluing maps of Lemma 3.3 are continuous along it. This is the coherence with ordinary end gluing. Families with strips glued at the output are families of the same kind. $\square$

B.4. Completed torus domains. Let a compact family on $W_{a,\infty}$ be given. By Lemma 8.6(i) it is in torus normal form beyond some r_0 after a continuous family of small gauge transformations. The truncation $\alpha + (1 - t + t\chi_R)b$ on the tail, with $\chi_R = 1$ for $r \leq R$ and 0 for $r \geq R + 1$, is continuous in the topology of Section 2, whose weight is independent of r , and fixes members that are α beyond R ; at $t = 1$ every member is α beyond $R + 1$. Such a configuration is the extension by α of a configuration on $W_{a,R+2}$ which equals α near the flat face, and the face is part of the matching seam. The deformations of Sections B.2–B.3 preserve this condition: the collar gauges and the chart cutoffs are the identity where the configuration and its limit are α , the widening gauges are supported away from the neck and the seam, the collar pullbacks fix configurations equal to α near the seam,

the seam term $\tilde{\psi}$ vanishes on the torus component, the horizontal lifts and the representatives are α on T , and the translation and the straightening bring the positive extension, which is α on T , to the neighbourhood of the face. Extending by α along the whole deformation gives Proposition 2.2(a) on $W_{a,\infty}$, ending in $\mathcal{T}_x$ with a representative which is α on T ; the constant pair is then α on the tail.

B.5. The localized several-input shift. On a triangle or folded triangle at positive collapse scale, and on a finite-width joined torus domain, the outgoing closed cut writes the gauge region as an ordinary cobordism followed by an output region M . For the joined torus domain this is Paper I, Lemma 10.18; for the triangles and folded triangles it is the cut of the proofs of Paper I, Theorems 9.10 and 11.8, through the collapsing region (Paper I, Lemmas 9.11 and 9.12). In each case M , with a smooth product collar at the cut, is identified with the complement of $(-\infty, -c_1] \times Y_\#$ in a special quintuple by a diffeomorphism of decorated manifolds, continuous in the width and in the scale on compact positive ranges; its compression bodies are those of the pair (D_0, D_1) , or of the marked pair of the triangle. The cut is $\{-c_1\} \times Y_\#$, with $c_1 > 5$. The steps of Sections B.2–B.3 are supported in $r \geq -5$, away from a collar of the cut, except for the translation. Replace the translation by the localized map: with χ nondecreasing, zero near $-c_1$ and one on $[-5, \infty)$, put $F_\kappa(r) = r + \kappa\chi(r)$ and

$$A_\kappa = F_\kappa^* \tilde{A}, \quad u_\kappa(r, t) = u(r + \kappa, t).$$

Since $F'_\kappa > 0$ this is a smooth configuration; it is a translation on $[-5, \infty)$, so the seam and the corners are treated as in Lemma B.3, and near the cut it is the identity, so it glues with the ordinary cobordism. For $\kappa = S + 4$ followed by the straightening and the collar widening, the region $r \geq -4$ carries a constant pair C^c , and the configuration is $A_E \# C^c$, with A_E an ordinary cobordism configuration ending at c on the parallel copy $\{-7/2\} \times Y_\#$ of the cut, whose collar $[-4, -3] \times Y_\#$ carries $\text{pr}^* c$, and with the original inputs unchanged. This is the deformation of Section 5, which proves Proposition 2.2(b). Relative and continuity statements follow as in Lemma B.3.

APPENDIX C. A SIGN CALCULATION FOR PROJECTIVE COMPOSITION

The conversion of a strictly commuting map F of degree f to a closed map is $F^r = F\varepsilon^f$. For another strictly commuting map G of degree g , the relation $\varepsilon^f G = (-1)^{fg} G\varepsilon^f$ gives

$$F^r G^r = (-1)^{fg} F G \varepsilon^{f+g} = (-1)^{fg} (FG)^r.$$

Thus this convention change is compatible with composition after projectivization. The factor is scalar, despite the grading operator used in converting each map.

For a strict homotopy $F - F' = dH + Hd$, where $|H| = f + 1$, the correctly converted homotopy is $H\varepsilon^f$. Indeed

$$\delta(H\varepsilon^f) = dH\varepsilon^f - (-1)^{f+1} H\varepsilon^f d = (dH + Hd)\varepsilon^f.$$

Using $H\varepsilon^{f+1}$ instead would give the wrong sign. This calculation is useful whenever a gauge composition homotopy is translated into the Hom complex convention.

Lastly, if $F_i - G_i = \delta K_i$ and the maps are closed, a homotopy between two composites is obtained by replacing one factor at a time and using (3.2). For example, for three factors one may take

$$K_3 F_2 F_1 + (-1)^{|G_3|} G_3 K_2 F_1 + (-1)^{|G_3|+|G_2|} G_3 G_2 K_1.$$

Applying δ telescopes to $F_3 F_2 F_1 - G_3 G_2 G_1$. This is the chain-level algebra used in Lemma 14.1.

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